TPL-01 · FREE TEMPLATES

Project controls execution plan

The plan that says which system is the source of truth for each field, and who is accountable when two systems disagree

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A project controls execution plan states how a specific project will be measured: which structures exist, what the baseline is, how progress is claimed and verified, what gets reported to whom and when, how change is controlled, which system is authoritative for each field, and where artificial intelligence may and may not touch the numbers. This template gives the section structure and, under every heading, the prompts you must answer — not prose to be adapted, but questions that expose the decisions most plans leave unmade.

About this document

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Project controls execution plan

In one paragraph. A project controls execution plan states how a specific project will be measured: which structures exist, what the baseline is, how progress is claimed and verified, what gets reported to whom and when, how change is controlled, which system is authoritative for each field, and where artificial intelligence may and may not touch the numbers. This template gives the section structure and, under every heading, the prompts you must answer — not prose to be adapted, but questions that expose the decisions most plans leave unmade.

Who this is for. Project controls managers and heads of project management office mobilising a project or a portfolio; cost engineers and planners inheriting a project with no written controls basis; and the project directors who will be asked to approve the plan.

1. When to use this

Write the plan during mobilisation, before the first baseline is approved and before the first report is issued. The plan is the thing the baseline and the reports are consistent with; producing it afterwards turns it into a description of whatever habits formed in the first two months.

Three other moments justify writing or rewriting one:

- **On taking over a project that has none.** The absence is itself the finding. Write the plan by documenting current practice honestly, marking every place where practice and intention differ, and presenting that gap list as the mobilisation backlog.
- **At a re-baseline.** A re-baseline changes what is measured against what. If the plan is not updated in the same approval, the next month's variances will be computed against a basis nobody has agreed.
- **When the delivery model changes** — a joint venture forms, a major package moves from remeasurable to lump sum, a second reporting entity appears. Each changes the source of truth for whole fields.

Do not use this template as a bid deliverable dressed as a plan. A controls execution plan that is written to be scored rather than to be followed is worse than none, because it creates a documented standard the project is then measured against and cannot meet.

2. How to complete it

Answer the prompts; do not adapt the prose. Every heading in §3 carries prompts rather than model text, because model text invites paraphrase and paraphrase hides the decisions. A prompt you cannot answer is a live issue for the mobilisation log, and writing "to be advised" against it is a legitimate and useful outcome — far better than a sentence that sounds settled and is not.

Name four things in every answer. The framework this series is written to treats detail as specificity: the artefact, the field, the frequency and the owner. "Progress is reviewed regularly" fails all four. "The control account manager submits the progress claim in the progress measurement sheet (`TPL-05`) by 12:00 on the second working day after cut-off; the cost engineer verifies the evidence reference on every line above 500 budget hours" passes.

Fill the definitions section first (§3.2). Most arguments in a controls function are definitional, not arithmetical. Two people disagreeing about the cost variance almost always agree about subtraction and disagree about whether an accrual is in the actual cost. Settle those in writing before the first report.

Decide the source of truth per field, not per system. Systems overlap. The useful statement is field-level: commitment value is authoritative in the procurement system; actual cost is authoritative in the finance ledger; earned value is authoritative in the controls workbook; forecast completion date is authoritative in the schedule tool. Then state what happens when two disagree — who reconciles, on what cycle, and what tolerance is accepted without escalation.

Get it approved by the person who will be asked to overrule it. A plan approved only inside the controls function has no authority in the month someone senior wants a number changed.

Using the tables. The tables in §3 are pipe-delimited Markdown. To move one into a spreadsheet, copy the block, paste it into a single column, split on the pipe character, and delete the alignment row.

3. The template

Copy from here. Numbering is designed for citation — a change request can reference "PCEP §3.5" and mean exactly one clause.

3.0 Document control

Field	Entry
Project / contract	
Plan version and date	
Prepared by (name, role)	
Reviewed by (name, role)	
Approved by (name, role, date)	
Supersedes	
Distribution	
Classification	
Next review trigger	

Prompts. What event — not what date — triggers the next review? Which of the client's own controls requirements does this plan sit beneath, and where are the two inconsistent?

3.1 Purpose and scope of the controls function

Prompts. Which contracts, packages, entities and joint-venture interests does this plan govern? What is explicitly outside it, and who controls those? Does the plan bind subcontractors, and through which contractual clause? Where the client has its own controls procedure, which prevails, and who has confirmed that in writing?

3.2 Definitions used on this project

Prompts. Define each of the following once, here, with the arithmetic where arithmetic applies: budget at completion (does it include the contingency reserve, or not — the choice governs every index reported in `TPL-07`); management reserve and who holds it; commitment; accrual; actual cost; per cent complete; earned value; milestone; float and who owns it; "approved change"; "authorised to proceed". For each, name the system that holds the value.

3.3 Structures and coding

Prompts. How many levels does the work breakdown structure have and at which level is the control account set? What is the sizing rule for a work package — by budget, by duration, or by both? Who owns the organisational breakdown structure and how is it mapped to the work breakdown structure? What is the cost breakdown structure and code of accounts, and how does it map to the general ledger? What is the resource breakdown structure? Who may create a new code, and against what test? Where are the structures published, and how is a change to a structure controlled? See `TPL-02` and `TPL-03`.

3.4 Baseline

Prompts. What documents constitute the baseline, at which version, frozen on what date? What is the performance measurement baseline and what sits outside it? Where is the contingency reserve held, who has custody, and what is the release test? Where is management reserve held and who authorises its release? Which schedule is the baseline schedule and how is it identified in the tool? What is measured against what — is a variance computed against the original baseline, the current approved baseline, or both, and in which report?

3.5 Change control

Prompts. What are the approval thresholds and who sits at each? What is the routing and the target turnaround at each stage? Which form is used and where does it live? Who maintains the register? What triggers a re-baseline and what is expressly forbidden as a reason for one? How are pending changes treated in the forecast — excluded, disclosed separately, or included with probability weighting? Who may draw on contingency, and what evidence must accompany the request? See `TPL-04`.

3.6 Progress measurement

Prompts. Which measurement techniques are permitted on this project, and which are prohibited? Where is the rule-of-credit library and who approves an entry in it? Who claims progress and who verifies it? What objective evidence is required at each credit step, and where is it filed? What is the cut-off calendar? How is level of effort treated and capped? What happens when a claimed quantity exceeds the budgeted quantity? See `TPL-05`.

3.7 Cost management

Prompts. What are the sources for commitments, actual costs and accruals, and what is the cut-off for each? What is the timesheet discipline and who enforces it? Who reconciles the controls workbook to the finance ledger, on what cycle, and what difference is tolerated without escalation? How are inter-company and joint-venture charges captured? How are currency and escalation handled, and on what basis are figures reported — nominal or constant prices?

3.8 Schedule management

Prompts. What schedule levels exist and who owns each? What is the update cycle and the statusing rule — actual dates only, or remaining duration overrides? What controls a change to logic or to calendars? Who owns float, and is it consumed first-come or allocated? What schedule quality check is run, by whom, and on what cadence? See `TPL-14`.

3.9 Forecasting

Prompts. Which estimate at completion methods are permitted, and who selects? Who prepares the forecast, who challenges it, and at what forum? How often is a bottom-up estimate to complete required rather than an index-based extrapolation? How is the selected method recorded and how is a change of method disclosed? See `TPL-08`.

3.10 Risk process

Prompts. Who owns the risk register and on what cadence is it reviewed? What are the probability and impact scales, and are they defined in absolute terms rather than adjectives? What triggers a quantitative schedule or cost risk analysis? How does risk exposure connect to the contingency position, and who makes that connection in writing? See `TPL-10` and `TPL-11`.

3.11 Reporting cadence and distribution

Report	Data date	Issue offset (working days)	Prepared by	Approved by	Distribution	Classification

Calculated column — issue date. In words: the issue date is the data date advanced by the stated number of working days, excluding weekends and the project holiday calendar. Spreadsheet, with the data date in `B2`, the offset in `C2` and a named range `Holidays`:

```
=IF(OR(B2="",C2=""),"",WORKDAY(B2,C2,Holidays))
```

Calculated column — working days actually taken. In words: the count of working days between the data date and the actual issue date, excluding the data date itself. With the actual issue date in `E2`:

```
=IF(OR(B2="",E2=""),"",NETWORKDAYS(B2,E2,Holidays)-1)
```

`NETWORKDAYS` counts both end dates, so subtract one to express elapsed working days.

Prompts. What is on each distribution list, by name and role rather than by team? What is the escalation route when a report is late, and at what point does lateness itself get reported? Which report is the single authoritative statement of position, so that a figure quoted from anywhere else is known to be indicative?

3.12 Systems and data

Data field	System of record	Owner	Update frequency	Feeds	Reconciled to	Tolerance

Prompts. For every field in the monthly report, which system is authoritative? Which integrations are automated and which are manual re-keying — and where a field is re-keyed, who checks it? What are the access levels and who approves them? What is the backup and archive arrangement, and who has tested a restore? Which data quality checks run, on what cadence, and who sees the exceptions?

3.13 Artificial intelligence: use and governance

The Institute's position is that artificial intelligence proposes and the professional disposes. Anything that reaches a report must be explainable, validated and owned by a competent human. This section makes that operational on one project.

Task	AI permitted?	Tool and version	Human validation gate	Validator (role)	Evidence retained	Data classification permitted

Prompts. Which controls tasks may be AI-assisted — narrative drafting, anomaly detection in cost ledgers, schedule logic screening, document search, first-pass risk identification? Which are prohibited outright, and why? For each permitted task, who is the named human who validates the output before it leaves the controls function, and what does validation consist of? What project data may be sent to a model hosted outside the project's own environment, and what may not? How is an AI-assisted output labelled in the report, so a reader knows what they are reading? What evidence is retained so the contribution of a model to a reported number can be reconstructed at audit? Who is notified when a tool's model version changes, and does the validation gate change with it? On what cadence is this register reviewed?

3.14 Roles, responsibilities and RACI

Activity	Controls manager	Cost engineer	Planner	Control account manager	Project director	Finance

Use R (responsible), A (accountable), C (consulted), I (informed). Exactly one A per row.

Calculated column — accountability check. In words: flag any activity row that does not carry exactly one accountable party. With the role columns in `B2:G2`:

```
=IF(COUNTIF(B2:G2,"A")=1,"OK","CHECK — one A per row")
```

Prompts. What are the delegation limits in value and in kind? What are the cover arrangements when the accountable person is unavailable, and are they named? Which decisions may the controls function make and which may it only recommend?

3.15 Interfaces

Prompts. Which internal functions does controls depend on for data, and what does each owe by when — finance, procurement, engineering, construction, commercial, human resources? Which external parties, and under which contractual reporting obligation? Where is the interface register? What is the agreed data exchange format and calendar for each? What happens when an input is late — does the report slip, or does it issue with a stated gap?

3.16 Assurance

Prompts. What does the function check on itself, how often, and who reviews the result? When is an independent review carried out and by whom? What is the health-check instrument and its cadence — see `TPL-15`. What audit trail exists for a number that changed between two reports?

3.17 Mobilisation and closeout

Prompts. What must exist by which date for the function to be operating — structures approved, baseline loaded, rules of credit signed off, first report issued? At closeout, what data is handed back, in what format, to whom, and by when? What is captured for benchmarking and lessons learned, and who owns that after the project team disperses? See `TPL-16`.

3.18 Appendices

Cut-off and reporting calendar · distribution matrix · forms and templates index · glossary of terms used on this project · list of open items from §3.2 onwards.

4. Worked fragment

Illustrative figures. An extract from a completed §3.11 and §3.14 for a fictional facility upgrade project. Dates use the calendar for 2026; the holiday calendar is assumed empty across the range, and the week is Monday to Friday.

§3.11 Reporting cadence — extract

Report	Data date	Issue offset (working days)	Prepared by	Approved by	Distribution	Classification
Weekly progress flash	Friday	+1	Planner	Controls manager	Project delivery team	Internal
Monthly project controls report	Last calendar day of month	+6	Controls manager	Project director	Client project manager, sponsor, function heads	Confidential
Cost-to-ledger reconciliation	Finance month-end close	+8	Cost engineer	Project accountant	Controls manager, finance business partner	Internal
Quarterly forecast review pack	Quarter end	+10	Controls manager	Project director and sponsor	Steering group	Confidential

Worked issue date for the monthly report at the May cut-off: the data date is Sunday 31 May 2026; six working days advance through Monday 1, Tuesday 2, Wednesday 3, Thursday 4 and Friday 5 June, then Monday 8 June. `=WORKDAY(DATE(2026,5,31),6)` returns **Monday 8 June 2026**. The May report in `TPL-06` carries that issue date.

§3.14 RACI — extract

Activity	Controls manager	Cost engineer	Planner	Control account manager	Project director	Finance	Check
Submit monthly progress claim	C	C	I	**A** / R	I	—	OK
Verify progress evidence	**A**	R	C	I	—	—	OK
Post accruals at cut-off	I	R	—	C	—	**A**	OK
Approve a change up to 250 budget units	C	R	C	I	**A**	I	OK
Select the estimate at completion method	**A**	R	C	C	C	I	OK

The check column applies the formula in §3.14 and returns `OK` because each row carries exactly one A. The row "post accruals at cut-off" is the one worth arguing about: accountability sits with finance because the ledger is the system of record for actual cost under §3.12, while the cost engineer does the work. Writing it the other way round is the most common source of a month-end reconciliation that never closes.

5. Common mistakes

Describing the discipline instead of the project. A section that explains what earned value is has been copied from a textbook. The plan is not a teaching document; it says what this project does, at which level, with which technique, verified by whom.

Leaving budget at completion undefined. Whether it includes the contingency reserve changes every variance, every index and every forecast in every report. Two organisations can both be right and the joint venture between them will publish two different cost performance indices for the same month.

A source-of-truth table by system rather than by field. "Cost is in the finance system, schedule is in the planning tool" leaves commitments, accruals, forecast dates and earned value unallocated — and those are precisely the fields that get argued about.

RACI with two accountable parties, or with none. A row with two As is a row where nobody decides. The check formula in §3.14 exists because this is very common and invisible at a glance.

A reporting calendar with no offsets. "Monthly report issued in the first week" is not a commitment anyone can plan around, and it silently permits the data date to drift towards the issue date until the report describes a position nobody recognises.

An artificial intelligence section that lists principles. "AI will be used responsibly" is not a control. The control is a named task, a named validator, a stated data classification and a retained evidence trail. If §3.13 could be pasted into any other project's plan unchanged, it does nothing.

Approval by the controls function only. The plan's purpose is to be quotable when someone senior asks for a number to be presented differently. That requires a signature from someone senior.

6. Adapting it

Safe to change. Section order, provided cross-references are updated. The role names, to match your organisation. The number of reports and their cadence. The approval thresholds, which are project-specific by nature. Adding sections — contract administration, document control, materials management, sustainability reporting — where the project needs them.

Change with care. Merging §3.12 (systems and data) into §3.3 (structures) loses the field-level source-of-truth statement, which is the section most often cited later. Merging §3.5 (change control) into §3.4 (baseline) tends to lose the routing and threshold detail.

Do not remove. The definitions section (§3.2), the field-level system of record table (§3.12), the accountability table (§3.14) and the artificial intelligence register (§3.13). Those four are what make the document a control rather than a description. A plan without them will read well and settle nothing.

On a small project, keep every heading and let the answers be short. A one-line answer under a heading is a decision recorded. A deleted heading is a decision not taken, and it will be taken later by whoever is in the room.

7. Completion checklist

- [ ] Every prompt in §3 has an answer or a dated open item with an owner
- [ ] Budget at completion, actual cost, accrual and per cent complete are defined with their arithmetic
- [ ] Every field in the monthly report appears in the §3.12 system-of-record table
- [ ] Reconciliation cycle and tolerance stated, with a named reconciler
- [ ] Every RACI row returns `OK` on the one-accountable check
- [ ] Every permitted AI task has a named human validator and a stated data classification
- [ ] Prohibited AI uses stated explicitly, not implied by omission
- [ ] Reporting calendar has data dates, working-day offsets and named approvers
- [ ] Contingency and management reserve custody and release tests stated
- [ ] Change thresholds and routing agreed by the approvers named in them
- [ ] Approved by the project director or equivalent, not only within the controls function
- [ ] Review trigger stated as an event, not a date

Related

- `TPL-02 — Work breakdown structure and WBS dictionary` — the structure §3.3 requires you to define
- `TPL-05 — Progress measurement and rules of credit sheet` — the instrument that delivers §3.6
- `TPL-06 — Monthly project controls report` — the output §3.11 schedules
- `TPL-15 — Project controls health check` — the assurance instrument referenced at §3.16
- `BPG-01 — Building a project controls function from zero` — the reasoning behind the sequence of §3
- `BPG-19 — Project controls assurance and health checks` — how to test whether the plan is being followed
- `AIG-08 — Governing AI on a project — the control framework` — the full treatment behind §3.13

Sources and standards

- PCI Canonical Facts (`docs/publication-framework/00-framework/CANONICAL-FACTS.md`), verified August 2026: the Institute's position on responsible, governed use of artificial intelligence, from which §3.13 takes its validation-gate structure.

This template is an original instrument. Where the section structure resembles established practice, that is because the underlying activities — baselining, change control, progress measurement, reporting — are common to the discipline. No third-party plan, form or checklist is reproduced, and no standard is quoted.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-02 · FREE TEMPLATES

Work breakdown structure and WBS dictionary

The hierarchy table, the dictionary entry form, and the roll-up formulas that keep them consistent

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A work breakdown structure decomposes the scope of a project into elements that can be budgeted, assigned and measured; the dictionary is where each element acquires a description, a deliverable, acceptance criteria, an owner and a budget. This template provides both — a hierarchy table with level, parent and roll-up columns that compute themselves, and a dictionary entry form with the ten fields that determine whether a work package can be argued about at handover.

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Work breakdown structure and WBS dictionary

In one paragraph. A work breakdown structure decomposes the scope of a project into elements that can be budgeted, assigned and measured; the dictionary is where each element acquires a description, a deliverable, acceptance criteria, an owner and a budget. This template provides both — a hierarchy table with level, parent and roll-up columns that compute themselves, and a dictionary entry form with the ten fields that determine whether a work package can be argued about at handover.

Who this is for. Project controls managers, cost engineers and planners setting up a project; and the control account managers who will be asked to own an element and should insist on a dictionary entry before they do.

1. When to use this

Build the structure before the estimate is loaded and before the schedule is coded, because both will inherit whatever the structure decides. Anything created afterwards has to be mapped back, and mapping is where scope quietly falls between two elements.

The structure earns its keep in four places, and if it is not serving all four it has been built for one audience only:

- **Budget.** Every currency unit of the estimate lands on exactly one budget-bearing element.
- **Schedule.** Every activity rolls up to exactly one element.
- **Assignment.** Every element at control-account level has one named owner.
- **Measurement.** Progress and earned value are reported against elements, not against activities.

Rebuild it when scope changes in kind rather than in quantity — a new facility, a new contract, a new delivery entity. Do not rebuild it because the numbering has become untidy; renumbering a live structure breaks every historical comparison in the project, and untidiness is cheaper than that.

2. How to complete it

Decompose by deliverable, not by activity or by discipline. The test at each level is whether the child elements together produce the parent and nothing else. "Piling and foundations" is a deliverable. "Civil engineering" is a discipline and will collect work from three unrelated deliverables. "Excavate, then pile, then pour" is a sequence and belongs in the schedule.

Stop decomposing when the element passes the sizing rule. Set that rule in the controls execution plan (`TPL-01` §3.3) and apply it without exception. A workable rule has two limbs: a work package should not exceed one reporting period's worth of measurable output, and it should not be so small that the effort of statusing it exceeds the value of knowing.

Set the control-account level deliberately. The control account is where budget, schedule and responsibility meet, and it is the level at which cost performance is reported. Setting it too high hides the variance inside an aggregate; too low produces indices computed on numbers too small to be stable.

Budget only budget-bearing elements. Enter values against work packages and planning packages only. Everything above them rolls up. If you enter a budget against a summary element as well, the total will double-count and the error will not be obvious because both numbers look plausible.

Store the identifier column as text. A structure identifier such as `1.2` is a decimal number to a spreadsheet unless the column is formatted as text, and the roll-up formulas below match on text. Formatting the column as text before typing anything is the single change that prevents most of the trouble people have with this sheet.

Write the dictionary entry when the element is created, not later. The entry takes fifteen minutes at creation and an argument at handover.

Using the tables. Copy a table block, paste into a single spreadsheet column, split on the pipe character, and delete the alignment row.

3. The template

3.1 Sheet 1 — the hierarchy

Col	Field	Type	Definition and entry rule
A	WBS ID	Text	Dot-separated numeric path, e.g. `1.2.3`. Format the column as text before entry. Never reused, never renumbered once budgets are loaded.
B	Level	Calculated	Depth in the hierarchy, where the project element is level 1.
C	Parent WBS ID	Calculated	The identifier one level up. Blank for the project element.
D	Element title	Text	A noun phrase naming the deliverable. Not a verb, not a discipline.
E	Element type	List	`Project` · `Summary` · `Control account` · `Work package` · `Planning package`
F	Budget-bearing	List	`Yes` for work packages and planning packages; `No` for everything else. Drives the roll-up.
G	Control account code	Text	The control account this element belongs to or is. Blank above control-account level.
H	Responsible party	Text	Named role in the organisational breakdown structure. One only.
I	Budget entered	Number	Cost budget. Enter only where column F is `Yes`; leave blank elsewhere.
J	Rolled-up budget	Calculated	The element's total budget, computed from the budget-bearing elements beneath it.
K	% of project budget	Calculated	The element's share of the project total.
L	Budgeted hours	Number	Direct labour hours, where hours are controlled separately from cost. Same entry rule as column I.
M	Dictionary status	List	`Not started` · `Drafted` · `Approved` · `Superseded`
N	Notes	Text	Anything a reader of the hierarchy needs that is not in the dictionary.

Calculated column B — Level. In words: the level is the number of dot separators in the identifier, plus one. Spreadsheet:

```
=IF(A2="","",LEN(A2)-LEN(SUBSTITUTE(A2,".",""))+1)
```

`SUBSTITUTE` removes the dots; the difference in length is the count of dots. `1.2.3` has two dots and is level 3.

Calculated column C — Parent WBS ID. In words: everything to the left of the last dot; blank if there is no dot. Spreadsheet:

```
=IF(ISERROR(FIND(".",A2)),"",LEFT(A2,FIND("~",SUBSTITUTE(A2,".","~",LEN(A2)-LEN(SUBSTI-
TUTE(A2,".",""))))-1))
```

The inner `SUBSTITUTE` replaces only the last dot with a tilde — the fourth argument selects which occurrence — so `FIND` can locate it. For `1.2.3` this returns `1.2`.

Calculated column J — Rolled-up budget. In words: if the element carries a budget itself, use it; otherwise sum the budgets of every budget-bearing element whose identifier begins with this element's identifier followed by a dot. Spreadsheet, over a 200-row sheet:

```
=IF($F2="Yes",$I2,SUMIFS($I$2:$I$200,$F$2:$F$200,"Yes",$A$2:$A$200,$A2&".*"))
```

The criterion `$A2&".*"` uses the wildcard `*`, which `SUMIFS` supports on text. Including the dot in the criterion is deliberate: it prevents `1.1` from matching `1.10`. The formula is level-independent — the same expression works at every level and never double-counts, because it sums only budget-bearing rows.

Calculated column K — % of project budget. In words: the element's rolled-up budget divided by the project total, which sits in the level-1 row. Spreadsheet, with the project row at row 2:

```
=IF($J$2=0,"",J2/$J$2)
```

Format as a percentage. Store percentages as decimal fractions with percentage formatting; do not type `50` into a cell formatted as a percentage, which stores 5,000 per cent.

Optional integrity check. In words: flag any row whose parent identifier does not exist in the sheet. Spreadsheet:

```
=IF(C2="","",IF(COUNTIF($A$2:$A$200,C2)=0,"ORPHAN — parent not in sheet",""))
```

3.2 Sheet 2 — the dictionary entry form

One form per element at control-account level and below. Reproduce the block for each element.

Field	Entry	Notes on completion
WBS ID		Must exist in Sheet 1
Element title		Identical to Sheet 1, column D
Level / parent		From Sheet 1, columns B and C
Element type		Work package, planning package or control account
Description of work		What is done, in three to six sentences. Written so a competent stranger could scope it.
Deliverable(s)		The physical or documentary output. Countable where possible.
Acceptance criteria		The test that decides whether the deliverable is complete. Objective, evidenced, and agreed by whoever accepts it.
Responsible party		Name and role. One accountable party, not a team.
Control account code		The account against which cost performance is reported
Cost breakdown structure codes		The codes to which cost in this element is charged — see `TPL-03`
Budget — cost		Currency, amount, and the basis on which it is stated
Budget — hours		Direct labour hours, if controlled
Basis of estimate reference		Document and version. Not a description; a reference.
Progress measurement technique		Which technique and which rule of credit applies — see `TPL-05`
Planned start / finish		From the baseline schedule, with the schedule activity identifiers
Predecessors and interfaces		What must exist before this element can proceed, and who owes it
Assumptions		Every assumption the budget and duration depend on. If it turns out false, this is the list that justifies a change request.
Exclusions		What a reader might reasonably expect to be here and is not, with the element identifier where it actually sits
Risk register references		Risks whose realisation would change this element
Approved by / date		The person who will be held to the budget
Revision		Version and change reason

The two fields people skip. Acceptance criteria and exclusions are the fields that do the work. An element with a description and a budget but no acceptance criteria can be declared complete by anyone with an opinion. An element with no exclusions absorbs, by default, everything adjacent to it that nobody else claimed.

4. Worked fragment

Illustrative figures. A fictional facility upgrade project. Currency is stated in generic currency units (CU) in thousands; the basis is nominal, single currency, with no escalation applied. Budgets are distributed control-account budget only; the contingency reserve is held outside the structure and is not shown here. Percentages are rounded to one decimal place, half away from zero.

4.1 Hierarchy extract

WBS ID	Level	Parent	Element title	Type	Budget-bearing	CA code	Responsible	Budget entered	Rolled-up budget	% of project
1	1		Facility upgrade project	Project	No		Project director		8,000	100.0 %
1.1	2	1	Civil works	Control account	No	CA-1000	Civil delivery manager		4,000	50.0 %
1.1.1	3	1.1	Site preparation and earthworks	Work package	Yes	CA-1000	Civil delivery manager	650	650	8.1 %
1.1.2	3	1.1	Piling and foundations	Work package	Yes	CA-1000	Civil delivery manager	1,900	1,900	23.8 %
1.1.3	3	1.1	Structural steel	Work package	Yes	CA-1000	Civil delivery manager	1,050	1,050	13.1 %
1.1.4	3	1.1	Civil supervision and quality assurance	Work package	Yes	CA-1000	Civil delivery manager	400	400	5.0 %
1.2	2	1	Mechanical works	Control account	No	CA-2000	Mechanical delivery manager		3,000	37.5 %
1.2.1	3	1.2	Equipment procurement	Work package	Yes	CA-2000	Procurement lead	1,800	1,800	22.5 %
1.2.2	3	1.2	Mechanical installation	Work package	Yes	CA-2000	Mechanical delivery manager	1,200	1,200	15.0 %
1.3	2	1	Controls and commissioning	Control account	No	CA-3000	Commissioning manager		1,000	12.5 %
1.3.1	3	1.3	Control system supply and configuration	Work package	Yes	CA-3000	Systems engineer	600	600	7.5 %
1.3.2	3	1.3	Commissioning and handover	Planning package	Yes	CA-3000	Commissioning manager	400	400	5.0 %

Verification of the roll-up. Civil works: 650 + 1,900 + 1,050 + 400 = 4,000. Mechanical works: 1,800 + 1,200 = 3,000. Controls and commissioning: 600 + 400 = 1,000. Project: 4,000 + 3,000 + 1,000 = 8,000, which matches the sum of the eight budget-bearing rows and is the figure the earned value sheet in `TPL-07` uses as its budget at completion.

Verification of the percentages. The level-3 shares of the project total sum to 100.0 per cent: 8.125 + 23.75 + 13.125 + 5.0 + 22.5 + 15.0 + 7.5 + 5.0 = 100.0. Displayed to one decimal place they read 8.1, 23.8, 13.1, 5.0, 22.5, 15.0, 7.5 and 5.0 — which sum visibly to 100.1 because of rounding. Do not force the displayed figures to add; footnote the rounding instead.

Element `1.3.2` is a planning package rather than a work package: the commissioning scope is budgeted but not yet decomposed, and it carries a date by which it must be. That is a legitimate state during execution and a serious one at closeout.

4.2 Dictionary entry extract — element 1.1.2

Field	Entry
WBS ID	1.1.2
Element title	Piling and foundations
Level / parent	3 / 1.1
Element type	Work package
Description of work	Install bored piles to the design schedule across the utilities building footprint, construct pile caps and ground beams, and complete backfill to formation level. Includes setting out, pile integrity testing and reinstatement of the working platform. Excludes the working platform itself, which is constructed under 1.1.1.
Deliverable(s)	240 bored piles installed and tested to the specified acceptance regime; 60 pile caps and associated ground beams cast and cured.
Acceptance criteria	Pile integrity test results issued and accepted for every pile; concrete cube results at 28 days meeting the specified characteristic strength for every pour; as-built setting-out survey issued and within the specified tolerance; no open non-conformance reports against the element.
Responsible party	Civil delivery manager (named individual)
Control account code	CA-1000
Cost breakdown structure codes	`FU1-10-CIV-10-SK`, `FU1-10-CIV-10-SR`, `FU1-10-CIV-20-BU`, `FU1-10-CIV-40-RM`
Budget — cost	CU 1,900 thousand, nominal, no escalation
Budget — hours	9,000 direct labour hours
Basis of estimate reference	Estimate BOE-CIV-004 rev C
Progress measurement technique	Piling: units complete on pile count. Pile caps: weighted milestone. See `TPL-05`.
Planned start / finish	6 April 2026 / 30 October 2026 (activities ACT-1020, ACT-1030)
Predecessors and interfaces	Working platform complete and surveyed (1.1.1); piling design issued for construction by engineering; ground investigation report accepted by the client.
Assumptions	No obstructions below formation level requiring removal; continuous access to the full footprint from the start date; single piling rig throughout; cost includes materials, plant and subcontract, while the hours figure is direct labour only.
Exclusions	Working platform (1.1.1). Ground improvement or obstruction removal, which is not in the baseline. Building slab, which is in 1.1.3.
Risk register references	R-014 unknown obstructions; R-022 piling rig availability
Approved by / date	Project director, 18 February 2026
Revision	Rev B — assumption on rig count added after estimate review

The first assumption in that entry is doing a great deal of work. When obstructions are found, the existence of the written assumption is the difference between a change request and an argument. The worked change request in `TPL-04` is the one that follows from it.

5. Common mistakes

Decomposing by discipline. A branch called "Civil", "Mechanical" and "Electrical" reads tidily and measures nothing, because a single deliverable is then split across three branches and no element can be declared complete. Discipline belongs in the cost breakdown structure (`TPL-03`), where it is a coding segment, not in the work breakdown structure, where it is a scope boundary.

Budgeting summary elements as well as work packages. The total doubles somewhere and the error is invisible because every individual number looks right. Column F exists to make this structurally impossible.

Renumbering a live structure. Every historical report, change request, risk reference and schedule code then points at the wrong element. If an identifier is wrong, retire it and create a new one; never reassign.

Identifiers stored as numbers. `1.10` and `1.1` become the same value, the roll-up wildcard stops matching, and the level formula returns nonsense. Format the column as text first.

A dictionary that repeats the title. "Description: piling and foundations works" adds nothing. If the description does not let a competent stranger scope the element, it is not a description.

Acceptance criteria written as adjectives. "Completed to a satisfactory standard" cannot be failed and therefore cannot be passed. Name the test, the document that records it and the person who signs it.

No exclusions. The element with no exclusions is the one that absorbs everything nobody else claimed, and the absorption is discovered at the point when the budget is already spent.

Planning packages that are never decomposed. A planning package is a promise to decompose later. Give each one a date and put that date on the mobilisation log, or it will still be a planning package at handover.

6. Adapting it

Safe to change. The number of levels. Identifier formats — alphanumeric segments work with the level and parent formulas provided the separator is a dot and the column is text. Adding columns for contract package, site or delivery entity. Splitting the hierarchy sheet by branch on a very large project, provided the roll-up ranges are widened to match.

Change with care. Moving the control-account level after budgets are loaded changes what every historical index was computed on. If you must, keep both and report the change explicitly for one period.

Do not remove. Column F (budget-bearing), because without it the roll-up double-counts; and the acceptance criteria and exclusions fields in the dictionary, because those are the two fields the document exists for.

On a small project, the hierarchy may be three levels and the dictionary a single sheet with one row per element and the same field names as column headings. The fields do not shrink; only the formatting does.

7. Completion checklist

- [ ] WBS ID column formatted as text before any entry
- [ ] Every level-3 and below element decomposes its parent completely and adds nothing else
- [ ] Sizing rule from `TPL-01` §3.3 applied without exception
- [ ] Control-account level set and stated
- [ ] Budgets entered only where budget-bearing is `Yes`
- [ ] Roll-up total equals the sum of budget-bearing rows, checked independently
- [ ] Orphan check returns no results
- [ ] Every control account has one named responsible party
- [ ] Every element at control-account level and below has a dictionary entry
- [ ] Every dictionary entry has objective acceptance criteria and explicit exclusions
- [ ] Every assumption that the budget depends on is written down
- [ ] Every planning package has a date by which it will be decomposed
- [ ] Structure agreed with the schedule owner and the cost breakdown structure owner before budgets load

---

Related

- `TPL-01 — Project controls execution plan` — where the sizing rule and control-account level are set
- `TPL-03 — Cost breakdown structure and code of accounts` — the coding dimension this structure does not carry
- `TPL-05 — Progress measurement and rules of credit sheet` — how the elements defined here are measured
- `BPG-02 — The work breakdown structure` — the reasoning behind decomposition choices
- `BPG-03 — Cost breakdown structure and the code of accounts` — how the two structures work together

Sources and standards

No external source is cited. This is an original instrument; the decomposition principles it applies are common to established project management and cost engineering practice, explained here in the Institute's own words, and no third-party structure, template or wording is reproduced.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-03 · FREE TEMPLATES

Cost breakdown structure and code of accounts

A five-segment coding structure, its mapping to the general ledger, and the rules for adding a code

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A cost breakdown structure decides what questions you will be able to answer about cost for the life of the project, because a cost can only be analysed along the dimensions it was coded with. This template gives a five-segment code of accounts — project, area, discipline, cost type and resource class — with each segment's values defined, a mapping column to the general ledger, formulas that build and validate the code, and the six rules that govern adding a new one.

About this document

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Cost breakdown structure and code of accounts

In one paragraph. A cost breakdown structure decides what questions you will be able to answer about cost for the life of the project, because a cost can only be analysed along the dimensions it was coded with. This template gives a five-segment code of accounts — project, area, discipline, cost type and resource class — with each segment's values defined, a mapping column to the general ledger, formulas that build and validate the code, and the six rules that govern adding a new one.

Who this is for. Cost engineers and cost controllers setting up or repairing a coding structure; project controls managers who must agree it with finance; and project accountants who will have to live with the mapping.

1. When to use this

Set the structure before the first commitment is raised. A cost that has been incurred against a code cannot be re-analysed along a dimension the code never carried; it can only be reclassified, and reclassification across a closed accounting period is an accounting event with its own approval requirements rather than a spreadsheet edit.

Use it in three situations:

- **Project set-up**, alongside the work breakdown structure (`TPL-02`) and before the estimate is loaded. The two structures are different questions: the work breakdown structure asks what scope, the cost breakdown structure asks what kind of cost. Every commitment carries one identifier from each.
- **Taking over a project whose reports cannot be broken down.** If nobody can tell you what proportion of spend to date is subcontract, the structure is the cause and no amount of reporting effort substitutes.
- **Before a portfolio comparison.** Two projects can only be compared on cost type or discipline if both coded for it. Retrofitting a comparison across projects that coded differently produces a number with no defensible meaning.

2. How to complete it

Design the segments backwards from the questions. List the cost questions the project must answer — to the client, to finance, to the estimating function for future benchmarking, to the tax and statutory reporting process. Each recurring question is a segment. A segment that answers no question anyone has asked is overhead on every transaction for the life of the project.

Keep segments orthogonal. Each segment answers a different question, and a value in one segment should never imply a value in another. If discipline `CIV` always appears with cost type `40`, one of the two is redundant and the structure is more granular than it is informative.

Fix the segment lengths and never vary them. Fixed-length segments allow the code to be parsed by position, which is what makes a flat code list from a finance system usable. Variable-length segments force every consumer to parse on the separator, and one missing separator breaks the whole extract.

Include a "not applicable" value in every segment. Plant hire has no meaningful skill class; a project-wide cost has no area. Without an explicit value such as `ZZ` or `00`, users will either leave the segment blank — which breaks fixed-length parsing — or invent a value.

Map every code to exactly one general ledger account before opening it. The mapping is the point where project reporting and statutory reporting are reconciled. A code with no mapping produces a cost that appears in the project report and cannot be found in the accounts, and that discrepancy is always discovered at the least convenient moment.

Do not encode an accounting policy decision in the structure. Whether an item is capitalised or expensed depends on the financial reporting framework the reporting entity applies and on that entity's own accounting policy, and it varies between jurisdictions and between entities. The mapping table records that determination and names who made it; it does not make it, and this template does not state what the answer should be.

Using the tables. Copy a table block, paste into a single spreadsheet column, split on the pipe character, and delete the alignment row.

3. The template

3.1 The code structure

Five segments, separated by hyphens, fixed length, total sixteen characters:

```
PPP-AA-DDD-TT-RR
 │   │   │   │  └── Resource class      2 characters
 │   │   │   └───── Cost type           2 digits
 │   │   └───────── Discipline          3 letters
 │   └───────────── Area                2 digits
 └───────────────── Project             3 characters
```

Example: `FU1-10-CIV-10-SK` reads as project FU1, area 10, civil discipline, direct labour, skilled craft.

3.2 Segment definitions

Segment 1 — Project (3 characters). The project, contract or delivery entity whose cost this is. Assigned centrally, not by the project. Needed even on a single-project sheet, because the sheet will eventually be consolidated.

Segment 2 — Area (2 digits). The physical, geographic or facility area in which the cost is incurred.

Value	Meaning
00	Project-wide; not attributable to an area
10	(define — e.g. utilities building)
20	(define — e.g. process area)
30	(define — e.g. external works and site infrastructure)
90	Temporary works and site establishment
ZZ	Not applicable

Segment 3 — Discipline (3 letters). The engineering or delivery discipline that owns the work. Suggested values: `CIV` civil, `STR` structural, `MEC` mechanical, `PIP` piping, `ELE` electrical, `ICA` instrumentation and control, `ARC` architectural, `PMC` project management and controls, `HSE` health, safety and environment, `ZZZ` not applicable.

Segment 4 — Cost type (2 digits). The nature of the cost. This is the segment finance cares about and the one most often left out.

Value	Cost type	Includes
10	Direct labour	Own-workforce labour charged to the project at a rate
20	Permanent materials	Materials and equipment forming part of the finished asset
30	Construction plant and equipment	Owned, hired or operated plant used to build, not delivered
40	Subcontract	Work executed under a subcontract, of any pricing basis
50	Site indirects	Site establishment, supervision, temporary facilities, site services
60	Expenses and other	Travel, permits, insurances charged to the project, sundry
70	Risk provision	Contingency held against identified risk. Never charged directly; released by change (`TPL-04`)
80	Escalation provision	Provision for price movement, where held separately

Segment 5 — Resource class (2 characters). The rate-bearing resource group within the cost type. The valid values depend on the cost type, which is deliberate and must be enforced by the code register rather than by a free-text field.

Cost type	Valid resource classes
10 Direct labour	`EN` engineering and technical staff · `SR` supervision · `SK` skilled craft · `SS` semi-skilled and general
20 Permanent materials	`BU` bulk materials · `TE` tagged equipment · `CS` consumables
30 Plant	`OP` operated hire · `DR` dry hire · `OW` owned plant
40 Subcontract	`LS` lump sum · `RM` remeasurable · `DT` daywork and time-and-materials
50, 60, 70, 80	`ZZ` not applicable

3.3 Sheet 1 — the code register

Col	Field	Type	Definition and entry rule
A	Project	Text	Segment 1
B	Area	Text	Segment 2. Store as text so leading zeros survive.
C	Discipline	Text	Segment 3
D	Cost type	Text	Segment 4. Store as text.
E	Resource class	Text	Segment 5
F	Full code	Calculated	The assembled sixteen-character code
G	Description	Text	Plain-language description of what belongs here
H	Budget	Number	Approved budget against this code
I	General ledger account	Text	The single account this code maps to
J	General ledger account name	Text	As it appears in the chart of accounts
K	Cost centre	Text	Where the receiving entity uses cost centres
L	Capital / expense determination	Text	The determination made, the policy relied on, and who made it. Not a project controls decision — see §2.
M	Opened by / date	Text	Who created the code and when
N	Effective from period	Text	The accounting period from which the code may be charged
O	Status	List	`Proposed` · `Open` · `Closed`
P	Length check	Calculated	Structural validation
Q	Duplicate check	Calculated	Uniqueness validation

Calculated column F — Full code. In words: the five segments joined by hyphens. Spreadsheet:

```
=IF(COUNTBLANK(A2:E2)>0,"",TEXTJOIN("-",TRUE,A2:E2))
```

Where `TEXTJOIN` is unavailable, use `=IF(COUNTBLANK(A2:E2)>0,"",A2&"-"&B2&"-"&C2&"-"&D2&"-"&E2)`.

Calculated column P — Length check. In words: a valid assembled code is exactly sixteen characters; anything else means a segment is the wrong length. Spreadsheet:

```
=IF(F2="","",IF(LEN(F2)=16,"OK","CHECK — segment length"))
```

Calculated column Q — Duplicate check. In words: flag a code that appears more than once in the register. Spreadsheet:

```
=IF(F2="","",IF(COUNTIF($F$2:$F$500,F2)>1,"DUPLICATE","OK"))
```

Roll-up by any segment. In words: sum the budget where the chosen segment column matches a given value. Spreadsheet, summing all civil budget:

```
=SUMIFS($H$2:$H$500,$C$2:$C$500,"CIV")
```

Always roll up on the segment columns, not on the assembled code. Where you receive a flat list of codes from a finance system with no segment columns, the segments can be recovered by position because the lengths are fixed — `LEFT(F2,3)` for project, `MID(F2,5,2)` for area, `MID(F2,8,3)` for discipline, `MID(F2,12,2)` for cost type and `MID(F2,15,2)` for resource class. Rebuild the segment columns from those and roll up on the columns; do not embed the extraction inside every summation.

3.4 Sheet 2 — the general ledger mapping summary

One row per general ledger account, for reconciliation with finance.

Col	Field	Definition
A	General ledger account	The account code
B	Account name	As in the chart of accounts
C	Cost breakdown structure codes mapped	List, or a count
D	Budget mapped	Sum of budget across the codes mapped to this account
E	Actual cost per project ledger	From the controls system
F	Actual cost per general ledger	From finance
G	Difference	Calculated
H	Difference %	Calculated
I	Explanation and owner	Required whenever column H exceeds the agreed tolerance

Calculated column G — Difference. In words: the project ledger figure less the general ledger figure. Spreadsheet: `=E2-F2`.

Calculated column H — Difference %. In words: the difference expressed as a proportion of the general ledger figure, which is the authoritative one. Spreadsheet:

```
=IF(F2=0,"",G2/F2)
```

Calculated column D — Budget mapped. In words: the total budget of every code mapped to this account. Spreadsheet, referencing Sheet 1:

```
=SUMIFS('Sheet 1'!$H$2:$H$500,'Sheet 1'!$I$2:$I$500,$A2)
```

3.5 Rules for adding a code

These belong in the controls execution plan (`TPL-01` §3.3) and are quoted here so the register carries them.

1. **A code is created by the cost controller on a written request**, never by a user typing a new value into a free-text field. The register in §3.3 is the only place codes exist.
2. **A new code is opened only when an existing code cannot carry the cost without losing a distinction somebody has asked for.** "It would be tidier" is not a reason; name the report the new code serves.

3. **Segment values are added, never redefined.** Changing what `20` means in the area segment invalidates every historical figure coded to it. If a value is wrong, close it and open a new one.
4. **A closed value is never reused.** Reuse makes two different things share a history.
5. **Every code maps to exactly one general ledger account before it is opened**, and the mapping is agreed with finance, not asserted by the project.
6. **A code is effective from a stated accounting period.** No retrospective charging into a closed period and no re-coding of a closed period without a documented reclassification approved through the same route as a journal entry.

4. Worked fragment

Illustrative figures. A fictional facility upgrade project. Currency is generic currency units (CU) in thousands, nominal basis, no escalation. The general ledger account numbers are an illustrative chart of accounts and are not a recommendation; every organisation's chart differs. Percentages are to two decimal places here because they are shares of a subtotal, and rounding them to one would obscure the check.

4.1 Code register extract — civil control account, area 10

Full code	Description	Budget	GL account	GL account name	Length check
FU1-10-CIV-10-SK	Civil direct labour — skilled craft	620	5100	Direct labour	OK
FU1-10-CIV-10-SR	Civil direct labour — supervision	180	5100	Direct labour	OK
FU1-10-CIV-20-BU	Civil permanent materials — bulk (concrete, reinforcement)	1,240	5200	Materials	OK
FU1-10-CIV-30-DR	Civil construction plant — dry hire	310	5300	Plant hire	OK
FU1-10-CIV-40-RM	Civil subcontract — remeasurable (piling)	1,450	5400	Subcontract	OK
FU1-10-CIV-50-ZZ	Civil site indirects	200	5500	Site indirects	OK
	Total — control account CA-1000	**4,000**			

Verification. 620 + 180 + 1,240 + 310 + 1,450 + 200 = 4,000, which is the budget of control account CA-1000 in the work breakdown structure at `TPL-02` §4.1 and the budget at completion for that control account in the earned value sheet at `TPL-07` §4.

Verification of the length check. `FU1-10-CIV-10-SK` is 3 + 1 + 2 + 1 + 3 + 1 + 2 + 1 + 2 = 16 characters, so column P returns `OK`. A code of any other length has a segment entered at the wrong width and will fail positional parsing downstream.

4.2 Roll-up by cost type — same extract

Cost type	Description	Budget	Share of control account
10	Direct labour	800	20.00 %
20	Permanent materials	1,240	31.00 %
30	Construction plant	310	7.75 %
40	Subcontract	1,450	36.25 %
50	Site indirects	200	5.00 %
	Total	**4,000**	**100.00 %**

Verification. Direct labour is 620 + 180 = 800. The shares are 800 ÷ 4,000 = 20.00 %, 1,240 ÷ 4,000 = 31.00 %, 310 ÷ 4,000 = 7.75 %, 1,450 ÷ 4,000 = 36.25 % and 200 ÷ 4,000 = 5.00 %, summing to 100.00 %.

This is the analysis the structure exists to produce, and it is the one that cannot be produced at all if the cost type segment was omitted at set-up. Note what it tells you that the work breakdown structure cannot: over a third of this control account is subcontract, so its cost performance will be driven by subcontract administration rather than by productivity, and the forecasting method chosen in `TPL-08` should reflect that.

5. Common mistakes

Building the cost breakdown structure as a copy of the work breakdown structure. They answer different questions. If the coding structure repeats the scope hierarchy, the project has one dimension of analysis where it needs two, and cost type — the dimension finance and estimating both want — is missing.

Omitting the cost type segment. It is the most commonly omitted and the most commonly needed. Without it there is no answer to what proportion of spend is labour, and therefore no basis for a productivity conversation or for a future estimate.

Variable-length segments. A structure where the project code is sometimes two characters and sometimes four cannot be parsed by position, and every downstream extract needs bespoke handling.

Codes stored as numbers. An area value of `01` becomes `1`, the assembled code fails the length check, and positional extraction returns the wrong segment.

A free-text code field on the requisition form. Within a month there will be three spellings of the same code and a reconciliation that never closes. Codes come from a controlled list.

One code, several general ledger accounts. As soon as a code can land in more than one account, the reconciliation in §3.4 stops being arithmetic and becomes an investigation.

Treating a capitalisation determination as a coding convention. Whether costs are capitalised or expensed follows the applicable financial reporting framework and the entity's accounting policy, varies between jurisdictions, and is finance's determination to make. The register records it and names who made it. A project controls function that decides it has created an accounting exposure it cannot see.

Charging directly to the risk provision code. Contingency is released into a control account by an approved change (`TPL-04`) and spent there. Charging costs straight to code type `70` destroys the audit trail between the risk that was identified and the money that was spent.

6. Adapting it

Safe to change. Segment lengths, provided they are fixed and the length check and positional extraction are updated to match. The values inside each segment. Adding a sixth segment — contract package, funding source, work-type — where a recurring question needs it. Reordering segments, provided you do it before any cost is incurred and update the extraction positions.

Change with care. Adding a segment to a live structure means every existing code is short and every historical comparison needs a default value. It is possible, but do it at a period boundary with a documented effective date, not mid-period.

Do not remove. The cost type segment, the general ledger mapping column, and the effective-from period. Those three are what make the structure reconcilable to the accounts rather than a private project taxonomy.

Where the client imposes a coding structure, adopt theirs as the reporting code and keep your own as the internal code, with the mapping held in the register as an additional column. Do not try to work in one structure that satisfies both; the compromise loses distinctions each side actually needs.

7. Completion checklist

- [ ] Every segment answers a question someone has actually asked
- [ ] Segment lengths fixed and documented; all segment columns formatted as text
- [ ] Every segment has an explicit "not applicable" value
- [ ] Valid resource classes constrained by cost type in the register, not left free-text
- [ ] Every code maps to exactly one general ledger account, agreed with finance
- [ ] Capital or expense determination recorded with the policy relied on and the person who made it
- [ ] Length check and duplicate check return `OK` on every row
- [ ] Roll-up by cost type reconciles to the control account budget
- [ ] Codes carry an effective-from accounting period
- [ ] The six rules in §3.5 are written into the controls execution plan and agreed with finance
- [ ] Requisition and timesheet systems draw the code from the controlled list, not from free text
- [ ] A reconciliation cycle and a tolerance are agreed before the first period close

---

Related

- `TPL-01 — Project controls execution plan` — where the coding rules and the reconciliation cycle are agreed

- `TPL-02 — Work breakdown structure and WBS dictionary` — the scope dimension this structure complements
- `TPL-07 — Earned value calculation sheet` — where coded actual cost becomes a performance measure
- `BPG-03 — Cost breakdown structure and the code of accounts` — the design reasoning in full
- `BPG-07 — Accruals and cut-off discipline` — why the effective-from period rule matters at close

Sources and standards

No external source is cited. The segment scheme, values and rules here are an original instrument. Chart of accounts numbers in §4 are illustrative and do not represent any organisation's chart. Accounting treatment referred to in §2 and §5 varies by financial reporting framework and by jurisdiction, and this document deliberately states no treatment as universal.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-04 · FREE TEMPLATES

Baseline change request

The form that records a decision rather than a receipt, and the register that tracks the position

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A baseline change request is the document that moves scope, budget or dates from one approved state to another, and the record that has to survive an audit two years later. This template gives the form — originator through to implementation record, including the alternatives considered and the funding source that determines whether budget at completion moves at all — and the register that tracks every request, the running approved position and the remaining contingency.

About this document

Document	TPL-04
Series	Free Templates
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Baseline change request

In one paragraph. A baseline change request is the document that moves scope, budget or dates from one approved state to another, and the record that has to survive an audit two years later. This template gives the form — originator through to implementation record, including the alternatives considered and the funding source that determines whether budget at completion moves at all — and the register that tracks every request, the running approved position and the remaining contingency.

Who this is for. Project controls managers and cost engineers who administer change; control account managers and package managers who raise it; and project directors and sponsors who approve it and will be asked later why.

1. When to use this

Raise a request whenever something would make the approved baseline no longer describe the work. That covers more than new scope:

- **Scope added, reduced or transferred** between elements or between parties.
- **A budget transfer** between control accounts, even where the total does not move. The total not moving is exactly why these go unrecorded, and why control account variances become uninterpretable.
- **A release from contingency** into a control account.
- **A correction to the baseline** — an estimating error, a quantity wrong in the take-off, a duplicated item.
- **A schedule change** that moves a contractual milestone or consumes float that another party has a claim on.
- **A change of measurement basis** — a different rule of credit, a different control-account level, a different forecasting method used for reporting. These change what the numbers mean without changing a single figure, and they are the least often recorded.

Do not raise one for a forecast movement. A forecast change is a report of what is expected; a baseline change is a decision about what is authorised. Confusing the two produces a project whose baseline follows its actuals, which is a project that cannot report a variance.

Do not raise one to make a variance disappear. The controls execution plan (`TPL-01` §3.5) should list that explicitly as a prohibited reason for a re-baseline, so the prohibition can be cited rather than argued.

2. How to complete it

One change, one request. Bundling three unrelated items into one request means the approver must accept or reject all three. The bundle is usually assembled around the one item that would not survive on its own.

Complete the impact fields even when the impact is nil. "Schedule impact: none — the affected activities carry 40 days of total float, path not driving" is an assessment. A blank field is an omission, and the difference matters when someone asks two years later whether the schedule effect was considered.

Name the funding source before you compute the revised budget. This is the field that determines whether budget at completion moves, and it is where most registers go wrong:

- **Contingency release** transfers budget that is already inside the budget at completion from the contingency reserve into a control account. Distributed budget rises, contingency falls, budget at completion does not move.
- **Additional authorised funding** — a client variation, an approved supplementary allocation, a release from management reserve — adds budget that was not previously inside the budget at completion. Budget at completion rises.
- **Trade-off within the baseline** moves budget between control accounts. Nothing changes at project level, but two control-account variances change, which is precisely why it must be recorded.

This depends on the definition of budget at completion agreed in the controls execution plan (`TPL-01` §3.2). Some organisations hold contingency inside it and some outside; both are defensible and the register must state which, because otherwise the same set of approved changes produces two different revised budgets.

Write the alternatives honestly. The alternatives field is the one that turns the form into a decision record. Record what else was considered, what each would have cost in money and time, and why it was rejected. "No alternative" is occasionally true and should then be justified in a sentence, because a change with genuinely no alternative is unusual.

State the estimate basis and its class. A cost impact assembled from firm quotations and one built from a rate applied to a guessed quantity are different objects and should not be approved on the same form without the difference visible.

Route by threshold and record the route. The thresholds live in `TPL-01` §3.5. The request records which threshold applied and who therefore had authority — not merely who signed.

Close the loop. A request is not complete when it is approved; it is complete when the baseline has been changed in the tools, the change verified, and the request marked implemented with the resulting baseline version recorded. Approved-but-not-implemented is the most dangerous state a change can sit in, because everyone believes the budget has moved and none of the reports agree.

Using the tables. Copy a table block, paste into a single spreadsheet column, split on the pipe character, and delete the alignment row.

3. The template

3.1 The form

Field	Entry	Completion note
Identification		
Request number		Sequential, never reused
Project / contract		
Date raised		
Originator — name, role, organisation		A person, not a function
Classification		
Change title		One line, specific enough to be recognised in the register
Change class		`Scope addition` · `Scope reduction` · `Scope transfer` · `Budget transfer` · `Estimate correction` · `Schedule only` · `Measurement basis`
Trigger		`Client instruction` · `Design development` · `Site condition` · `Risk realised` · `Error or omission` · `Regulatory or consent` · `Third-party`
Risk register reference		Where the change is a realised risk, the risk it realises
Description and justification		
Description of change		What changes, in the baseline's own terms. Reference the affected element identifiers.
Why it cannot be absorbed		The test the request must pass before impact is even assessed
Justification		Why the change should be approved, on the project's objectives rather than on convenience
Affected baseline elements		
WBS elements affected		Identifiers from `TPL-02`
Control accounts affected		
Cost breakdown structure codes affected		Codes from `TPL-03`; note any new code required
Schedule activities affected		Activity identifiers
Cost impact		
Direct cost		
Indirect cost		
Total cost impact		Calculated — see §3.2
Currency and basis		Currency, nominal or constant prices, and whether escalation is included
Estimate basis and class		Quotations, measured rates, parametric, judgement — and the estimate class
Estimate prepared by		
Schedule impact		
Activities affected and duration change		

Field	Entry	Completion note
Effect on total float of the affected path		Float before, float after
Is the affected path driving?		Yes / no, and which path is
Effect on contractual milestones		Named milestones, dates before and after
Effect on forecast completion		
Other impacts		
Scope impact		What the project will and will not now deliver
Quality, safety and environmental impact		
Risk impact		Risks created, closed or changed, with register references
Interface impact		Other parties affected and whether they have been consulted
Options		
Alternatives considered		Each option, its cost and time effect, and why it was rejected
Consequence of doing nothing		Required. If it is genuinely nothing, the change is not needed.
Funding		
Funding source		`Contingency release` · `Additional authorised funding` · `Trade-off within baseline`
Effect on budget at completion		Calculated — see §3.2
Effect on remaining contingency		Calculated — see §3.2
Approval		
Threshold applied and approval route		From `TPL-01` §3.5
Reviewed by — name, role, date, comment		
Decision		`Approved` · `Approved with conditions` · `Rejected` · `Deferred` · `Withdrawn`
Conditions of approval		
Approved value		Where different from the requested value, and why
Decision authority — name, role, date		
Implementation record		
Baseline version before / after		Tool-level version identifiers, not descriptions
Cost baseline updated — by / date		
Schedule baseline updated — by / date		
Registers updated — risk, change, code register		

Field	Entry	Completion note
Verification — by / date		A second person confirms the baseline now matches the approved change
Implemented status		`Not started` · `In progress` · `Implemented` · `Verified`

3.2 Calculated fields on the form

Total cost impact. In words: direct cost plus indirect cost. With direct in `B20` and indirect in `B21`: `=N(B20)+N(B21)`. Wrapping in `N()` treats a blank as zero rather than propagating text.

Effect on budget at completion. In words: the budget at completion moves only where the funding source is additional authorised funding; a contingency release or an internal trade-off leaves it unchanged. With the funding source in `B30` and the approved value in `B31`:

```
=IF(B30="Additional authorised funding",N(B31),0)
```

Effect on remaining contingency. In words: contingency falls only where the change is funded by a contingency release.

```
=IF(B30="Contingency release",-N(B31),0)
```

3.3 The register

One row per request. This is the sheet that answers "where do we stand".

Col	Field	Type	Definition and entry rule
A	Request number	Text	Matches the form
B	Date raised	Date	
C	Title	Text	
D	Originator	Text	
E	Class	List	As on the form
F	Trigger	List	As on the form
G	Status	List	`Draft` · `Submitted` · `Under review` · `Approved` · `Rejected` · `Deferred` · `Withdrawn` · `Implemented`
H	Funding source	List	`Contingency release` · `Additional authorised funding` · `Trade-off within baseline` · blank until decided
I	Cost impact requested	Number	
J	Cost impact approved	Number	Blank unless status is `Approved` or `Implemented`
K	Schedule impact requested (days)	Number	
L	Schedule impact approved (days)	Number	
M	Decision date	Date	Blank while open
N	Decision authority	Text	
O	Days open	Calculated	
P	Cumulative approved cost	Calculated	
Q	Revised budget at completion	Calculated	
R	Contingency remaining	Calculated	
S	Baseline version implemented	Text	
T	Implementation verified — by / date	Text	

Hold three control values in fixed cells above the table: original budget at completion in `$C$1`, opening contingency in `$C$2`, and the reporting cut-off date in `$C$3`.

Calculated column O — Days open. In words: calendar days from the date raised to the decision date; where no decision has been made, to the reporting cut-off. Spreadsheet:

```
=IF(B5="","",IF(M5="",$C$3-B5,M5-B5))
```

Use the cut-off cell rather than `TODAY()`. `TODAY()` recalculates every time the file is opened, so a register issued with the monthly report will report different ageing next week than it did on the day it was approved.

Calculated column P — Cumulative approved cost. In words: the running total of approved cost impacts down the register. Spreadsheet, with the expanding range anchored at the first data row:

```
=SUMIFS($J$5:J5,$G$5:G5,"Approved")+SUMIFS($J$5:J5,$G$5:G5,"Implemented")
```

Two terms are needed because an implemented change is still an approved change; a single criterion would drop rows as they progress to `Implemented`.

Calculated column Q — Revised budget at completion. In words: the original budget at completion plus only those approved changes funded by additional authorised funding. Spreadsheet:

```
=$C$1+SUMIFS($J$5:J5,$G$5:G5,"Approved",$H$5:H5,"Additional authorised funding")+SUM-
IFS($J$5:J5,$G$5:G5,"Implemented",$H$5:H5,"Additional authorised funding")
```

Calculated column R — Contingency remaining. In words: opening contingency less every approved contingency release to this point.

```
=$C$2-SUMIFS($J$5:J5,$G$5:G5,"Approved",$H$5:H5,"Contingency release")-SUM-
IFS($J$5:J5,$G$5:G5,"Implemented",$H$5:H5,"Contingency release")
```

Summary block for the monthly report. Approved this period, approved to date, pending value, rejected to date, contingency drawn to date, contingency remaining, and pending value as a proportion of contingency remaining. That last figure is the one that gets a sponsor's attention:

```
=IF(R_latest=0,"",pending_value/R_latest)
```

4. Worked fragment

Illustrative figures. A fictional facility upgrade project. Currency is generic currency units (CU) in thousands, nominal basis, no escalation, rounded to the nearest thousand. The reporting cut-off is 31 May 2026. Control values: original budget at completion CU 8,100 thousand, comprising distributed control-account budget of CU 7,700 thousand and a contingency reserve of CU 400 thousand. On this project the contingency reserve sits inside budget at completion, as recorded in the controls execution plan.

4.1 Register extract

Ref	Raised	Title	Status	Funding	Cost requested	Cost approved	Sched. req. (days)	Decision date	Days open	C a
BCR-011	03 Mar 26	Revised electrical room layout	Implemented	Contingency release	120	100	0	18 Mar 26	15	1
BCR-012	14 Mar 26	Client-instructed additional metering	Implemented	Additional authorised funding	200	200	0	02 Apr 26	19	3
BCR-013	21 Apr 26	Correction to steel tonnage in the estimate	Rejected	—	145	—	0	06 May 26	15	3
BCR-014	12 May 26	Ground improvement — obstructions at pile positions	Under review	Contingency release (proposed)	295	—	+9	—	19	3

Verification of the days open. BCR-011: 3 to 18 March is 15 days. BCR-012: 14 March to 2 April is 17 days remaining in March plus 2 in April, so 19 days. BCR-013: 21 to 30 April is 9 days plus 6 in May, so 15 days. BCR-014 is undecided, so it ages to the cut-off: 12 to 31 May is 19 days.

Verification of the budget position. BCR-011 is a contingency release: distributed budget rises from 7,700 to 7,800, contingency falls from 400 to 300, and budget at completion is unchanged at 8,100. BCR-012 is additional authorised funding: distributed budget rises from 7,800 to 8,000, contingency stays at 300, and budget at completion rises to 8,300. BCR-013 was rejected and changes nothing. The distributed figure of CU 8,000 thousand is the budget at completion used for performance measurement in TPL-07 and for forecasting in TPL-08; the CU 300 thousand of contingency is held outside the control accounts and is not earned against.

The position the register is really reporting. BCR-014 requests CU 295 thousand from a contingency reserve of CU 300 thousand. If approved, CU 5 thousand of contingency remains, against 52.5 per cent of the distributed budget still to be earned — the earned value sheet at TPL-07 shows earned value at 47.5 per cent of budget at completion at this cut-off. That sentence, not the total of the register, is what belongs on the first page of the monthly report.

4.2 Form extract — BCR-014

Field	Entry
Request number	BCR-014
Date raised	12 May 2026
Originator	Civil delivery manager (named)
Change title	Ground improvement at obstructed pile positions
Change class	Scope addition
Trigger	Site condition
Risk register reference	R-014 unknown obstructions — realised
Description of change	Excavate and remove buried obstructions at 22 pile positions in the utilities building footprint and replace with engineered fill to formation level before piling resumes at those positions. Affects WBS 1.1.2, control account CA-1000.
Why it cannot be absorbed	The work is outside the scope described in the dictionary entry for 1.1.2, whose stated assumption is that no obstructions requiring removal exist below formation level. There is no allowance in the work package for it.
Justification	Piling cannot proceed at the affected positions and the obstructions cannot be piled through. The alternative locations assessed do not satisfy the structural design.
Direct cost	260
Indirect cost	35
Total cost impact	295
Currency and basis	CU thousands, nominal, no escalation
Estimate basis and class	Subcontractor quotation for excavation and disposal against a measured obstruction survey; fill quantities from the survey at contract rates. Firm quotation for the majority; measured rates for the remainder.
Schedule impact	ACT-1020 piling extended by 9 calendar days. Total float on that path 12 days before, 3 days after. The path is not currently driving; the driving path runs through mechanical installation.
Effect on contractual milestones	None
Risk impact	Closes R-014 as realised. Opens R-031: further obstructions outside the surveyed area, unquantified.
Alternatives considered	(1) Relocate the affected piles — rejected: the structural design does not permit the required offsets, and redesign was assessed at 6 weeks and greater cost. (2) Pile deeper through the obstructions — rejected: the piling subcontractor will not warrant integrity and the specified acceptance regime could not be met. (3) Defer the affected positions to the end of the piling sequence — rejected: it does not avoid the work and it would move the path onto the driving path.
Consequence of doing nothing	22 piles cannot be installed; the foundation to the north-east quadrant cannot be completed; the structural steel sequence stops at that quadrant within approximately four weeks.
Funding source	Contingency release (proposed)
Effect on budget at completion	Nil — the release moves budget already inside budget at completion from the contingency reserve into CA-1000

Field	Entry
Effect on remaining contingency	Reduces remaining contingency from 300 to 5
Decision	Under review at the reporting cut-off

The alternatives block is what makes this document defensible. Two years later, the question will not be whether CU 295 thousand was a reasonable price for the work — it will be whether the project should have been doing that work at all. Only the alternatives field answers that.

5. Common mistakes

Approving without a funding source. The change is approved, the budget moves, and nobody can say whether budget at completion changed. Six months on, two reports quote different revised budgets and both can trace their arithmetic.

No record of budget transfers. Because the project total does not move, transfers are treated as administration. They are the reason a control account with a large favourable variance sits next to one with a large adverse variance and neither manager recognises their own numbers.

Impact fields left blank rather than assessed as nil. A blank is indistinguishable from an oversight. Write the nil assessment and the grounds for it.

A cost impact with no estimate basis. A firm quotation and a judgement are both legitimate inputs and must not be approved on the same form without the difference visible to the approver.

Approved but never implemented. The register says the budget is 8,300; the cost tool still says 8,100. Both are quoted in the same meeting. The implementation record and the verification field exist for this, and the register should be reviewed for approved-not-implemented rows every period.

Contingency spent without a change request. Contingency drawn by adjusting a control account budget directly leaves no link between the risk that was identified and the money that was spent, which is exactly the link a sponsor asks for when contingency runs low.

Changing the baseline to remove a variance. A re-baseline that resets earned value to actual cost produces a cost performance index of exactly one, and a project that has lost its ability to forecast. The prohibition belongs in the controls execution plan so that citing it is not an argument.

A register with no ageing. Without days open, a request that has sat under review for eleven weeks looks the same as one raised on Tuesday, and slow decisions are a cost the project never sees.

6. Adapting it

Safe to change. Field order and grouping. The class and trigger lists, which should match how your organisation analyses change. Adding fields for contract clause reference, client change number, insurance recovery, or delay-analysis method where they apply. Splitting the register by originating party on a project with several contractors.

Change with care. Merging the request into the change order log (TPL-12) works only where every baseline change is also a contract variation. On most projects it is not: baseline changes include internal transfers, corrections and contingency releases that never reach the contract. Keep both and cross-reference by number.

Do not remove. Alternatives considered, consequence of doing nothing, funding source, and the implementation and verification record. Those four are the difference between a decision record and a receipt.

On a small project, the form can be a single page and the register a single sheet — but keep every field, and let the answers be short. A one-line alternatives entry is a decision recorded.

7. Completion checklist

- [ ] One change per request, with a title recognisable in the register
- [ ] Affected element, control account, code and activity identifiers all listed
- [ ] Every impact field completed, including those assessed as nil, with the grounds stated
- [ ] Estimate basis and class stated, and the preparer named
- [ ] Schedule impact states float before and after, and whether the path is driving
- [ ] Alternatives considered recorded with their cost and time effects and the reason for rejection
- [ ] Consequence of doing nothing stated
- [ ] Funding source selected, and the effect on budget at completion computed from it
- [ ] Approval route matches the threshold in the controls execution plan
- [ ] Approved value recorded where it differs from the requested value, with the reason
- [ ] Baseline versions before and after recorded at implementation
- [ ] Implementation verified by a second person
- [ ] Register ages open requests to the reporting cut-off, not to today's date
- [ ] Contingency remaining and pending value reported together on the same line

---

Related

- TPL-01 – Project controls execution plan — where thresholds, routing and the budget definition are set
- TPL-06 – Monthly project controls report — where the change position is reported
- TPL-12 – Change order log — the contractual variation record this cross-references
- BPG-04 – Baselining and baseline change control — the reasoning behind the controls here
- BPG-10 – Contingency and management reserve — custody, release tests and drawdown discipline
- BPG-11 – Change orders and variations — the commercial treatment of client-instructed change

Sources and standards

No external source is cited. This is an original instrument. The distinction between contingency reserve and management reserve, and the identity that a contingency release does not move budget at completion, follow the budget definitions taught in the Institute's Body of Knowledge and stated in the Institute's master formula sheet; they are applied here in the Institute's own words, with the caveat in §2 that organisations legitimately differ on whether contingency sits inside budget at completion.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-05 · FREE TEMPLATES

Progress measurement and rules of credit sheet

Five techniques, one rule-of-credit library, and a roll-up that weights by budget rather than averaging

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

Per cent complete is an assertion, and the sheet that produces it is where the assertion is either evidenced or invented. This template gives the measurement sheet and the rule-of-credit library behind it: five techniques with the arithmetic for each, a defined credit step for every claim, the evidence required before credit is released, and a roll-up that weights activities by budget instead of averaging their percentages.

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Progress measurement and rules of credit sheet

In one paragraph. Per cent complete is an assertion, and the sheet that produces it is where the assertion is either evidenced or invented. This template gives the measurement sheet and the rule-of-credit library behind it: five techniques with the arithmetic for each, a defined credit step for every claim, the evidence required before credit is released, and a roll-up that weights activities by budget instead of averaging their percentages.

Who this is for. Cost engineers and planners who produce the monthly progress position; control account managers who claim it; and project controls managers who have to defend it when the earned value it produces is challenged.

1. When to use this

Build the sheet before the first progress claim, not after the first argument. The measurement technique for an activity is a decision that belongs with the baseline: it is recorded in the work breakdown structure dictionary (`TPL-02`) when the element is created, and the rule of credit is approved before any credit is claimed against it.

Use it every reporting period, at the cut-off defined in the controls execution plan (`TPL-01` §3.6). Its output feeds three places:

- **Earned value**, where per cent complete multiplied by the budget at completion of the element gives the earned value used in `TPL-07`.
- **The monthly report**, where period movement is the number that tells a reader whether the project is accelerating or stalling.
- **Payment**, on projects where interim application is driven by measured progress. Where that is the case, say so on the sheet, because it changes the incentives on every claim.

The one time not to use it is on an activity whose output is genuinely unmeasurable and whose budget is material. There is no technique that rescues that; the answer is to decompose the activity until part of it becomes measurable, or to accept level of effort and state plainly that the element earns to plan.

2. How to complete it

Choose the technique from the work, not from convenience. The five techniques, and the condition each requires:

Technique	Use when	Per cent complete is
Units complete	Output is countable and homogeneous, and the total quantity is known	Quantity installed ÷ budgeted quantity
Incremental milestone	One work item passes through a fixed sequence of steps	The cumulative credit of the steps completed, in order
Weighted milestone	A deliverable is reached through discrete events that may complete out of order	The sum of the credits of the milestones achieved
Level of effort	The work is support or supervision, with no output of its own	Elapsed duration ÷ planned duration
Apportioned effort	The work varies directly with another activity and has no independent output	The per cent complete of the named base activity

Distinguish incremental from weighted milestone properly. Both assign credit at defined points. The difference is whether out-of-order achievement is permitted. Incremental milestone is a sequence: a step cannot be credited before its predecessor, and the sheet should refuse it. Weighted milestone is a set: each milestone carries a weight and may be achieved in any order. Choosing incremental where the work is genuinely non-sequential forces claimants to under-report; choosing weighted where the work is sequential lets them claim the easy milestones first.

Level of effort must be capped and must be small. Level of effort earns to plan by construction, so it generates no cost variance from schedule causes and never reports a schedule problem. Two disciplines follow: cap it at 100 per cent so it cannot keep earning past its planned finish, and keep it to a small share of the control account. If most of a control account is level of effort, its cost performance index is measuring almost nothing.

Apportioned effort must name its base activity. Not a discipline, not a control account — one activity, recorded on the sheet. Its per cent complete is then a formula referencing that row, so it cannot drift.

Write the rules of credit before the first claim, and freeze them. Changing a credit step mid-project changes reported progress without any work being done. If a rule must change, treat it as a measurement basis change under `TPL-04` and report the effect separately for one period.

Require objective evidence at every credit step. The evidence reference on the sheet is a document number, a survey, a test certificate, an inspection release — not a name. "Confirmed by the site manager" is not evidence; it is the same assertion again.

Roll up by weighting, never by averaging. The rolled-up per cent complete of a control account is total earned budget hours divided by total budget hours, not the mean of the activity percentages. This is the single most common arithmetic error in the discipline and §4.3 shows what it costs.

Store percentages as decimal fractions with percentage formatting. Type `0.55` into a cell formatted as a percentage, not `55`. A sheet that mixes the two produces credit totals of 5,500 per cent and roll-ups that are silently wrong by a factor of a hundred.

Using the tables. Copy a table block, paste into a single spreadsheet column, split on the pipe character, and delete the alignment row.

3. The template

3.1 Sheet 1 — the measurement sheet

Col	Field	Type	Definition and entry rule
A	Activity ID	Text	Matches the schedule activity identifier
B	Activity description	Text	
C	WBS element / control account	Text	From `TPL-02`
D	Technique	List	`UC` units complete · `IM` incremental milestone · `WM` weighted milestone · `LOE` level of effort · `AE` apportioned effort
E	Rule of credit reference	Text	The rule identifier in Sheet 2. Required for `IM` and `WM`.
F	Base activity (apportioned effort only)	Text	The activity ID whose per cent complete this follows
G	Unit of measure	Text	For `UC` — piles, cubic metres, welds, drawings, tonnes
H	Budgeted quantity	Number	For `UC` — the total scope in the stated unit
I	Budgeted hours	Number	The weight used in the roll-up. Cost may be used instead, but not both in one sheet.
J	Quantity to date / credit basis	Number	`UC`: quantity installed. `IM`/`WM`: cumulative credit from Sheet 2. `LOE`: elapsed duration. `AE`: leave blank.
K	Planned duration (LOE only)	Number	Same unit as column J for level-of-effort rows
L	Per cent complete this period	Calculated	See below
M	Per cent complete last period	Number	Carried forward from the previous issue of the sheet
N	Period movement	Calculated	
O	Earned quantity	Calculated	For `UC` rows — a cross-check
P	Earned hours	Calculated	The figure the roll-up uses
Q	Evidence reference	Text	Document, survey, certificate or inspection release number
R	Claimed by / verified by	Text	Two names, not one
S	Exception flag	Calculated	

Calculated column L — Per cent complete. In words, by technique:

- **Units complete** — quantity installed divided by budgeted quantity.
- **Incremental and weighted milestone** — the cumulative credit of the milestones achieved, brought in from Sheet 2.
- **Level of effort** — elapsed duration divided by planned duration, capped at 100 per cent.
- **Apportioned effort** — the per cent complete of the named base activity.

Spreadsheet, as one expression covering all five:

```
=IF($D2="UC",IF($H2=0,"",$J2/$H2),
 IF(OR($D2="IM",$D2="WM"),IF($J2="","",$J2),
 IF($D2="LOE",IF($K2=0,"",MIN(1,$J2/$K2)),
 IF($D2="AE",IFERROR(INDEX($L$2:$L$500,MATCH($F2,$A$2:$A$500,0)),""),""))))
```

Every division is guarded against a zero denominator. The apportioned-effort branch looks up the base activity's per cent complete by activity identifier; the `IFERROR` returns blank when the base activity cannot be found, which is itself a defect worth seeing. If you prefer to keep the branches separate, split the sheet by technique — the arithmetic is identical.

Calculated column N — Period movement. In words: this period's per cent complete less last period's, expressed in percentage points.

```
=IF(OR($L2="",$M2=""),"",$L2-$M2)
```

Calculated column O — Earned quantity. In words: per cent complete multiplied by the budgeted quantity. For a units-complete row this must equal the quantity to date, which makes it a free check that the technique and the entry agree.

```
=IF(OR($L2="",$H2=""),"",$L2*$H2)
```

Calculated column P — Earned hours. In words: per cent complete multiplied by the budgeted hours.

```
=IF(OR($L2="",$I2=""),"",$L2*$I2)
```

Calculated column S — Exception flag. In words: flag a per cent complete above 100 per cent, a backwards movement, or a milestone row with no rule reference.

```
=IF($L2="","",IF($L2>1,"OVER 100 % — quantity growth: raise a change or re-baseline the quantity",
 IF($N2<0,"NEGATIVE MOVEMENT — explain",
 IF(AND(OR($D2="IM",$D2="WM"),$E2=""),"NO RULE OF CREDIT",""))))
```

Do not cap a units-complete row at 100 per cent. A quantity that has exceeded its budget is real information — it means the take-off was wrong or the scope has grown — and capping it hides the very thing the sheet should surface. Level of effort is different and is capped, because it has no quantity to grow.

Roll-up to a control account. In words: total earned hours divided by total budgeted hours across the activities in the account. Spreadsheet, over the sheet's data range:

```
=IF(SUMIF($C$2:$C$500,$C2,$I$2:$I$500)=0,"",SUMIF($C$2:$C$500,$C2,$P$2:$P$500)/SUMIF($C$2:$C$500,$C2,$I$2:$I$500))
```

Earned value from the roll-up. In words: the control account's rolled-up per cent complete multiplied by its budget at completion. That figure is the earned value input to `TPL-07`, whose treatment of the resulting measures this document does not repeat.

3.2 Sheet 2 — the rule-of-credit library

One block per rule. Rules are shared across activities of the same type; that is the point of a library.

Col	Field	Type	Definition
A	Rule ID	Text	e.g. `ROC-CIV-03`
B	Technique	List	`IM` or `WM`
C	Applies to	Text	The work type this rule measures
D	Step / milestone	Text	The event that releases credit
E	Sequence number	Number	For `IM`, the order that must be observed. Blank for `WM`.
F	Credit	Number	The credit released by this step, as a decimal fraction
G	Cumulative credit	Calculated	For `IM` only
H	Objective evidence required	Text	The document or test that proves the step
I	Released by (role)	Text	Who confirms the evidence
J	Achieved this period	List	`Yes` / `No`
K	Date achieved	Date	
L	Approved by / date	Text	Approval of the rule itself, not of a claim

Calculated column G — Cumulative credit. In words: the running total of credit down the sequence.

```
=IF($E2="","",SUMIFS($F$2:F2,$A$2:A2,$A2))
```

Rule total validation. In words: the credits within a rule must total exactly 100 per cent.

```
=IF(ROUND(SUMIF($A$2:$A$200,$A2,$F$2:$F$200),6)=1,"OK","CHECK — credits must total 100 %")
```

`ROUND` to six decimal places prevents a false failure from floating-point representation, which is otherwise a genuine nuisance on rules with thirds.

Claimed credit for a rule. In words: for weighted milestone, the sum of the credits of milestones marked achieved. For incremental milestone, the cumulative credit at the highest sequence number achieved, which enforces the ordering.

Weighted milestone:

```
=SUMIFS($F$2:$F$200,$A$2:$A$200,$A2,$J$2:$J$200,"Yes")
```

Incremental milestone:

```
=IF(COUNTIFS($A$2:$A$200,$A2,$J$2:$J$200,"Yes")=0,0,SUMIFS($F$2:$F$200,$A$2:$A$200,$A2,$E$2:$E$200,"<="&MAXIFS($E$2:$E$2
```

The incremental expression deliberately credits everything up to the highest achieved step, so a claim that skips a step credits the skipped step too — which is the correct behaviour for a sequence, and makes an out-of-order claim visible rather than profitable. Where `MAXIFS` is unavailable, sort the rule by sequence and use a lookup on the last `Yes`.

4. Worked fragment

Illustrative figures. Control account CA-1000, civil works, on a fictional facility upgrade project. The period is May 2026, cut-off 31 May 2026. The weighting basis is budgeted direct labour hours. Currency is not used on this sheet; the resulting earned value is expressed in generic currency units (CU) thousands in `TPL-07`. Percentages are shown to one decimal place, rounded half away from zero; hours to whole hours.

4.1 The measurement sheet

Activity	Description	Tech.	Rule	Unit	Budget qty	Budget hrs	Qty / basis	% this period	% last period	Movement (pp)	Earn hrs
ACT-1010	Bulk excavation	UC	—	m³	12,000	2,000	9,600	80.0 %	68.0 %	+12.0	1,60
ACT-1020	Bored piling	UC	—	piles	240	7,800	108	45.0 %	37.5 %	+7.5	3,51
ACT-1030	Pile caps and ground beams	IM	ROC-CIV-03	—	—	1,200	50.0 %	50.0 %	25.0 %	+25.0	600
ACT-1040	Structural steel erection	WM	ROC-STR-01	—	—	6,000	60.0 %	60.0 %	20.0 %	+40.0	3,60
ACT-1090	Civil supervision	LOE	—	months	—	2,000	11 of 20	55.0 %	50.0 %	+5.0	1,10
ACT-1095	Steel inspection and quality assurance	AE	base ACT-1040	—	—	800	—	60.0 %	20.0 %	+40.0	480
	Control account CA-1000					**19,800**		**55.0 %**	**35.1 %**	**+19.9**	**10,8**

Verification of the activity arithmetic. Excavation: 9,600 ÷ 12,000 = 0.800; earned hours 0.800 × 2,000 = 1,600. Piling: 108 ÷ 240 = 0.450; earned hours 0.450 × 7,800 = 3,510. Pile caps: cumulative credit 50.0 per cent from rule `ROC-CIV-03` below; earned hours 0.500 × 1,200 = 600. Steel: achieved credit 60.0 per cent from rule `ROC-STR-01` below; earned hours 0.600 × 6,000 = 3,600. Supervision: 11 ÷ 20 = 0.550; earned hours 0.550 × 2,000 = 1,100. Inspection: follows ACT-1040 at 60.0 per cent; earned hours 0.600 × 800 = 480.

Verification of the roll-up. Budgeted hours: 2,000 + 7,800 + 1,200 + 6,000 + 2,000 + 800 = 19,800. Earned hours: 1,600 + 3,510 + 600 + 3,600 + 1,100 + 480 = 10,890. Per cent complete = 10,890 ÷ 19,800 = 0.550 = **55.0 per cent**. Last period's earned hours were 1,360 + 2,925 + 300 + 1,200 + 1,000 + 160 = 6,945, giving 6,945 ÷ 19,800 = 0.3508 = 35.1 per cent, so the movement is 19.9 percentage points.

Earned value. Control account CA-1000 has a budget at completion of CU 4,000 thousand (`TPL-02` §4.1). Earned value = 0.550 × 4,000 = **CU 2,200 thousand**, which is the earned value entered for CA-1000 in `TPL-07` §4.

A feature, not a fault. The 40-point movement on ACT-1040 is what milestone techniques do: nothing was credited while the frame was being erected, and 40 per cent landed the day the primary frame was signed off. Milestone progress is lumpy by construction. If a reader needs a smoother curve, the answer is more milestones with smaller weights, not a technique that credits partial achievement of a milestone.

4.2 The rules of credit behind the two milestone rows

`ROC-CIV-03` **— pile caps and ground beams, incremental milestone.** Sequential; a step cannot be credited before its predecessor.

Seq	Step	Credit	Cumulative	Objective evidence	Achieved
1	Setting out complete and checked	10.0 %	10.0 %	Setting-out survey record	Yes
2	Blinding placed	15.0 %	25.0 %	Pour record	Yes
3	Reinforcement fixed and inspected	25.0 %	50.0 %	Reinforcement inspection release	Yes
4	Concrete placed	40.0 %	90.0 %	Pour record and delivery tickets	No
5	Stripped, cured and survey accepted	10.0 %	100.0 %	28-day cube results and as-built survey	No
	Total	**100.0 %**			

Credit claimed: cumulative credit at the highest achieved sequence, step 3 → **50.0 per cent**. Verification: 10.0 + 15.0 + 25.0 = 50.0.

`ROC-STR-01` **— structural steel erection, weighted milestone.** Milestones may be achieved in any order; credit is the sum of those achieved.

Milestone	Credit	Objective evidence	Achieved
Steel delivered to site and receipt-inspected	20.0 %	Delivery notes and receipt inspection record	Yes
Primary frame erected and bolted	40.0 %	Erection sign-off sheet	Yes
Secondary steel erected	15.0 %	Erection sign-off sheet	No
Frame aligned, plumbed and surveyed	15.0 %	Survey report	No
Grouted and final inspection released	10.0 %	Inspection release	No
	100.0 %		

Credit claimed: 20.0 + 40.0 = **60.0 per cent**. Both rule totals return `OK` on the validation formula.

4.3 What averaging would have cost

The six activity percentages are 80.0, 45.0, 50.0, 60.0, 55.0 and 60.0. Their simple average is 350.0 ÷ 6 = 58.3 per cent. The correct budget-weighted figure is 55.0 per cent. Applied to the control account's budget at completion of CU 4,000 thousand, the average would report earned value of 0.5833 × 4,000 = CU 2,333 thousand against the correct CU 2,200 thousand — **CU 133 thousand of earned value that has not been earned**, and a cost performance index overstated by the same margin.

The error is small here only because the budgets are not wildly unequal. Take two activities: one of 100 budget hours at 90 per cent complete, and one of 9,900 budget hours at 10 per cent. The simple average is (90 + 10) ÷ 2 = 50.0 per cent. The weighted answer is (0.90 × 100 + 0.10 × 9,900) ÷ 10,000 = (90 + 990) ÷ 10,000 = **10.8 per cent**. The averaging method reports a project nearly half complete that has barely started.

5. Common mistakes

Averaging activity percentages. See §4.3. It is the error that survives longest because the answer always looks plausible.

Level of effort used because measurement is inconvenient. Level of effort earns to plan and therefore never reports a problem. A control account that is mostly level of effort has a cost performance index that measures the passage of time.

Level of effort with no cap. Past its planned finish it keeps earning and eventually reports more than 100 per cent complete, which then propagates into earned value greater than budget at completion.

Apportioned effort pointing at a discipline rather than an activity. It stops being a formula and becomes an opinion, and it drifts.

Milestone credits that do not total 100 per cent. The activity can never reach complete, or it reaches complete before it is. The validation formula in §3.2 exists because this is common and invisible.

Credit released without evidence. Once a claim can be made on assertion, the sheet stops being a measurement instrument and becomes a negotiation. The evidence reference column is the whole control.

Changing a rule of credit mid-project. Reported progress moves without any work being done. Treat it as a measurement basis change under `TPL-04` and disclose the effect for one period.

Capping a units-complete row at 100 per cent. Quantity growth is information: it means the take-off was wrong or the scope has grown, and either way somebody needs to know. Flag it; do not hide it.

One person claiming and verifying. The two-name requirement in column R is not bureaucracy. It is the only structural control on a number that determines both reported performance and, on many projects, payment.

Backwards movement passed over in silence. Progress that reduces is either an error being corrected or a claim being withdrawn. Both need an explanation in the same period, not a quiet restatement.

6. Adapting it

Safe to change. Weighting by cost instead of hours, provided you use one basis consistently — a sheet that weights some activities by hours and others by cost produces a roll-up with no meaning. Adding columns for subcontractor claim reference, payment application line, or physical location. Splitting the sheet by control account on a large project. Adding techniques that are variants of those here, such as fixed-formula start/finish rules, which are incremental milestone with two steps.

Change with care. Moving an activity from one technique to another mid-project. It is sometimes right — a planning package decomposed into measurable work should move from level of effort to units complete — but it changes reported progress without work being done, so it is a measurement basis change and belongs in the change register.

Do not remove. The evidence reference, the two-name claim and verification, the credit-total validation, and the budget-weighted roll-up. Those four are the reason the output can be defended.

Where the client's own progress measurement differs, run both and reconcile them on the sheet as an additional column pair. Do not adopt a single method that satisfies neither; the reconciliation itself is usually the most valuable thing on the page.

7. Completion checklist

- [ ] Every activity has a technique recorded in the dictionary before its first claim
- [ ] Every milestone activity references an approved rule of credit
- [ ] Every rule of credit totals exactly 100 per cent and returns `OK` on the validation
- [ ] Every credit step names the objective evidence and who releases it
- [ ] Level of effort is capped, and its share of each control account is known and stated
- [ ] Every apportioned-effort row names one base activity by identifier
- [ ] Percentages stored as decimal fractions in percentage-formatted cells
- [ ] Earned quantity equals quantity to date on every units-complete row
- [ ] Exception flags cleared or explained: over 100 per cent, negative movement, missing rule
- [ ] Roll-up computed as earned hours ÷ budgeted hours, not as an average of percentages
- [ ] Every row carries an evidence reference and two names
- [ ] Period movement reviewed activity by activity before the sheet is issued
- [ ] Resulting earned value reconciled to the figure entered in the earned value sheet

---

Related

- `TPL-01 — Project controls execution plan` — where permitted techniques and the cut-off calendar are set
- `TPL-02 — Work breakdown structure and WBS dictionary` — where an element's technique is first recorded
- `TPL-06 — Monthly project controls report` — where period movement is reported and explained
- `TPL-07 — Earned value calculation sheet` — what the earned value produced here is used for

- `BPG-06 — Progress measurement and rules of credit` — the design reasoning behind technique selection
- `BPG-08 — Earned value in practice` — how a measured per cent complete becomes a defensible index

Sources and standards

- PCI Master Formula Sheet (`docs/downloads/master-formula-sheet.md`), August 2026: the earned value identity used at §4.1 to convert a rolled-up per cent complete into earned value. Published under the credential's retired code; the credential is PCL-AI.

The five techniques named here are common to established cost engineering practice and are described in the Institute's own words. No third-party sheet, rule set or wording is reproduced; the rules of credit in §4.2 are illustrative and original.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-06 · FREE TEMPLATES

Monthly project controls report

A section structure that opens with the decisions required, and a one-page executive summary that survives being read alone

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A monthly project controls report exists to cause decisions, and most of them fail because the decisions are on page eleven. This template inverts the order: the decisions required come first, then a one-page executive summary that is true when read alone, then status, a variance narrative that names cause, action and owner, look-ahead, risk movement, the change position and the cash position — each with the fields defined and the calculated ones given in words and as spreadsheet expressions.

About this document

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Monthly project controls report

In one paragraph. A monthly project controls report exists to cause decisions, and most of them fail because the decisions are on page eleven. This template inverts the order: the decisions required come first, then a one-page executive summary that is true when read alone, then status, a variance narrative that names cause, action and owner, look-ahead, risk movement, the change position and the cash position — each with the fields defined and the calculated ones given in words and as spreadsheet expressions.

Who this is for. Project controls managers who produce the report; project directors who present it; and sponsors, steering group members and client project managers who have to act on it in the twenty minutes before the meeting.

1. When to use this

Use it for the recurring report that goes to people who can change the project's direction — sponsor, steering group, client. Its cadence, data date and issue offset are fixed in the controls execution plan (`TPL-01` §3.11) and it is assembled after the progress measurement sheet (`TPL-05`), the earned value sheet (`TPL-07`) and the forecast comparison (`TPL-08`) have closed, in that order.

It is not the right instrument for two adjacent jobs. A weekly progress flash to the delivery team is shorter, narrower and does not need an executive summary. A live dashboard answers "what is happening now"; this report answers "what has changed, why, and what needs deciding" — see `BPG-15` for how the two divide.

The test of whether it is working is not whether it is read. It is whether the decisions listed in §1 of the report were taken by the dates the report asked for. Track that, and a report that has stopped causing decisions becomes visible before it becomes a habit.

2. How to complete it

Write section 1 last and put it first. The decisions required are the output of the analysis, not the introduction to it. Each entry names the decision, the person who must take it, the date by which, the consequence of not taking it by then, and the section that supports it. A decision with no date is a discussion topic.

Make the executive summary true standing alone. Assume it will be forwarded without the rest. Every figure in it carries its units and basis, every judgement carries its ground, and nothing in it depends on a caveat that appears later.

Write the variance narrative as cause, action, owner, date, expected effect. Five fields. A narrative that restates the number is not a narrative: "cost variance of CU (220) thousand driven by adverse cost performance" says nothing that the table above it did not. "Piling output averaged 3.1 piles per shift against an estimate basis of 4.0, from harder-than-anticipated ground in the north-east quadrant" says something a reader can act on.

Report movement, not only position. A cost performance index of 0.955 means little without last period's. Movement is what tells a reader whether the project is deteriorating, stabilising or recovering, and it is what a monthly cadence exists to show.

Report the forecast with its range and its selected method. A single estimate at completion with no method named and no range invites a reader to treat it as fact. The comparison in `TPL-08` produces both.

Do not carry colour alone. Where a status indicator is used, pair it with a word — `Behind`, `On plan`, `Ahead` — so the report survives monochrome printing, projection and readers who do not distinguish the colours.

Cite rather than restate. The definitions of the earned value measures belong to `TPL-07`, the rules of credit to `TPL-05`, the forecast methods to `TPL-08`. The report carries the numbers and the meaning; it does not re-teach the arithmetic every month.

Close section 14 honestly. A data quality statement that says which figures are estimated, which cut-offs did not align, and what is not yet in the numbers costs half a page and buys the report its credibility. It is also the section a reader remembers when something later turns out to have been missing.

Using the tables. Copy a table block, paste into a single spreadsheet column, split on the pipe character, and delete the alignment row.

3. The template

3.0 Control block

Field	Entry
Project / contract	
Reporting period	
Data date	
Issue date	
Report version	
Prepared by / reviewed by / approved by	
Distribution	
Classification	
Basis: currency, scale, nominal or constant prices	

3.1 Decisions required this period

The first page. Nothing precedes it.

Ref	Decision required	Decision owner	Required by	Consequence if not taken by that date	Supporting section

Calculated column — days remaining. In words: calendar days from the report's data date to the date the decision is required.

```
=IF(OR($D2="",$B$2=""),"",$D2-$B$2)
```

Reference the data date cell, not `TODAY()`. A report is a fixed statement of a position, and a field that recalculates when the file is reopened makes the issued report disagree with itself.

3.2 Executive summary — one page

Written to be read alone. The block below is the whole page.

Element	Content
Position in one sentence	What state the project is in, and the single most important thing about it
Overall status	Schedule, cost, risk, change and cash, each with a word and, if used, a colour
Headline figures	Budget at completion, per cent complete, cost and schedule performance indices with last period's, selected estimate at completion with its range, forecast completion date with the variance in days
What changed this period	Three items at most, each one sentence, each stating the movement
What to worry about	One item. The thing that will matter in three months and is cheapest to act on now.
Decisions required	The references from §3.1, listed, with their dates
Basis and rounding	Units, currency, scale, data date and rounding rule

3.3 Health, safety, environment and quality

Field	Entry
Reportable incidents this period / cumulative	
Observations and near misses	
Open non-conformances, and any over the agreed ageing threshold	
Environmental or consent issues	
Anything requiring a decision	Cross-reference to §3.1

Report this first among the status sections and report it factually. It is not a controls output, and the report is not the place to interpret it — but a controls report that omits it signals a set of priorities.

3.4 Progress and schedule status

Field	Entry
Physical per cent complete this period / last period / movement	From `TPL-05`
Planned per cent complete at the data date	
Milestone table	Below
Critical path description	Which path is driving, and whether it changed this period
Float position	Total float on the driving path; whether it moved
Schedule health	Result of the last schedule quality review — see `TPL-14`

Milestone	Baseline date	Forecast / actual date	Variance (days)	Contractual?	Movement since last period

Calculated column — variance in days. In words: forecast date less baseline date, in calendar days; a positive figure is late.

```
=IF(OR($B2="",$C2=""),"",$C2-$B2)
```

3.5 Cost status

Carry the control-account table from `TPL-07` and add movement. Do not restate the formulas; cite the sheet.

Control account	BAC	PV	EV	AC	CV	SV	CPI	CPI last period	SPI	SPI last period	% complete

Beneath it, the reconciliation to total authorised budget: measured total, undistributed budget, contingency reserve, total.

3.6 Variance narrative

One block per control account breaching the reporting threshold set in `TPL-01`. Five fields, all required.

Field	Entry rule
Control account and variance	The figure, so the block stands alone
Cause	The physical or commercial reason. Not a restatement of the number. Where more than one cause applies, quantify each.
Action	What is being done, specifically
Owner	A named person
By when	A date
Expected effect	What the reader should see in next period's numbers if the action works. This is what makes the narrative testable next month.
Status of last period's action	Done, in progress, or abandoned — and if abandoned, why

That last row is the one that changes behaviour. A narrative that is never checked against the previous month's promise becomes a monthly essay.

3.7 Forecast

Field	Entry
Selected estimate at completion and method	From `TPL-08`
Range across methods	Lowest and highest, with the spread as a percentage of budget at completion
Variance at completion	
Movement since last period, and what caused it	Performance, scope, or a change of method — say which
To-complete performance index to budget at completion, against the achieved cost performance index	The credibility test
Forecast completion date and its basis	
Remaining contingency, and pending change requests against it	From `TPL-04`
Can the forecast be absorbed?	The single sentence a sponsor needs

3.8 Look-ahead

Period	Key activities	Milestones due	Planned value in the period	Resource or access constraints	What could stop it
Next period					
Following two periods					

State planned value for the coming period explicitly. It is the figure against which next month's earned value will be judged, and printing it in advance makes next month's schedule variance a shared expectation rather than a surprise.

3.9 Risk movement

Field	Entry
Risks opened this period	Reference, description, probability, impact, expected value
Risks closed	Reference, and whether closed by mitigation, by realisation or by expiry
Risks changed	Reference, what moved, and why
Top exposures	The largest few by expected value
Total expected value of the register	
Exposure against remaining contingency	The comparison, with the caveat below

Calculated column — expected value. In words: probability multiplied by impact, in the same currency and scale as the rest of the report.

```
=IF(OR($B2="",$C2=""),"",$B2*$C2)
```

State the caveat every time: a sum of expected values is not a confidence level and is not the amount of contingency required. It is an arithmetic aggregate of point estimates, useful for ranking and for movement, and it says nothing about the distribution of outcomes. Where the project needs that, it needs a quantitative analysis — see TPL-11.

3.10 Change position

From the register in TPL-04.

Field	Entry
Approved this period — number and value	
Approved to date — number and value	
Pending — number and value, with the oldest and its age	
Rejected or withdrawn to date	
Revised budget at completion	
Contingency drawn to date / remaining	
Pending value against remaining contingency	
Approved but not yet implemented	The list. This should normally be empty.

3.11 Cash position

Field	Entry
Value certified this period / cumulative gross	
Retention held	
Net certified for payment	Calculated
Cash received cumulative	
Receivables outstanding	Calculated
Uncertified work in progress	Calculated
Project cash invested	Calculated
Days sales outstanding	Calculated
Forecast net cash movement next period	From `TPL-09`
Payment issues and disputes	

Net certified for payment. In words: gross certified value less retention held.

```
=N(B2)-N(C2)
```

Receivables outstanding. In words: net certified for payment less cash received.

```
=N(D2)-N(E2)
```

Uncertified work in progress. In words: actual cost incurred less gross certified value — work done and paid for by the project but not yet certified by the client.

```
=N(F2)-N(B2)
```

Project cash invested. In words: actual cost incurred less cash received. It should equal uncertified work in progress plus retention held plus receivables outstanding, and checking that identity every month finds errors in the certification data.

```
=N(F2)-N(E2)
```

Days sales outstanding. In words: receivables outstanding divided by the average daily certified value for the period. With receivables in `G2`, value certified in the period in `H2` and days in the period in `I2`:

```
=IF(OR(N(H2)=0,N(I2)=0),"",G2/(H2/I2))
```

State the base period. Days sales outstanding computed on a single month is volatile on a project with lumpy certification, and it is a different figure from one computed on a rolling three months. Whichever you use, use it every month.

Note in the report that certified value is not necessarily recognised revenue. Whether, when and how much revenue is recognised depends on the financial reporting framework the reporting entity applies and on its accounting policy, and it varies between jurisdictions and entities. The controls report states what has been certified and what has been received; it does not state the accounting result.

3.12 Interfaces and dependencies

Interface / party	What is owed	To whom	Due	Status	Effect if late

3.13 Actions and decisions log

Ref	Action or decision	Owner	Raised	Due	Status	Note

Carry every item forward until it is closed. An item that disappears without being closed teaches readers that the log is decorative.

3.14 Data quality and basis statement

Field	Entry
Cut-off dates actually applied to each input	
Where cut-offs did not align, and the effect	
Figures that are estimated or accrued rather than actual	
Known omissions	
Changes of method or basis this period	
Where artificial intelligence assisted in preparing this report, and who validated the output	Per the register in `TPL-01` §3.13

3.15 Appendices

Full control-account table · full change register · full risk register extract · schedule extract · glossary of terms used in this report.

4. Worked fragment

Illustrative figures. The executive summary page and the cash block for the fictional facility upgrade project used throughout this series.

- **Currency and scale:** generic currency units (CU), thousands.
- **Basis:** nominal, single currency, no escalation.
- **Reporting period:** May 2026. **Data date:** 31 May 2026. **Issue date:** 8 June 2026, six working days after the data date per `TPL-01` §4.
- **Rounding:** amounts to the nearest CU thousand; indices to three decimal places; percentages to one decimal place, half away from zero. Adverse figures in parentheses.

4.1 Section 1 — decisions required

Ref	Decision required	Owner	Required by	Consequence if not taken by that date	Section
D-01	Approve or reject BCR-014, ground improvement at 22 obstructed pile positions, CU 295 thousand from contingency	Project sponsor	19 Jun 2026	Piling cannot resume at those positions. Structural steel erection in the north-east quadrant stops from approximately 17 Jul 2026, at which point the delay moves onto the driving path.	§3.10, §3.6
D-02	Confirm the funding route for the forecast variance at completion of CU (430) thousand, given remaining contingency of CU 300 thousand and BCR-014 pending against it	Project sponsor with finance	30 Jun 2026	The forecast overrun continues to be reported against contingency that would not cover it if BCR-014 is approved	§3.7
D-03	Approve the proposed recovery plan for the 21-day forecast slip, or accept a forecast practical completion of 21 May 2027	Project director	30 Jun 2026	Recovery options requiring long-lead procurement close at the end of June	§3.4

4.2 Section 2 — executive summary

Position in one sentence. The project is 47.5 per cent complete, running about five per cent over budget on the work performed and forecasting practical completion 21 days later than the contractual date, and it does not have enough contingency to cover both the pending change and the forecast overrun.

Status. Schedule: **Behind**. Cost: **Behind**. Risk: **Deteriorating**. Change: **Elevated**. Cash: **On plan**.

Headline figures. Budget at completion CU 8,000 thousand (distributed), plus CU 300 thousand contingency, total authorised CU 8,300 thousand. Per cent complete 47.5 per cent against 34.0 per cent last period. Cost performance index 0.955, from 0.968 last period. Schedule performance index 0.950, from 0.961. Selected estimate at completion CU 8,430 thousand by the bottom-up method, within a range across methods of CU 8,180 to CU 8,610 thousand. Variance at completion CU (430) thousand, being 5.4 per cent of budget at completion. Forecast practical completion 21 May 2027, 21 days after the contractual date of 30 April 2027.

What changed this period. (1) The structural steel primary frame was signed off, releasing 40 per cent credit on the largest civil activity and lifting the control account from 35.1 to 55.0 per cent complete. (2) Buried obstructions were confirmed at 22 pile positions, realising risk R-014 and generating change request BCR-014 at CU 295 thousand. (3) The forecast moved from CU 8,180 to CU 8,430 thousand as the piling productivity shortfall was accepted as structural rather than transient — a change of forecast method from Method 2 to Method 4, disclosed at §3.7.

What to worry about. Contingency. Remaining contingency is CU 300 thousand. BCR-014 requests CU 295 thousand of it. The forecast variance at completion is CU (430) thousand, and the expected value of the open risk register is CU 275 thousand. Those three figures cannot all be funded from what is left, and the decision about which of them is funded is better taken in June than in November.

Decisions required. D-01 by 19 June 2026 · D-02 by 30 June 2026 · D-03 by 30 June 2026.

Basis. All figures in generic currency units, thousands, nominal, single currency, no escalation. Data date 31 May 2026. Amounts to the nearest thousand; indices to three decimal places; percentages to one decimal place.

That page is the whole report for most of its readers, and everything in it is either a figure from a sheet elsewhere in this series or a judgement with its ground stated in the same sentence.

4.3 Section 11 — cash position

Field	Value	Derivation
Gross value certified — cumulative	3,600	Input
Gross value certified — this period	620	Input
Retention held (5 per cent of gross certified)	180	3,600 × 0.05
Net certified for payment	3,420	3,600 − 180
Cash received — cumulative	3,150	Input
Receivables outstanding	270	3,420 − 3,150
Actual cost incurred (from `TPL-07`)	3,980	Input
Uncertified work in progress	380	3,980 − 3,600
Project cash invested	830	3,980 − 3,150
Days sales outstanding (May base, 31 days)	13.5 days	270 ÷ (620 ÷ 31)

Verification of the identity. Project cash invested should equal uncertified work in progress plus retention held plus receivables outstanding: 380 + 180 + 270 = 830, which matches 3,980 − 3,150 = 830. The identity holding is the check that the certification and receipt data are consistent with the cost ledger; when it fails, one of the three inputs is stated to a different cut-off.

Verification of days sales outstanding. Average daily certified value in May = 620 ÷ 31 = 20.0. Receivables 270 ÷ 20.0 = 13.5 days. Reported on a single-month base, which is stated because a rolling three-month base would give a different figure from the same receivables.

The retention percentage is a term of the illustrative contract, not a norm, and whether certified value corresponds to recognised revenue depends on the applicable financial reporting framework and the entity's accounting policy.

5. Common mistakes

Opening with status. The reader who has twenty minutes reads what is in front of them. If page one is a status summary, the decisions on page eleven do not get taken, and the report has failed while looking complete.

An executive summary that is not true alone. A summary that says "cost performance index 0.955" with the qualification that two control accounts are excluded, appearing on page nine, is a defect. It will be forwarded, and the qualification will not.

A variance narrative that restates the number. "Unfavourable variance due to higher than expected costs" is the number in words. The reader needs the physical cause and its size.

No owner, or a function as owner. "Procurement" cannot be asked how it is going. A person can.

No status on last period's action. Without it, the narrative section becomes a monthly essay that nobody is accountable for. One row fixes it.

Position without movement. Every headline figure needs last period's beside it. A project that is behind and recovering and one that is behind and deteriorating look identical in a single-period report, and they require opposite decisions.

A single forecast figure with no method and no range. It reads as fact, and the assumption behind it disappears. See `TPL-08`.

Status carried by colour alone. It fails in monochrome, on a projector, and for a proportion of every audience. Pair the colour with a word, always.

Expected value read as a contingency requirement. A register expected value of CU 275 thousand does not mean CU 275 thousand of contingency is enough. It is a sum of point estimates and carries no confidence level.

A report that never says what it does not know. The data quality section costs half a page and is what distinguishes a report that can be relied on from one that merely looks complete.

Late enough that it describes history. A report six weeks after its data date describes a project that no longer exists. If the offset cannot be met, shorten the report rather than delay it.

6. Adapting it

Safe to change. Section order below §3.2 — health and safety, progress, cost and cash can be sequenced to suit the audience. Adding sections for procurement status, quality metrics, workforce and resourcing, sustainability reporting, or community and stakeholder matters. Adding client-specific appendices. Reporting at package level as well as project level on a large programme.

Change with care. Merging the decisions section into the executive summary. It saves a page and usually loses the required-by dates, which are the part that makes the section work. If you merge them, keep the dates in a table, not in prose.

Do not remove. Section 1 and its dates; the movement figures beside every headline; the cause, action, owner and expected effect in the variance narrative; the status of last period's action; and the data quality statement. Those five are what make the report an instrument rather than a record.

On a small project, this can be four pages: decisions, executive summary, a combined progress and cost section with the variance narrative, and cash with change and risk. Do not drop the decisions page. It is the reason the report exists.

Where the client mandates a format, produce theirs and keep this one internally for the month it takes to notice which fields the mandated format omits. Usually it is the movement figures and the status of last period's actions.

7. Completion checklist

- [ ] Decisions section is first, with an owner, a required-by date and a stated consequence for each
- [ ] Executive summary reads true with nothing else attached
- [ ] Every headline figure carries last period's figure beside it
- [ ] Units, currency, scale, basis, data date and rounding stated on the summary page

- [ ] Every control account over the reporting threshold has a variance block with all five fields
- [ ] Status of last period's actions recorded, including any abandoned
- [ ] Forecast reported with its method, its range and the credibility test
- [ ] Contingency remaining reported alongside pending change value and forecast variance at completion
- [ ] Risk expected value reported with the caveat that it is not a contingency requirement
- [ ] Approved-but-not-implemented changes listed, or confirmed as none
- [ ] Cash identity checks: cash invested equals uncertified work plus retention plus receivables
- [ ] Days sales outstanding states its base period
- [ ] No status carried by colour alone
- [ ] Data quality statement completed, including any AI assistance and who validated it
- [ ] Issued within the working-day offset committed in the controls execution plan

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Related

- `TPL-01 — Project controls execution plan` — where cadence, distribution and thresholds are fixed
- `TPL-04 — Baseline change request` — the source of the change position in §3.10
- `TPL-05 — Progress measurement and rules of credit sheet` — the source of the progress position
- `TPL-07 — Earned value calculation sheet` — the source of the cost position and the indices
- `TPL-08 — Estimate at completion scenario comparison` — the source of the forecast and its range
- `TPL-09 — Cash flow forecast` — the forward cash position referenced in §3.11
- `TPL-10 — Risk register` — the source of the risk movement in §3.9
- `BPG-14 — Monthly reporting that gets read` — why the order of sections is the whole design
- `BPG-15 — Dashboards and data visualisation for controls` — how a live dashboard and this report divide the work

Sources and standards

- PCI Master Formula Sheet (`docs/downloads/master-formula-sheet.md`), August 2026: the expected monetary value and days sales outstanding formulas used in §3.9 and §3.11. Published under the credential's retired code; the credential is PCL-AI.

This is an original instrument. The retention percentage, certification figures and dates in §4 are illustrative and do not describe any real project. Revenue recognition and accounting treatment vary by financial reporting framework and by jurisdiction, and no treatment is presented here as universal.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-07 · FREE TEMPLATES

Earned value calculation sheet

Four inputs per control account, eight derived measures, and the rounding rules that stop two people getting different answers

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

An earned value sheet takes four entered numbers per control account — budget at completion, planned value, earned value and actual cost — and derives everything else by arithmetic. This template gives the control-account table with every derived measure defined in words and as a spreadsheet expression, the guard against dividing by zero on each one, the units and rounding rules that make two people's answers agree, and the roll-up rule that stops indices being averaged.

About this document

Document	TPL-07
Series	Free Templates
Version	1.0
Status	draft
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Earned value calculation sheet

In one paragraph. An earned value sheet takes four entered numbers per control account — budget at completion, planned value, earned value and actual cost — and derives everything else by arithmetic. This template gives the control-account table with every derived measure defined in words and as a spreadsheet expression, the guard against dividing by zero on each one, the units and rounding rules that make two people's answers agree, and the roll-up rule that stops indices being averaged.

Who this is for. Cost engineers who build the sheet; control account managers who are asked to explain the indices it produces; and project controls managers who have to reconcile it with the schedule and the ledger.

1. When to use this

Produce the sheet every reporting period, at the cut-off set in the controls execution plan (`TPL-01` §3.7), immediately after the progress measurement sheet (`TPL-05`) closes and the actual cost cut-off has been applied. It sits between those two inputs and the monthly report (`TPL-06`) and the forecast (`TPL-08`), and it should be produced in that order, because every downstream number depends on it.

Two conditions must hold before the output means anything.

The three cumulative measures must be to the same data date and the same scope. Planned value from a schedule statused on the 31st, earned value from a progress claim cut off on the 25th and actual cost from a ledger closed on the 28th produce indices that describe nothing. Where the cut-offs genuinely cannot align, the actual cost figure must include an accrual to the data date, and the sheet must say so.

Actual cost must include commitments received but not yet invoiced. A ledger figure alone reports the invoicing cycle, not the work. The cost performance index of a project whose subcontractor invoices quarterly will look excellent for two months and terrible in the third, and none of the three months will be true.

Do not produce the sheet for a control account that is less than roughly fifteen per cent complete. The indices are ratios of small numbers and move violently on small absolute changes; reporting a cost performance index of 0.62 on an account that has earned three per cent of its budget invites a conversation about a problem that may not exist.

2. How to complete it

Enter four numbers per control account; compute the rest. Budget at completion, planned value, earned value and actual cost. If anything else on the sheet is typed rather than computed, someone will eventually type a figure that does not follow from the inputs, and the sheet will lose the property that makes it trustworthy.

Know where each input comes from. Budget at completion comes from the approved baseline as amended by the change register (`TPL-04`). Planned value comes from the time-phased baseline in the schedule. Earned value comes from the progress measurement sheet (`TPL-05`), as rolled-up per cent complete multiplied by budget at completion. Actual cost comes from the ledger plus accruals, reconciled to the code register (`TPL-03`). Four inputs, four different owners, and the reconciliation of each is a named responsibility in the controls execution plan.

Measure only distributed budget. Contingency and any other undistributed budget carry no planned value, no earned value and no actual cost. Include a row for them so the sheet reconciles to the total authorised budget, but compute no measures on that row — indices on a row with zero earned value are either zero or undefined and mean nothing either way.

Roll up by summing the inputs, never by averaging the indices. The project cost performance index is total earned value divided by total actual cost. It is not the mean of the control-account indices, and it is not a budget-weighted mean of them either — summing the inputs gives the budget weighting automatically and exactly.

Fix the units and the rounding, and put them on the sheet. State the currency, the scale (units, thousands, millions), the basis (nominal or constant prices, and whether escalation is included), and the rounding rule for each type of figure. Two people working from the same inputs disagree far more often about rounding than about arithmetic.

Round for display only; compute at full precision. A sheet that rounds an index to three decimal places and then uses the rounded value in a forecast introduces an error that grows with the size of the project. Every downstream formula should reference the input cells, not the displayed index.

Show adverse figures in parentheses if that is your house convention, but do it with number formatting, not by typing brackets. A cell containing `(220)` as text is text, and every formula that touches it fails or, worse, treats it as zero.

Using the tables. Copy a table block, paste into a single spreadsheet column, split on the pipe character, and delete the alignment row.

3. The template

3.1 The control-account table

Col	Field	Type	Definition and entry rule
A	Control account ID	Text	From `TPL-02`
B	Control account title	Text	
C	Control account manager	Text	The person accountable for the variance, not the person who typed the numbers
D	Budget at completion (BAC)	Input	Approved distributed budget for the account, including approved changes to the cut-off
E	Planned value (PV)	Input	Cumulative budgeted cost of work scheduled to the data date
F	Earned value (EV)	Input	Cumulative budgeted cost of work performed to the data date, from `TPL-05`
G	Actual cost (AC)	Input	Cumulative cost incurred to the data date, including accruals
H	Cost variance (CV)	Calculated	
I	Cost variance %	Calculated	
J	Schedule variance (SV)	Calculated	
K	Schedule variance %	Calculated	
L	Cost performance index (CPI)	Calculated	
M	Schedule performance index (SPI)	Calculated	
N	% complete	Calculated	
O	% of budget spent	Calculated	
P	Variance cause and owner	Text	Required on any row breaching the reporting threshold

3.2 The derived measures

Data rows begin at row 2.

Cost variance. In words: earned value less actual cost. A negative figure means more was spent than the work performed was budgeted to cost.

```
=IF(OR($F2="",$G2=""),"",$F2-$G2)
```

Cost variance per cent. In words: cost variance as a proportion of earned value — the base is what was earned, not what was budgeted, because the question is how much the work performed overran.

```
=IF(N($F2)=0,"",$H2/$F2)
```

Schedule variance. In words: earned value less planned value, expressed in currency. It is a measure of work not done, not of time; a project can carry a large negative schedule variance and still finish on time if the missing work is off the critical path.

```
=IF(OR($F2="",$E2=""),"",$F2-$E2)
```

Schedule variance per cent. In words: schedule variance as a proportion of planned value — the base is what was planned, because the question is how far behind the plan the work is.

```
=IF(N($E2)=0,"",$J2/$E2)
```

Cost performance index. In words: earned value divided by actual cost. Below one means the work performed cost more than it was budgeted to cost.

```
=IF(N($G2)=0,"",$F2/$G2)
```

Schedule performance index. In words: earned value divided by planned value. Below one means less work has been performed than was scheduled by this date.

```
=IF(N($E2)=0,"",$F2/$E2)
```

Per cent complete. In words: earned value as a proportion of budget at completion.

```
=IF(N($D2)=0,"",$F2/$D2)
```

Per cent of budget spent. In words: actual cost as a proportion of budget at completion. Reported beside per cent complete because the gap between the two is the cost variance stated as a proportion, and some readers see it faster that way.

```
=IF(N($D2)=0,"",$G2/$D2)
```

`N()` returns zero for a blank or text cell, so the guard holds whether the input is zero, empty or mistyped. `IFERROR` would also suppress the error, but it suppresses every error, including the ones that signal a broken reference; prefer an explicit guard on the denominator.

3.3 Roll-up and reconciliation

Roll-up. In words: sum the four inputs across the control accounts, then apply the same formulas to the totals. Every index in the total row must reference the summed inputs, never the individual indices.

```
D_total  =SUM($D$2:$D$20)     E_total  =SUM($E$2:$E$20)
F_total  =SUM($F$2:$F$20)     G_total  =SUM($G$2:$G$20)
```

Reconciliation to the authorised budget. Below the measured total, add rows that carry budget but no performance measurement:

Row	Content
Measured total — distributed budget	Sum of the control accounts above
Undistributed budget	Budget allocated but not yet in a control account
Contingency reserve	Held against identified risk; released by change (`TPL-04`)
Total authorised budget	The figure that must agree with the change register

State on the sheet whether the contingency reserve is inside or outside budget at completion on this project, because both conventions are in use and the same set of facts produces two different totals under them. The convention is fixed once, in `TPL-01` §3.2.

3.4 Units, rounding and presentation

State these on the face of the sheet:

Item	Rule
Currency and scale	Named currency and scale, e.g. thousands. Currency-neutral sheets use a generic unit.
Basis	Nominal or constant prices; whether escalation and exchange effects are included
Data date	One date, applying to all three cumulative measures
Amounts	Rounded to the stated scale; never rounded before summing
Indices	Three decimal places throughout, or two throughout — state which and never mix
Percentages	One decimal place; rounding half away from zero
Adverse figures	Parentheses by number format, not by typed brackets
Precision	Display rounding only; all downstream formulas reference unrounded cells

On the index convention: the Institute's master formula sheet rounds indices to two decimal places, while worked examples in the publication framework are shown to three. Either is defensible. Three decimal places is preferable on a large project, because at two decimal places a cost performance index of 0.955 and one of 0.954 both display as 0.95 while differing by tens of thousands of currency units in the resulting forecast. Choose one convention per project, write it on the sheet, and apply it everywhere.

3.5 Reporting thresholds

Set a threshold in the controls execution plan and apply it here, so that the variance narrative in `TPL-06` is written for a defined set of rows rather than for whichever ones caught someone's eye.

```
=IF($L2="","",IF(OR(ABS(N($I2))>threshold_pct,ABS(N($H2))>threshold_value),"EXPLAIN",""))
```

A threshold with both a percentage and an absolute limb is the workable form: percentage alone flags trivial amounts on small accounts, and value alone misses a large proportional overrun on a small one.

4. Worked fragment

Illustrative figures. A fictional facility upgrade project at the May 2026 cut-off.

- **Currency and scale:** generic currency units (CU), thousands.
- **Basis:** nominal, single currency, no escalation or exchange effects included.
- **Data date:** 31 May 2026, applying to planned value, earned value and actual cost alike.
- **Rounding:** amounts to the nearest CU thousand; indices to three decimal places; percentages to one decimal place, half away from zero. Adverse figures in parentheses.
- **Contingency convention on this project:** the contingency reserve sits inside budget at completion and outside the control accounts, as recorded in the controls execution plan.

Earned value for CA-1000 is the figure produced in `TPL-05` §4.1: a rolled-up 55.0 per cent complete against a budget at completion of CU 4,000 thousand.

CA	Title	BAC	PV	EV	AC	CV	CV %	SV	SV %	CPI	SPI	% compl.	% s
CA-1000	Civil works	4,000	2,400	2,200	2,420	(220)	(10.0 %)	(200)	(8.3 %)	0.909	0.917	55.0 %	6 %
CA-2000	Mechanical works	3,000	1,200	1,260	1,200	60	4.8 %	60	5.0 %	1.050	1.050	42.0 %	4 %
CA-3000	Controls and commissioning	1,000	400	340	360	(20)	(5.9 %)	(60)	(15.0 %)	0.944	0.850	34.0 %	3 %
	Measured total	**8,000**	**4,000**	**3,800**	**3,980**	**(180)**	**(4.7 %)**	**(200)**	**(5.0 %)**	**0.955**	**0.950**	**47.5 %**	**4 %**
	Contingency reserve	300	—	—	—	—	—	—	—	—	—	—	—
	Total authorised budget	**8,300**											

Verification, control account by control account.

CA-1000: CV = 2,200 − 2,420 = (220); CV % = (220) ÷ 2,200 = (10.0 %); SV = 2,200 − 2,400 = (200); SV % = (200) ÷ 2,400 = (8.3 %); CPI = 2,200 ÷ 2,420 = 0.909; SPI = 2,200 ÷ 2,400 = 0.917; % complete = 2,200 ÷ 4,000 = 55.0 %; % spent = 2,420 ÷ 4,000 = 60.5 %.

CA-2000: CV = 1,260 − 1,200 = 60; CV % = 60 ÷ 1,260 = 4.8 %; SV = 1,260 − 1,200 = 60; SV % = 60 ÷ 1,200 = 5.0 %; CPI = 1,260 ÷ 1,200 = 1.050; SPI = 1,260 ÷ 1,200 = 1.050; % complete = 1,260 ÷ 3,000 = 42.0 %; % spent = 1,200 ÷ 3,000 = 40.0 %.

CA-3000: CV = 340 − 360 = (20); CV % = (20) ÷ 340 = (5.9 %); SV = 340 − 400 = (60); SV % = (60) ÷ 400 = (15.0 %); CPI = 340 ÷ 360 = 0.944; SPI = 340 ÷ 400 = 0.850; % complete = 340 ÷ 1,000 = 34.0 %; % spent = 360 ÷ 1,000 = 36.0 %.

Verification of the totals. BAC 4,000 + 3,000 + 1,000 = 8,000. PV 2,400 + 1,200 + 400 = 4,000. EV 2,200 + 1,260 + 340 = 3,800. AC 2,420 + 1,200 + 360 = 3,980. Then CV = 3,800 − 3,980 = (180); CV % = (180) ÷ 3,800 = (4.7 %); SV = 3,800 − 4,000 = (200); SV % = (200) ÷ 4,000 = (5.0 %); CPI = 3,800 ÷ 3,980 = 0.955; SPI = 3,800 ÷ 4,000 = 0.950; % complete = 3,800 ÷ 8,000 = 47.5 %; % spent = 3,980 ÷ 8,000 = 49.8 % (49.75 % before rounding).

Why the total is not the average. The mean of the three cost performance indices is (0.909 + 1.050 + 0.944) ÷ 3 = 0.968, against the correct 0.955. The averaging method flatters the project because it gives the small, well-performing mechanical account the same weight as the large, poorly performing civil account. The gap widens as the accounts become less equal in size.

What the sheet is actually saying. The project has earned 47.5 per cent of its budget and spent 49.8 per cent of it. CA-3000 has the worst schedule performance index at 0.850 but the smallest budget, so it contributes least to the project total; CA-1000 carries both the largest budget and the worst cost performance, and is therefore where the variance narrative in `TPL-06` and the forecast in `TPL-08` should concentrate. The forecast at completion is not derived here — the choice of method is a judgement, and it belongs to `TPL-08`.

5. Common mistakes

Mixed data dates. Earned value cut off on the 25th, actual cost on the 28th and planned value statused on the 31st produce indices that describe no real state of the project. The most common form is actual cost with no accrual, which reports the invoicing cycle rather than the work.

Averaging indices to roll up. See §4. Sum the inputs; the weighting then happens by itself.

Computing indices on undistributed budget or contingency. There is no earned value to divide by, so the result is either zero or an error, and both look like a finding.

Reading schedule variance as time. A schedule variance of CU (200) thousand does not mean twenty days late. It means the work performed was budgeted at CU 200 thousand less than the work scheduled. The translation to time requires either the schedule itself or an earned-schedule calculation, and both are separate exercises.

Publishing indices on a barely started account. At three per cent complete the index is a ratio of two small numbers and swings on rounding. Suppress it or annotate it.

Rounding before summing. Rounding each control account to the nearest thousand and then adding gives a total that differs from the total of the unrounded figures. Sum first, round for display.

Typed brackets for negatives. A cell containing the text `(220)` is not the number −220. Use number formatting.

Chasing a favourable variance no harder than an adverse one. A cost performance index of 1.05 on CA-2000 is as much a measurement question as one of 0.909. Common causes are earned value claimed ahead of the evidence, actual cost not yet accrued, and scope quietly moved out of the account — and all three reverse in a later period.

A sheet with typed values in the derived columns. The moment one derived figure is entered by hand, the sheet stops being a calculation and becomes a claim, and no reader can tell which cells are which.

6. Adapting it

Safe to change. Adding period columns beside the cumulative ones, provided the two are visually distinguished — period indices are volatile and are often the more useful diagnostic. Adding a prior-period block so movement is visible on the face of the sheet. Adding columns for hours as well as cost, on projects that control both. Reporting at work-package level beneath the control account.

Change with care. Adding earned-schedule measures. They answer the time question that schedule variance does not, but they require the time-phased planned value curve and a separate interpolation, and half-implementing them produces a number that looks like time and is not.

Do not remove. The reconciliation to total authorised budget, the statement of the contingency convention, and the units and rounding block. Without them the sheet cannot be reconciled to anything and two readers will legitimately compute different answers from it.

Where the client requires a different index convention — two decimal places, a different variance base, percentages against budget rather than earned value — run theirs on the client report and keep this one as the internal sheet, with the conversion stated. Do not change the internal convention mid-project; the comparability of your own history is worth more than the convenience.

7. Completion checklist

- [ ] One data date, applying to planned value, earned value and actual cost alike, stated on the sheet
- [ ] Actual cost includes accruals for work received and not yet invoiced
- [ ] Earned value reconciles to the roll-up in the progress measurement sheet
- [ ] Budget at completion reconciles to the baseline plus approved changes in the change register
- [ ] Only the four inputs are typed; every other column is a formula
- [ ] Every division guarded against a zero or blank denominator
- [ ] Totals computed by summing inputs, not by averaging indices
- [ ] Undistributed budget and contingency shown, with no measures computed on them
- [ ] Sheet reconciles to total authorised budget
- [ ] Currency, scale, basis, data date and rounding rules stated on the face of the sheet
- [ ] Index convention — two or three decimal places — stated and applied consistently
- [ ] Accounts below the materiality threshold for meaningful indices annotated or suppressed
- [ ] Every row breaching the reporting threshold has a named cause and owner before issue

Related

- `TPL-05 — Progress measurement and rules of credit sheet` — where the earned value input is produced
- `TPL-06 — Monthly project controls report` — where these measures are narrated and explained
- `TPL-08 — Estimate at completion scenario comparison` — the forecast this sheet deliberately does not make
- `BPG-06 — Progress measurement and rules of credit` — why the earned value input is the contested one
- `BPG-08 — Earned value in practice` — interpreting the indices on a real control account

Sources and standards

- PCI Master Formula Sheet (`docs/downloads/master-formula-sheet.md`), August 2026: the definitions of cost variance, schedule variance, the cost and schedule performance indices, and the budget identities used in §3.3. Published under the credential's retired code; the credential is PCL-AI.

The measures here are common to established earned value practice and are stated in the Institute's own words. No third-party sheet, table or wording is reproduced.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-08 · FREE TEMPLATES

Estimate at completion scenario comparison

One input block, four methods side by side, and a recommendation field that forces you to say why

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

Four standard ways of computing an estimate at completion will give four different answers from the same inputs, because each encodes a different assumption about whether the variance suffered so far will happen again. This template puts them side by side from one input block, computes variance at completion and the to-complete performance index for each, names the assumption each one encodes, and requires the user to record which was selected and why.

About this document

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Estimate at completion scenario comparison

In one paragraph. Four standard ways of computing an estimate at completion will give four different answers from the same inputs, because each encodes a different assumption about whether the variance suffered so far will happen again. This template puts them side by side from one input block, computes variance at completion and the to-complete performance index for each, names the assumption each one encodes, and requires the user to record which was selected and why.

Who this is for. Cost engineers and project controls managers who prepare the forecast; project directors and sponsors who approve it; and anyone who has been asked why the estimate at completion moved and needs an answer better than "the index changed".

1. When to use this

Produce the comparison every period in which a forecast is reported, immediately after the earned value sheet (`TPL-07`) closes. Producing a single estimate at completion by one method and reporting it as the forecast conceals the judgement that was made; producing four and selecting one makes the judgement visible and reviewable.

Three moments make it more than a routine:

- **The first period in which the cost performance index departs materially from one.** That is the period in which the project decides, usually without noticing, whether it believes the variance is structural or a one-off. The sheet forces the decision to be recorded.
- **Before a request for additional funding or a contingency release.** The forecast is the basis of the request, and the spread between methods is the honest expression of its uncertainty.
- **When the selected method changes.** A change of method moves the forecast without anything happening on site. It must be disclosed as a method change, with the previous method's answer shown alongside for one period.

Do not use index-based methods early. Below roughly fifteen to twenty per cent complete the indices are ratios of small numbers, and extrapolating them across the remaining eighty per cent produces a forecast with a wide error and a confident appearance. Early in a project, the bottom-up method is the only one worth reporting.

2. How to complete it

Enter the input block once. Budget at completion, planned value, earned value and actual cost, all to the same data date and all from the earned value sheet. Everything else on this sheet is computed from them. The definitions and interpretation of the indices belong to `TPL-07` and are not restated here; they are computed in the input block so that this sheet stands alone.

State which budget at completion you are using. The performance measurement baseline — distributed control-account budget — is the correct base for an index-based forecast, because it is the only budget against which earned value has been measured. If contingency sits inside your budget at completion, take it out for this calculation and show the remaining contingency separately below the forecast. A forecast that quietly absorbs contingency has removed a decision from the person entitled to make it.

Compute the bottom-up estimate to complete separately, and properly. It is the only method that is not an extrapolation, and it is the only one that can see a change in the work ahead. Ask each control account manager for a re-estimate of the remaining work at current rates, with the same estimate basis discipline you would apply to a change request (`TPL-04`): quantity, rate, source, and the assumptions the number depends on. A bottom-up figure assembled by scaling the remaining budget is not a bottom-up figure.

Read the to-complete performance index as a credibility test, not as a target. It states the cost performance the remaining work must achieve for a given endpoint to be met. Compare it with the cost performance index actually achieved. When the required performance is materially better than anything the project has yet demonstrated, the endpoint is an aspiration and the sheet says so in one number.

Use the three arithmetic self-checks in §3.4. Each index-based method has a to-complete performance index that must come out to a specific value. If yours does not, the sheet has an error in it.

Compute at full precision, round for display. Using a cost performance index rounded to three decimal places rather than the underlying ratio moves the first method's answer in the worked example by CU 2 thousand. On a larger project the same error is material.

Fill the recommendation block before the forecast leaves the room. The methods do not choose between themselves. The block requires the selected method, the reason it was selected over the others, what would change the choice, and who challenged it.

Using the tables. Copy a table block, paste into a single spreadsheet column, split on the pipe character, and delete the alignment row.

3. The template

3.1 Input block

Hold the inputs in a single column so every method references the same cells.

Cell	Field	Type	Entry rule
B1	Project / control account	Text	The level at which this forecast is made
B2	Data date	Date	Must match the earned value sheet
B3	Budget at completion (BAC)	Input	Distributed performance measurement baseline
B4	Planned value (PV)	Input	Cumulative to the data date
B5	Earned value (EV)	Input	Cumulative to the data date
B6	Actual cost (AC)	Input	Cumulative to the data date, including accruals
B7	Cost performance index (CPI)	Calculated	`=IF(N($B$6)=0,"",$B$5/$B$6)`
B8	Schedule performance index (SPI)	Calculated	`=IF(N($B$4)=0,"",$B$5/$B$4)`
B9	Bottom-up estimate to complete	Input	Re-estimate of the remaining work, assembled from the control accounts
B10	To-complete performance index to BAC	Calculated	See §3.3
B11	Remaining contingency	Input	From the change register (`TPL-04`)
B12	Currency, scale and basis	Text	e.g. CU thousands, nominal, no escalation

3.2 The method comparison table

Four rows, one per method. Data rows 15 to 18.

Col	Field	Type	Definition
A	Method	Text	
B	Assumption this method encodes	Text	Pre-filled below; do not paraphrase it away
C	Estimate at completion	Calculated	
D	Variance at completion	Calculated	
E	Variance at completion %	Calculated	
F	To-complete performance index to this estimate	Calculated	
G	Evidence supporting this assumption	Text	What is true about this project that makes it right
H	Evidence against	Text	What is true about this project that makes it wrong

Method 1 — Budget at completion divided by the cost performance index. In words: the remaining work will be delivered at the same cost efficiency as the work already done. Formula: `EAC = BAC ÷ CPI`. Spreadsheet, written so it never divides by a rounded or blank index:

```
=IF(OR(N($B$5)=0,N($B$6)=0),"",$B$3*$B$6/$B$5)
```

Since the cost performance index is earned value divided by actual cost, dividing budget at completion by it is the same as multiplying budget at completion by actual cost and dividing by earned value. The second form evaluates at full precision and guards both denominators.

Method 2 — Actual cost plus the remaining budget. In words: the variance to date was a one-off, and the remaining work will be delivered at the budgeted rate. Formula: `EAC = AC + (BAC − EV)`. Spreadsheet:

```
=IF(OR($B$3="",$B$5="",$B$6=""),"",$B$6+($B$3-$B$5))
```

Method 3 — Actual cost plus the remaining budget divided by the cost and schedule indices compounded. In words: the remaining work will be affected both by the cost efficiency achieved so far and by the schedule pressure implied by working behind plan. Formula: `EAC = AC + (BAC − EV) ÷ (CPI × SPI)`. Spreadsheet:

```
=IF(OR(N($B$4)=0,N($B$5)=0,N($B$6)=0),"",$B$6+($B$3-$B$5)/(($B$5/$B$6)*($B$5/$B$4)))
```

All three denominators are guarded: the two indices need non-zero actual cost and planned value, and the compound product needs non-zero earned value.

Method 4 — Actual cost plus a bottom-up estimate to complete. In words: historical performance does not describe the work remaining, so the remaining work has been re-estimated. Formula: `EAC = AC + ETC`. Spreadsheet:

```
=IF($B$9="","",$B$6+$B$9)
```

Variance at completion (column D). In words: budget at completion less the estimate at completion. Negative means a forecast overrun.

```
=IF(NOT(ISNUMBER(C15)),"",$B$3-C15)
```

Variance at completion per cent (column E). In words: variance at completion as a proportion of budget at completion.

```
=IF(OR(NOT(ISNUMBER(D15)),N($B$3)=0),"",D15/$B$3)
```

To-complete performance index to this estimate (column F). In words: the cost performance the remaining work must achieve for this estimate to be met — remaining budgeted work divided by remaining forecast money.

```
=IF(OR(NOT(ISNUMBER(C15)),C15=$B$6),"",($B$3-$B$5)/(C15-$B$6))
```

The `ISNUMBER` test is what makes the guard safe: a blank estimate cell is text, and subtracting actual cost from it would produce an error that propagates across the row.

3.3 To-complete performance index to budget at completion

In words: the cost performance the remaining work must achieve to finish within the original budget. Formula: `TCPI to BAC = (BAC − EV) ÷ (BAC − AC)`. Spreadsheet, in cell `B10`:

```
=IF(OR($B$3="",$B$3=$B$6),"",($B$3-$B$5)/($B$3-$B$6))
```

Where actual cost has already reached or passed budget at completion the denominator is zero or negative and the measure stops being meaningful — the guard returns blank, and the honest report is that the budget cannot be met, not a negative index.

Compare `B10` with `B7`. The ratio of the two is how much better than achieved performance the remaining work must run:

```
=IF(OR(N($B$7)=0,$B$10=""),"",$B$10/$B$7-1)
```

3.4 The three arithmetic self-checks

Each index-based method implies a specific to-complete performance index. These are identities, not coincidences, and they are the fastest way to find an error in the sheet.

Method	Its to-complete performance index must equal	Why
1 — `BAC ÷ CPI`	the cost performance index	Substituting `EAC = BAC × AC ÷ EV` gives `EAC − AC = AC × (BAC − EV) ÷ EV`, so the index reduces to `EV ÷ AC`
2 — `AC + (BAC − EV)`	exactly 1.000	`EAC − AC = BAC − EV`, so numerator and denominator are identical
3 — `AC + (BAC − EV) ÷ (CPI × SPI)`	the cost index multiplied by the schedule index	`EAC − AC = (BAC − EV) ÷ (CPI × SPI)`, so the ratio is `CPI × SPI`

Method 4 has no identity, because its estimate to complete is entered rather than derived. That is the point of it.

3.5 The recommendation block

Field	Entry
Method selected	
Why this method rather than the other three	Required. Name the cause of the variance to date and say whether it applies to the work remaining. A reason that would be true on any project is not a reason.
Evidence relied on	Estimate basis, quotations, productivity records, subcontract position
Previous period's method and estimate	If the method has changed, both are reported for one period
Range reported	Lowest and highest of the four, as the honest expression of uncertainty
What would change this choice	The specific observation that would move the forecast next period
Remaining contingency	From the change register
Variance at completion against remaining contingency	Whether the forecast can be absorbed
Prepared by / date	
Challenged by / date	A second person, not the preparer
Approved by / date	
Next review	

4. Worked fragment

Illustrative figures. The same fictional facility upgrade project as `TPL-07`, at the same cut-off.

- **Currency and scale:** generic currency units (CU), thousands.
- **Basis:** nominal, single currency, no escalation.
- **Data date:** 31 May 2026.
- **Budget at completion used:** CU 8,000 thousand, the distributed performance measurement baseline. The contingency reserve of CU 300 thousand sits outside it and is reported separately below.
- **Rounding:** amounts to the nearest CU thousand; indices to three decimal places; percentages to one decimal place, half away from zero. Adverse figures in parentheses. All arithmetic performed at full precision.

Input block.

Field	Value
Budget at completion	8,000
Planned value	4,000
Earned value	3,800
Actual cost	3,980
Cost performance index	0.955
Schedule performance index	0.950
Bottom-up estimate to complete	4,450
To-complete performance index to budget at completion	1.045
Remaining contingency	300

Indices verified: CPI = 3,800 ÷ 3,980 = 0.955 (0.954774 unrounded); SPI = 3,800 ÷ 4,000 = 0.950; TCPI to BAC = (8,000 − 3,800) ÷ (8,000 − 3,980) = 4,200 ÷ 4,020 = 1.045.

Method comparison.

Method	Assumption encoded	EAC	VAC	VAC %	TCPI to this EAC
1 — BAC ÷ CPI	Cost efficiency achieved to date continues across all remaining work	8,379	(379)	(4.7 %)	0.955
2 — AC + (BAC − EV)	The variance to date was a one-off; remaining work runs at budget	8,180	(180)	(2.3 %)	1.000
3 — AC + (BAC − EV) ÷ (CPI × SPI)	Remaining work is affected by both cost efficiency and schedule pressure	8,610	(610)	(7.6 %)	0.907
4 — AC + bottom-up ETC	History does not describe the work ahead; it has been re-estimated	8,430	(430)	(5.4 %)	0.944

Verification of each method.

Method 1: 8,000 ÷ 0.954774 = 8,378.95, so CU 8,379 thousand. Equivalently 8,000 × 3,980 ÷ 3,800 = 8,378.95. Variance at completion = 8,000 − 8,378.95 = (378.95), shown as (379); (378.95) ÷ 8,000 = (4.7 %). Using the rounded index of 0.955 instead would give 8,000 ÷ 0.955 = 8,376.96, CU 2 thousand adrift — which is why the formula in §3.2 avoids the index cell.

Method 2: 3,980 + (8,000 − 3,800) = 3,980 + 4,200 = 8,180. Variance at completion = 8,000 − 8,180 = (180); (180) ÷ 8,000 = (2.25 %), displayed as (2.3 %). Note that this variance at completion equals the cost variance in `TPL-07`, which it must: the method assumes no further variance is incurred.

Method 3: CPI × SPI = 0.954774 × 0.950 = 0.907035. Remaining budgeted work ÷ that product = 4,200 ÷ 0.907035 = 4,630.47. Estimate at completion = 3,980 + 4,630.47 = 8,610.47, so CU 8,610 thousand. Variance at completion = 8,000 − 8,610.47 = (610.47), shown as (610); (610.47) ÷ 8,000 = (7.6 %).

Method 4: 3,980 + 4,450 = 8,430. Variance at completion = 8,000 − 8,430 = (430); (430) ÷ 8,000 = (5.4 %). The bottom-up estimate to complete of 4,450 sits CU 250 thousand — 6.0 per cent — above the remaining budgeted work of 4,200.

Verification of the self-checks. Method 1: 4,200 ÷ (8,378.95 − 3,980) = 4,200 ÷ 4,398.95 = 0.955, equal to the cost performance index. Method 2: 4,200 ÷ (8,180 − 3,980) = 4,200 ÷ 4,200 = 1.000. Method 3: 4,200 ÷ (8,610.47 − 3,980) = 4,200 ÷ 4,630.47 = 0.907, equal to CPI × SPI. All three identities hold, so the sheet is arithmetically sound.

The credibility test. Finishing within the original budget requires a to-complete performance index of 1.045 against a cost performance index of 0.955 achieved so far — a ratio of 1.094, meaning the remaining work would have to be delivered about **nine per cent more efficiently than anything the project has yet demonstrated**, having so far run about five per cent worse than budget. Nothing in the record supports that, so budget at completion is not a forecast; it is a target.

The range. The four methods span CU 8,180 to CU 8,610 thousand, a spread of CU 430 thousand, or 5.4 per cent of budget at completion. That spread is the honest statement of forecast uncertainty at this data date, and it belongs in the monthly report next to whichever single figure is selected.

Recommendation, as it would be written.

Method selected: Method 4, bottom-up, at CU 8,430 thousand.

Why this method rather than the other three. The cost variance to date is concentrated in CA-1000 and has two identified causes: piling productivity below the estimate basis, and rework at the pile cap interface. The piling cause is structural and will persist across the remaining 132 piles, so Method 2 — which assumes the variance was a one-off — is rejected. The rework cause has been closed by a revised inspection sequence and will not recur, so Method 1, which projects the whole variance to date across all remaining work including the mechanical account that is currently ahead, overstates it. Method 3 additionally loads the schedule performance index onto the forecast; the schedule variance here arises in CA-3000, whose work is largely still ahead and is not resource-constrained, so compounding is not supported. The control account managers have re-estimated the remaining work at CU 4,450 thousand against a remaining budget of CU 4,200 thousand, with the piling shortfall priced at current measured rates.

What would change this choice. If piling output over June does not recover to the re-estimated rate, the bottom-up basis fails and Method 1 becomes the defensible forecast. If mechanical installation begins to consume float, Method 3 becomes relevant.

Funding position. Forecast variance at completion of CU (430) thousand against remaining contingency of CU 300 thousand — a shortfall of CU 130 thousand — before BCR-014, which itself requests CU 295 thousand from that contingency and is under review (`TPL-04` §4.1). If BCR-014 is approved, CU 5 thousand of contingency remains against a forecast overrun of CU 430 thousand. That is the decision the sponsor is being asked for, and it belongs on page one of the monthly report.

5. Common mistakes

Reporting one method as though it were the answer. The single figure hides the assumption. Report the selected figure with the range and the reason.

Using an index-based method too early. Below roughly fifteen to twenty per cent complete the extrapolation is built on a ratio of small numbers, and its confident appearance is the dangerous part.

Using a rounded index in the formula. Small on this example, material on a large project, and always avoidable by referencing the input cells.

Applying the compound method by default. Multiplying by the schedule performance index assumes that being behind schedule will cost money on the remaining work. Sometimes it will — acceleration, out-of-sequence working, extended preliminaries. Sometimes the schedule variance sits in work that is not resource-constrained and costs nothing extra. Compounding without that argument is not conservatism; it is an unexamined assumption dressed as prudence.

A "bottom-up" estimate that is the remaining budget with a factor applied. That is Method 1 with extra steps, and it inherits none of the diagnostic value.

Treating the to-complete performance index as a target. It is a statement of what is required, not an instruction. Circulating it as a productivity goal is how a forecast becomes a negotiation.

Forgetting that a contingency release changes budget at completion. When a contingency release moves budget into a control account, budget at completion for measurement rises and every index, variance and forecast is computed on the new base. Recompute; do not carry the previous period's forecast forward.

Changing method quietly. The forecast moves and nothing happened. Disclose the change, show both answers for one period, and say what caused the change of view.

Comparing the forecast overrun with total contingency rather than remaining contingency, or ignoring pending change requests against it. The relevant number is what is left after everything already approved and everything currently in the approval queue.

6. Adapting it

Safe to change. Running the sheet at control-account level as well as at project level, which is usually more informative — a project-level forecast averages away the account where the problem is. Adding methods: a weighted compound variant that blends the cost and schedule indices in some other proportion is legitimate provided the weighting is stated and justified on the sheet rather than inherited from a spreadsheet somebody was given. Adding an optimistic and pessimistic bottom-up case to widen the range deliberately.

Change with care. Adding a probabilistic forecast from a cost risk analysis. It answers a different question — the distribution of outcomes rather than a point estimate under an assumption — and placing it in the same table invites readers to compare a percentile with a deterministic figure as though they were the same kind of number. If you add it, label it separately.

Do not remove. The assumption column, the to-complete performance index column, the range, and the recommendation block's "why this method rather than the others" field. Without those four the sheet becomes a menu of numbers from which the most convenient can be chosen.

Where the client mandates a method, compute theirs and report it as the contractual forecast, and keep this comparison as the internal position with the difference reconciled. Do not let a mandated method become the project's own view of itself.

7. Completion checklist

- [ ] Inputs match the earned value sheet exactly, to the same data date
- [ ] Budget at completion used is the distributed performance measurement baseline, stated as such

- [ ] Contingency shown separately, not absorbed into the forecast
- [ ] All four methods computed from the same input cells
- [ ] Every division guarded, and every estimate cell tested with `ISNUMBER` before subtraction
- [ ] Formulas reference unrounded inputs, not displayed indices
- [ ] The three arithmetic self-checks in §3.4 all hold
- [ ] To-complete performance index to budget at completion computed and compared with the achieved index
- [ ] Bottom-up estimate to complete assembled from control accounts with a stated estimate basis
- [ ] Assumption column completed for every method, with evidence for and against
- [ ] Method selected, with a reason specific to this project's variance causes
- [ ] Range reported alongside the selected figure
- [ ] Variance at completion compared with remaining contingency after pending changes
- [ ] Forecast challenged by a second person and approved, both named and dated
- [ ] Any change of method disclosed, with the previous method's answer shown for one period

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Related

- `TPL-07 — Earned value calculation sheet` — the source of every input, and where the indices are defined
- `TPL-04 — Baseline change request` — the contingency position the forecast must be read against
- `TPL-06 — Monthly project controls report` — where the selected forecast and its range are reported
- `BPG-09 — Estimate at completion — choosing and defending a method` — the judgement in full
- `BPG-10 — Contingency and management reserve` — how a forecast overrun and contingency relate

Sources and standards

- PCI Master Formula Sheet (`docs/downloads/master-formula-sheet.md`), August 2026: the estimate at completion, variance at completion and to-complete performance index formulas used throughout. Published under the credential's retired code; the credential is PCL-AI.

The four methods are common to established earned value practice and are stated here in the Institute's own words. No third-party sheet, table or wording is reproduced. The identities in §3.4 are derived from the formulas themselves and are verified in §4.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-09 · FREE TEMPLATES

Cash flow forecast

A period-by-period model of the distance between work done, value certified and money received

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A working cash flow forecast that separates the three things most forecasts merge: the value of work executed, the value certified for payment, and the cash that actually arrives. It models retention withheld and released, advance-payment recovery, other deductions and contractual payment terms, and it derives the receipt period from the due date rather than assuming one. Complete it and you will know your peak funding requirement and the period it occurs in.

About this document

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Cash flow forecast

In one paragraph. A working cash flow forecast that separates the three things most forecasts merge: the value of work executed, the value certified for payment, and the cash that actually arrives. It models retention withheld and released, advance-payment recovery, other deductions and contractual payment terms, and it derives the receipt period from the due date rather than assuming one. Complete it and you will know your peak funding requirement and the period it occurs in.

Who this is for. Cost engineers, project controls managers, commercial managers and quantity surveyors who own the forecast; and the project directors and finance business partners who have to fund the gap it reveals.

1. When to use this

Use it at three moments, and it does a different job at each.

At tender or sanction, to price the cost of funding the work. The peak funding requirement this sheet produces is a real cost — it is financed, and someone pays the finance charge. A tender that has not calculated it has priced the work and not the job.

Monthly, alongside the cost report, as the forecast the treasury function actually uses. The cost report answers "what will this cost". This sheet answers "when do we need the money, and how much of it".

Whenever the payment mechanics change — a variation to the payment terms, a change in the retention regime, an advance payment agreed or a certification dispute that stops the certified value tracking the executed value. Each of those moves the cash curve without moving the cost forecast at all, which is precisely why a project can be reporting a healthy margin while it cannot pay its subcontractors.

Do not use it as a substitute for the cost forecast. It takes the cost forecast as an input and converts it into money and timing. If the underlying forecast is wrong, this sheet will produce a confidently wrong cash curve. `BPG-13 — Cash flow forecasting` sets out the method; this is the instrument.

2. How to complete it

2.1 Set the period basis first

Decide the period — calendar month, four-week accounting period, or quarter — and use the same period in the cost report, the payment application cycle and this sheet. Two different period bases in one project is the single most common cause of a cash forecast that never reconciles.

Enter the **period-end date** for every period in the model, including a **period 0** row that carries the opening position: any advance payment received, mobilisation spend, and opening balances.

Every lookup in this template keys off the period-end dates, so they must be complete, ascending, and one per row.

State the **currency and the base date**. If more than one currency is in play, run one sheet per currency and consolidate outside this template — converting inside it hides the exposure rather than showing it.

2.2 Set the parameters

Put these on a separate sheet named `Parameters` and define each as a named range. Every formula below refers to them by name.

Name	What it is	Where it comes from
`Contract_Value`	The current contract sum, updated for agreed change	The contract and the change order log
`Retention_Rate`	Proportion of each gross certified valuation withheld, as a decimal	The contract
`Retention_Cap`	The maximum retention that may be held at any one time, as a currency amount	The contract, usually a percentage of `Contract_Value`
`Recovery_Rate`	Proportion of each gross certified valuation applied to recovering the advance payment	The contract
`Cert_Lag`	Whole periods between work being executed and its value being certified	The valuation cycle in the contract
`Payment_Terms_Days`	Days from the certification date to the contractual due date	The contract

Two notes on these. `Retention_Cap` is a currency amount, not a percentage, because the cap and the rate are usually expressed against different bases and mixing them is a common error. `Payment_Terms_Days` is in calendar days here; if your contract counts working days, replace the addition in column Q with `WORKDAY(B3, Payment_Terms_Days, Holidays)`.

Payment regimes, retention rules and any statutory protection over payment differ by contract and by jurisdiction. This template models the mechanics; which mechanics apply to you is a matter for your contract and the law governing it.

2.3 Work through the columns in order

The sheet runs left to right in the order the money moves: what was planned, what was built, what was certified, what was deducted, what is due, when it lands, and what is left. Complete a period fully before moving to the next. The field definitions are in §3.

Two columns deserve attention before you start. **Column G, gross value certified**, defaults to the value of work executed but should be overridden with the certified figure the moment it is known — the gap between those two numbers is the earliest warning you get of a commercial problem. **Column I** reports that gap cumulatively; a figure below 100 % means you have built work you have not yet been paid for on paper, quite apart from the payment terms.

3. The template

Header row in row 1. Row 2 is period 0 and is entered as values. Formulas begin in row 3 and fill down.

3.1 Input columns

Col	Field	What goes in it
A	Period number	Integer, starting at 0 for the opening row. Unique, ascending, no gaps
B	Period ending	The date the period closes and the valuation is cut off. Ascending, one per row
C	Planned value in period	Value of work the baseline says will be executed in the period, at contract rates
E	Work executed in period	Value of work actually executed in the period at contract rates; forecast for future periods
G	Gross value certified in period	The value the certifier has agreed in the period, before deductions. Defaults to the formula in §3.2; override with the certified figure once known
K	Retention released in period	Retention returned in the period, per the contractual release events
O	Other deductions in period	Contra-charges, back-charges, liquidated damages, agreed set-offs. One line each in a supporting note
S	Other receipts in period	Advance payment, milestone lump sums outside the valuation cycle, insurance recoveries
U	Cash out in period	Payments made in the period: supply chain, labour, plant, staff, overheads, finance charges

3.2 Calculated columns

Formulas are written for row 3 and fill down. Where a formula uses a whole-column reference such as `$A:$A`, that is deliberate: `MATCH` and `COUNTIF` return positions relative to the start of the range, and using whole columns keeps the position and the sheet row identical.

Col	Field	Formula in words	Spreadsheet expression
D	Cumulative planned value	Running total of planned value from period 0 to this period	`=SUM($C$2:C3)`
F	Cumulative work executed	Running total of work executed	`=SUM($E$2:E3)`
G	Gross value certified (default)	The work executed in the period `Cert_Lag` periods earlier; zero if that period does not exist	`=IFERROR(INDEX($E:$E,MATCH($A3-Cert_Lag,$A:$A,0)),0)`
H	Cumulative gross certified	Running total of gross certified value	`=SUM($G$2:G3)`
I	Certified as a proportion of executed	Cumulative certified divided by cumulative executed; blank while nothing has been executed	`=IF(F3=0,"",H3/F3)`
J	Retention withheld in period	The retention rate applied to the gross certified value, but never more than the headroom left under the cap	`=MIN(Retention_Rate*G3,MAX(0,Retention_Cap-L2))`
L	Retention balance held	Last period's balance, plus what was withheld, less what was released	`=L2+J3-K3`
M	Advance recovery in period	The recovery rate applied to the gross certified value, but never more than the advance still outstanding	`=MIN(Recovery_Rate*G3,N2)`
N	Advance balance outstanding	Last period's outstanding advance less this period's recovery	`=N2-M3`
P	Net certified payable in period	Gross certified, less retention withheld, plus retention released, less advance recovery, less other deductions	`=G3-J3+K3-M3-O3`
Q	Payment due date	The period-end date plus the contractual payment period	`=B3+Payment_Terms_Days`
R	Receipt period	The number of the first period whose end date falls on or after the due date	`=IFERROR(INDEX($A$2:$A$200,COUNTIF($B$2:$B$200,"<"&$Q3)+1),"")`
T		Every net payable amount whose receipt	`=SUMIF($R$2:$R$200,$A3,$P$2:$P$200)+S3`

Col	Field	Formula in words	Spreadsheet expression
	Cash in — receipts in period	period is this period, plus other receipts	
V	Net cash in period	Cash in less cash out	`=T3-U3`
W	Cumulative net cash	Last period's cumulative position plus this period's net movement	`=W2+V3`

Two summary cells sit outside the table:

Cell	Field	Formula in words	Spreadsheet expression
—	Peak funding requirement	The largest negative cumulative cash position over the model, expressed as a positive number; zero if the position never goes negative	`=MAX(0,-MIN($W$2:$W$200))`
—	Period of peak funding	The period number at which that position occurs	`=IFERROR(INDEX($A$2:$A$200,MATCH(MIN($W$2:$W$200),$W$2:$W$200,0)),"")`

Column R is the mechanism that makes this template worth using. It does not assume a lag in whole periods — it computes the contractual due date and then asks which period that date lands in. Thirty-day terms on a month-end certificate do not mean the money arrives next month, and the worked fragment in §4 shows why.

3.3 Pasting it into a spreadsheet

Copy the header line below into cell A1, then use the text-to-columns tool with the pipe character as the delimiter. Freeze row 1, format columns B and Q as dates, and apply the formulas from §3.2 to row 3 before filling down.

```
Period|Period ending|Planned value in period|Cumulative planned value|Work executed in period|Cumulative work executed|Gross value certified in period|Cumulative gross certified|Certified / executed (cum)|Retention withheld|Retention released|Retention balance held|Advance recovery|Advance balance outstanding|Other deductions|Net certified payable|Payment due date|Receipt period|Other receipts|Cash in|Cash out|Net cash in period|Cumulative net cash
```

For use in Markdown, split the same columns into two blocks that share the period key — a valuation block (A to P) and a cash block (A, B and Q to W). That is how the worked fragment below is laid out, and it reads far better on a page than twenty-three columns in one table.

4. Worked fragment

Illustrative figures. Currency-neutral units, rounded to whole units. Monthly periods with calendar month-end cut-offs, period 0 ending 31 January 2027. Contract value 10,000,000. Retention 5 % of gross certified value, capped at 250,000. Advance payment of 500,000 received in period 0, recovered at 20 % of gross certified value. Certification lag zero periods. Payment terms 30 calendar days from the period-end certification date. No other deductions and no retention released in the periods shown.

Valuation block

Period	Ending	Planned value	Cum planned	Executed	Cum executed	Gross certified	Cum certified	Cert / exec	Retention withheld	Retent held
0	31 Jan 27	0	0	0	0	0	0	—	0	0
1	28 Feb 27	450,000	450,000	400,000	400,000	400,000	400,000	100.0 %	20,000	20,000
2	31 Mar 27	750,000	1,200,000	700,000	1,100,000	640,000	1,040,000	94.5 %	32,000	52,000
3	30 Apr 27	850,000	2,050,000	900,000	2,000,000	960,000	2,000,000	100.0 %	48,000	100,00
4	31 May 27	1,050,000	3,100,000	1,100,000	3,100,000	1,100,000	3,100,000	100.0 %	55,000	155,00

Cash block

Period	Ending	Due date	Receipt period	Other receipts	Cash in	Cash out	Net cash	Cumulative net cash
0	31 Jan 27	2 Mar 27	2	500,000	500,000	120,000	380,000	380,000
1	28 Feb 27	30 Mar 27	2	0	0	340,000	−340,000	40,000
2	31 Mar 27	30 Apr 27	3	0	300,000	600,000	−300,000	−260,000
3	30 Apr 27	30 May 27	4	0	480,000	780,000	−300,000	−560,000
4	31 May 27	30 Jun 27	5	0	720,000	940,000	−220,000	−780,000

The substitutions.

Period 2 retention withheld: `MIN(0.05 × 640,000, MAX(0, 250,000 − 20,000)) = MIN(32,000, 230,000) = 32,000`. The cap has not yet bitten, so the rate governs.

Period 4 advance recovery: `MIN(0.20 × 1,100,000, 100,000) = MIN(220,000, 100,000) = 100,000`. Here the cap does bite — only 100,000 of the advance remains outstanding, so recovery stops there. Total recovered across the four periods is `80,000 + 128,000 + 192,000 + 100,000 = 500,000`, exactly the advance.

Period 4 net payable: `1,100,000 − 55,000 + 0 − 100,000 − 0 = 945,000`.

Period 2 receipt period: the certificate is dated 28 February 2027 and the due date is `28 Feb + 30 days = 30 March 2027`. The first period-end on or after 30 March is 31 March, which is period 2 — so the money from period 1's work arrives in period 2. Period 2's own certificate, dated 31 March, is due `31 Mar + 30 days = 30 April`, and 30 April is period 3's end date, so it lands in period 3.

Cumulative net cash at period 4: `380,000 − 340,000 − 300,000 − 300,000 − 220,000 = −780,000`. The peak funding requirement over the fragment is therefore `MAX(0, −(−780,000)) = 780,000`, occurring in period 4.

Read the two blocks together. By period 4 the cumulative value of work executed, 3,100,000, is exactly equal to the cumulative planned value. On every progress measure this project is on plan. It is also 780,000 out of pocket, and 945,000 of certified value is still unpaid — it does not arrive until period 5. Nothing has gone wrong. This is what 30-day payment terms, 5 % retention and a 20 % advance recovery do to a project that is performing exactly as intended, and it is the number that should have been financed at tender.

A cross-check worth running every month: gross certified less retention held less advance recovered should equal cumulative net payable. Here, `3,100,000 − 155,000 − 500,000 = 2,445,000`, and the net payable column sums to `300,000 + 480,000 + 720,000 + 945,000 = 2,445,000`.

5. Common mistakes

Forecasting cash from the cost curve. The most common error is to take the cost forecast, shift it by a month and call it cash. That treats certification as automatic, retention as invisible and the advance as free money. Each of the three is a separate mechanism with its own timing, and all three are in this sheet because all three move the answer.

Assuming the payment lag in whole periods. "Thirty-day terms means next month" is wrong more often than it is right, as §4 shows. Compute the due date, then find the period it falls in.

Modelling the advance as income. An advance payment is a loan against future work, recovered from your own valuations. It flatters period 0 and it takes the money back exactly when the spend rate is highest. Column N exists so that the outstanding balance is always visible.

Letting the retention cap and the retention rate drift apart. The rate applies to each valuation; the cap applies to the balance held. If you only model the rate you will over-withhold at the top of the job; if you only model the cap you will under-withhold at the bottom. The `MIN`/`MAX` construction in column J handles both, and it is worth testing by entering a very large certified value in a scratch row and checking that the retention balance stops at the cap.

Certified value silently tracking executed value. Leaving column G on its default formula for periods that have already been certified turns a control into a mirror. Column I goes to 100 % and stays there, and the disallowed valuation nobody has resolved never appears. Override G with the certified figure every period, and treat any month where the gap widens as a commercial escalation, not a data-entry issue.

Ignoring the payment behaviour you actually observe. The contract says one thing; the record of when money has historically arrived may say another. Model the contractual position in this sheet, and if actual receipts run consistently later, add a second scenario using the observed pattern and report the difference. Quietly building the slippage into the contractual model destroys your ability to make the case for it.

Forgetting the tail. In the fragment, 945,000 arrives in period 5 — after the last period shown. A model that stops at practical completion misses the final account, the retention release and the defects period, which between them can be the difference between a profitable job and a funded one.

6. Adapting it

Safe to change. The period basis, the currency, the number of periods, the column headings, and the addition of columns for anything your contract deducts or adds. If you invoice against milestones rather than measured progress, replace column E with milestone values and set `Cert_Lag` to reflect the certification cycle. If you hold retention from your own supply chain, add a mirror block for retention you withhold — it is cash you hold, and it belongs in column U's netting.

Safe to add. A financing block, converting the peak funding requirement into an interest cost at your facility's rate. A currency-exposure block if you have receipts and payments in different currencies. A sensitivity block that re-runs the model with certification at 90 % of executed value and payment one period later, which is a more honest downside than a single-point forecast.

Do not change. The separation of the three quantities — executed, certified, received. Merging any two of them removes the only thing this sheet does that a cost report does not. And do not remove the period 0 row: every lookup in the template depends on the period sequence being complete from the opening position.

6.1 Before you issue it

- Period-end dates are complete, ascending and one per row, with no gaps in the period numbers.
- Every parameter on the `Parameters` sheet has been read out of the contract, not assumed.
- Column G has been overridden with actual certified values for every period already certified.
- Column I has been read, and any period below 100 % has a named owner and a resolution date.
- The retention balance in column L stops at `Retention_Cap` when tested with a large valuation.
- The advance balance in column N reaches zero and does not go negative.
- The cross-check reconciles: cumulative gross certified, less retention held, less advance recovered, equals cumulative net payable.
- The peak funding requirement and the period it occurs in have been stated to whoever funds it.
- The tail beyond the last period shown — final account, retention release, defects period — is either in the model or explicitly noted as excluded.
- The currency, base date, period basis and rounding are written on the face of the sheet.

Related

- `BPG-13 — Cash flow forecasting` — the method behind this instrument, including curve shapes and how to build the first forecast when there is no history
- `BPG-07 — Accruals and cut-off discipline` — why the cut-off date governs both the cost report and this sheet, and what goes wrong when they differ
- `BPG-06 — Progress measurement and rules of credit` — where the value of work executed in column E legitimately comes from
- `TPL-12 — Change order log` — the source of the changes that must reach column C before this forecast is complete
- `TPL-06 — Monthly project controls report` — where the peak funding requirement and the cash position are reported

Sources and standards

This is an original instrument developed by the Institute. It reproduces no third-party template, form or worked example. The retention, advance-recovery and payment-term mechanics it models are general commercial practice described here in the Institute's own words; the mechanics that bind any particular project are those in its contract, under its governing law. No accounting, tax or statutory payment treatment is presented as universal, and none should be inferred.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-10 · FREE TEMPLATES

Risk register

A register that separates cause, event and effect — and connects the score to the money

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A risk register built so that each entry can be acted on rather than admired. Cause, event and effect are three separate fields, not one sentence. Probability and impact are scored against a scale the project agrees and writes down, cost and schedule impact are assessed separately, and every risk carries a named owner, dated actions, a post-response assessment and an explicit link to contingency. The scoring scale is supplied as a convention to be agreed, not as a universal truth.

About this document

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Risk register

In one paragraph. A risk register built so that each entry can be acted on rather than admired. Cause, event and effect are three separate fields, not one sentence. Probability and impact are scored against a scale the project agrees and writes down, cost and schedule impact are assessed separately, and every risk carries a named owner, dated actions, a post-response assessment and an explicit link to contingency. The scoring scale is supplied as a convention to be agreed, not as a universal truth.

Who this is for. Risk managers, project controls managers, cost engineers and planners who maintain the register; the risk owners named in it; and the project managers and sponsors who have to decide what to fund.

1. When to use this

Open it at the point the project scope is stable enough to be described and keep it open until close-out. Three uses justify the effort, and a register that serves none of them is administration.

To decide where to spend money and attention. A register earns its place when it changes what somebody does this week. That is why response actions, owners and dates are in the same row as the score.

To feed the quantification. The cost risk model and the quantitative schedule risk analysis both draw their discrete events from here. If the register is loose, the model is precise nonsense. The mapping to the schedule analysis is `TPL-11 — Quantitative schedule risk analysis input sheet`.

To justify contingency. Contingency held without a stated basis is a number that will be taken away by the first person who asks where it came from. Columns AL to AN make the link explicit, risk by risk.

It is not an issues log. An issue has happened; a risk has not. Mixing them is the fastest way to make a register unreadable, because the two need different fields and different meetings. Keep a separate issues log and cross-reference it in column AR when a risk is realised.

2. How to complete it

2.1 Write the risk statement in three fields, not one

The cause, the event and the effect are separate columns on purpose. Written as one sentence they collapse, and what collapses is always the part that would have told you what to do.

- **Cause** (column H) is a condition that exists now, or will exist. It is a fact, not a possibility. The ground investigation covered only the northern half of the site.
- **Event** (column I) is the uncertain occurrence. It is the only part of the row that carries a probability. Obstructions are found in the southern foundations zone.

- **Effect** (column J) is the consequence on a project objective, described so that it can be measured. Additional excavation and disposal, and delay to the foundations completion milestone.

The discipline pays for itself immediately. "Bad weather" is a cause and cannot be scored. "Delay to handover" is an effect and will appear against a dozen unrelated events. If you cannot fill all three fields, you do not yet have a risk — you have a worry, and the honest thing is to log it as one and go and find out which of the three parts is missing.

Two tests are worth applying to every new entry. If the cause is already certain and the event has already occurred, it is an issue. If two rows share the same event, they are one risk with two effects, and merging them will change the score.

2.2 Agree the scoring scale before the first row

The scale below is a convention, not a universal truth. There is no correct probability–impact scale. Different scales are defensible, they produce different rankings of the same risks, and none of them is more right than another in the abstract. What matters is that the project agrees one scale, writes it down, uses it consistently, and states it on the face of every report that quotes a score. A register scored against two different scales — the usual result of a change of risk manager — is worse than no register, because it looks comparable and is not.

Agree the scale at the same meeting that appoints the risk owners, record it on a `Scales` sheet, and do not change it mid-project without rescoring every open risk.

Probability bands. Lower bound inclusive, so that a value falling exactly on a boundary has one unambiguous home.

Score	Band	Midpoint used for expected value
1	0 % to under 10 %	5 %
2	10 % to under 30 %	20 %
3	30 % to under 50 %	40 %
4	50 % to under 70 %	60 %
5	70 % and above	85 %

Record the probability as a percentage in column N and let the score calculate. Where an analyst has a considered percentage, use it directly for expected value rather than the band midpoint — the midpoints exist for rows where only a band was ever elicited.

Cost impact bands. Illustrative calibration. Expressed as a proportion of the approved project budget so that the scale is currency-neutral. Replace with the thresholds your project agrees.

Score	Band
1	Under 0.1 % of the approved budget
2	0.1 % to under 0.5 %
3	0.5 % to under 2 %
4	2 % to under 5 %
5	5 % and above

Calibrating to the budget makes registers comparable across projects of different sizes. It also means that a score of 5 on a small project is not the same amount of money as a score of 5 on a large one, which matters the moment a portfolio function aggregates registers. If that aggregation is going to happen, agree absolute currency thresholds instead and say so.

Schedule impact bands. Illustrative calibration. Working days of delay to the affected contractual milestone, not to the activity.

Score	Band
1	Under 5 working days
2	5 to under 10
3	10 to under 20
4	20 to under 40
5	40 and above

Severity is the probability score multiplied by the higher of the two impact scores, giving a range of 1 to 25. Taking the higher rather than the sum is a deliberate convention: a risk that is severe on either axis is severe, and adding cost and schedule scores flatters risks that are mildly bad at everything over risks that are catastrophic at one thing. Other conventions are defensible — separate cost and schedule severities, or a weighted combination. Pick one, state it, and never run two in one register.

2.3 Score twice: before the response and after it

The pre-response assessment is what the project faces if nothing further is done. The post-response assessment is what it faces once the actions in columns W to Z have been completed — not once they have been proposed. Until an action is complete, the post-response score is a forecast, and the register should be read that way.

The difference between the two, priced in column AJ and tested against the cost of the response in column AK, is the argument for spending the money. It is a prompt for a conversation and not a decision rule: a response can be worth buying at a ratio well below 1 when the risk threatens something the project cannot absorb, such as a milestone with a monetary consequence or a consent that cannot be re-applied for.

2.4 Set the review cadence when you open the row

Column AQ is a date, not a frequency. A risk with a proximity three months away and a next review in six months is not being managed. Set the next review inside the proximity window, and make the risk owner — not the risk manager — accountable for the review.

3. The template

Header row in row 1; data from row 2. Formulas below are written for row 2 and fill down. The scoring thresholds live on a sheet named `Scales`: column B holds the five probability lower bounds, column C the five cost lower bounds and column D the five schedule lower bounds, each starting at 0 in row 2 and ascending to row 6.

3.1 Input columns

Col	Field	What goes in it
A	Risk ID	Permanent, never reused, e.g. `RSK-014`
B	Date raised	The date the entry was created
C	Raised by	Named individual
D	Status	Open · In treatment · Closed · Realised · Retired
E	Category	From the project's agreed taxonomy — design, ground, consents, supply chain, resource, interface, commercial, weather, security, digital and data
F	WBS or control account reference	Where the effect lands in the cost and schedule structures
G	Threat or opportunity	Threat · Opportunity. Opportunities use the same fields with the sign reversed
H	Cause	The condition that exists. A fact, stated without hedging
I	Event	The uncertain occurrence. The only part carrying a probability
J	Effect	The consequence on an objective, described so it can be measured
K	Objective affected	Cost · Schedule · Scope · Quality · Safety · Environment · Consent · Reputation
L	Risk owner	The named individual who can actually influence the event, not the person who wrote the row
M	Proximity	The date or window in which the event could occur
N	Pre-response probability	Percentage, as a decimal
P	Pre-response cost impact	Currency, the most likely cost consequence if the event occurs
R	Pre-response schedule impact	Working days of delay to the affected milestone if the event occurs
V	Response strategy	Threats: Avoid · Reduce · Transfer · Accept. Opportunities: Exploit · Enhance · Share · Accept
W	Response actions	What will be done, specifically enough to be verifiable
X	Action owner	Named individual, one per action; split the row if there are several
Y	Action due date	A date, not a month
Z	Action status	Not started · In progress · Complete · Abandoned
AA	Cost of response	The money and resource the actions consume
AB	Post-response probability	Percentage, as a decimal, once the actions are complete
AD	Post-response cost impact	Currency
AF	Post-response schedule impact	Working days
AL	In the quantitative model?	Yes · No, with the model reference. Says whether this risk is represented in the cost or schedule simulation
AM	Contingency treatment	Held in contingency · Included in the cost forecast · Not provided · Transferred

Col	Field	What goes in it
AN	Amount provided	Currency amount actually carried, in contingency or in the forecast
AO	Residual accepted by	The named person who has accepted the post-response position, and the date
AP	Date last reviewed	
AQ	Date of next review	Inside the proximity window
AR	Closure or realisation outcome	What actually happened, recorded at closure. This is what makes the register useful to the next project

3.2 Calculated columns

Col	Field	Formula in words	Spreadsheet expression
O	Pre-response probability score	The band the probability falls into, using lower-bound-inclusive thresholds	`=IF($N2="","",IFERROR(MATCH($N2,Scales!$B$2:$B$6,1),""))`
Q	Pre-response cost impact score	The band the cost impact falls into	`=IF($P2="","",IFERROR(MATCH($P2,Scales!$C$2:$C$6,1),""))`
S	Pre-response schedule impact score	The band the schedule impact falls into	`=IF($R2="","",IFERROR(MATCH($R2,Scales!$D$2:$D$6,1),""))`
T	Pre-response severity	Probability score multiplied by the higher of the two impact scores	`=IF(OR($O2="",$Q2="",$S2=""),"",$O2*MAX($Q2,$S2))`
U	Pre-response expected cost	Probability multiplied by cost impact	`=IF(OR($N2="",$P2=""),"",$N2*$P2)`
AC	Post-response probability score	As column O, on the post-response probability	`=IF($AB2="","",IFERROR(MATCH($AB2,Scales!$B$2:$B$6,1),""))`
AE	Post-response cost impact score	As column Q, on the post-response cost impact	`=IF($AD2="","",IFERROR(MATCH($AD2,Scales!$C$2:$C$6,1),""))`
AG	Post-response schedule impact score	As column S, on the post-response schedule impact	`=IF($AF2="","",IFERROR(MATCH($AF2,Scales!$D$2:$D$6,1),""))`
AH	Post-response severity	Post-response probability score multiplied by the higher of the two post-response impact scores	`=IF(OR($AC2="",$AE2="",$AG2=""),"",$AC2*MAX($AE2,$AG2))`
AI	Post-response expected cost	Post-response probability multiplied by post-response cost impact	`=IF(OR($AB2="",$AD2=""),"",$AB2*$AD2)`
AJ	Expected cost avoided	Pre-response expected cost less post-response expected cost	`=IF(OR($U2="",$AI2=""),"",$U2-$AI2)`
AK	Expected value ratio of the response	Expected cost avoided divided by the cost of the response; reported as text where no response cost has been recorded	`=IF($AJ2="","",IF(N($AA2)=0,"No response cost recorded",$AJ2/$AA2))`

Two summary cells:

Field	Formula in words	Spreadsheet expression
Sum of pre-response expected cost	Total expected cost across all open threats	`=SUMIFS($U:$U,$D:$D,"Open",$G:$G,"Threat")`
Proportion of severity 12 or above with a complete response	Of the risks scoring 12 or more, the share whose actions are complete; blank if there are none	`=IF(COUNTIF($T:$T,">=12")=0,"",COUNTIFS($T:$T,">=12",$Z:$Z,"Complete")/COUNTIF($T:$T,">=12"))`

The expected cost figures are not contingency. Summing expected cost across a register produces a mean, and a mean is not a number you can hold. It ignores correlation, it ignores that a discrete risk either happens in full or does not happen at all, and it says nothing about the tail you are actually funding against. Use the register to feed a simulation and set contingency from the distribution — see `BPG-10 — Contingency and management reserve`.

3.3 Pasting it into a spreadsheet

Copy the header line into cell A1 and split on the pipe character. Format columns N, AB as percentages and B, M, Y, AP, AQ as dates, then apply the §3.2 formulas to row 2 and fill down.

```
Risk ID|Date raised|Raised by|Status|Category|WBS or control account|Threat or opportunity|Cause|Event|Effect|Objective affected|Risk owner|Proximity|Pre probability|Pre probability score|Pre cost impact|Pre cost score|Pre schedule impact (wd)|Pre schedule score|Pre severity|Pre expected cost|Response strategy|Response actions|Action owner|Action due|Action status|Cost of response|Post probability|Post probability score|Post cost impact|Post cost score|Post schedule impact (wd)|Post schedule score|Post severity|Post expected cost|Expected cost avoided|Response value ratio|In quantitative model?|Contingency treatment|Amount provided|Residual accepted by|Last reviewed|Next review|Closure outcome
```

In Markdown, split it into three blocks that share the risk ID: the statement block (A to M), the assessment block (N to AK) and the treatment block (AL to AR). Three readable tables beat one unreadable one.

4. Worked fragment

Illustrative figures. Currency-neutral units. Schedule impacts in working days against the affected contractual milestone. This project has agreed absolute cost bands with lower bounds of 50,000 · 150,000 · 400,000 · 1,000,000, and the illustrative schedule bands from §2.2.

Statement block

Risk ID	Cause	Event	Effect	Objective	Owner	Proximity
RSK-014	The ground investigation covered only the northern half of the site	Obstructions are found in the southern foundations zone	Additional excavation and disposal, and delay to the foundations completion milestone	Cost, Schedule	Construction manager	Foundations window, periods 4 to 7
RSK-021	The permit authority's published determination period is longer than the period assumed in the baseline	The discharge consent is granted later than the baseline date	Commissioning cannot start, delaying the takeover milestone	Schedule	Consents lead	Periods 9 to 12

Assessment block

Risk ID	Pre prob	Score	Pre cost	Score	Pre sched	Score	Severity	Expected cost	Post prob	Score	Post cost	Score
RSK-014	40 %	3	180,000	3	15	3	9	72,000	30 %	3	90,000	2
RSK-021	50 %	4	60,000	2	30	4	16	30,000	50 %	4	60,000	2

The substitutions.

RSK-014 pre-response severity: probability 40 % falls in the band from 30 % to under 50 %, so the score is 3. Cost impact 180,000 is at or above 150,000 and below 400,000, so the score is 3. Schedule impact 15 working days is at or above 10 and below 20, so the score is 3. Severity is `3 × MAX(3, 3) = 9`. Expected cost is `0.40 × 180,000 = 72,000`.

RSK-014 post-response: the response is to reduce — commission a supplementary ground investigation across the southern zone before foundation works begin, and pre-agree a rate for obstruction removal. Note what each part of that does. The investigation does not change the ground; it changes what is known, which allows the design and the sequence to be adjusted, which is what moves the probability of the effect from 40 % to 30 %. The pre-agreed rate is what moves the cost impact, from 180,000 to 90,000. Post severity is `3 × MAX(2, 2) = 6` and post expected cost is `0.30 × 90,000 = 27,000`. Expected cost avoided is `72,000 − 27,000 = 45,000` against a response cost of 25,000, giving a ratio of `45,000 ÷ 25,000 = 1.8`.

Notice that the probability score did not move even though the probability did: a ten-point reduction sits entirely inside one band. That is a property of every banded scale and it is worth knowing about before somebody asks why the response achieved nothing. Bands are for sorting; the underlying percentage is for arithmetic.

RSK-021 shows why severity takes the higher of the two impact scores rather than the sum. Probability 50 % scores 4. Cost impact 60,000 scores 2. Schedule impact 30 working days scores 4. Severity is `4 × MAX(2, 4) = 16`, second only to nothing else on this fragment — and correctly so, because a delayed takeover milestone is the damage, not the extended preliminaries.

It also shows the limits of expected cost as a test. The response — escalate the application, agree a commissioning sequence that does not depend on the consent, and re-plan so the consent-dependent work sits off the critical path — leaves the cost impact untouched at 60,000 and the probability untouched at 50 %, because none of it changes how long the authority takes. Expected cost avoided is `30,000 − 30,000 = 0` and the ratio is 0.0. On an expected-cost test the response is worthless. On the schedule it halves severity from 16 to 8. Any register that scores only cost will systematically underprice schedule-driven risk, and this is what that looks like in a single row.

5. Common mistakes

One sentence instead of three fields. "Risk of delay due to late design" contains a cause, an event and an effect in a form where none of them can be scored, owned or acted on. The three-column discipline is the whole reason this register produces different behaviour from the last one.

Issues logged as risks. If the event has already happened, the probability field is meaningless and the row will sit at severity 25 for the rest of the project, crowding out everything that can still be influenced. Move it to the issues log and record the realisation outcome in column AR.

Effects logged as risks. "Cost overrun" and "delay to handover" are the effects of dozens of events. A register organised by effect cannot tell anyone what to do on Monday.

The register owner as the risk owner. The risk manager owns the process. The risk owner must be the person with the authority to change the outcome. When column L fills up with the same name, the register has become somebody's private document.

Post-response scores recorded as though the actions were done. This is the most damaging error in the template, because it makes the register report a position the project does not hold. Column Z exists to prevent it: read the post-response score only in the light of the action status beside it.

Scores that never move. A register in which last month's scores are this month's scores has not been reviewed; it has been reopened. Column AP dates the review and column AQ commits to the next one, and both should be audited.

Changing the scale mid-project. Rescoring against a new scale without rescoring the closed and dormant rows produces a register in which a 12 from March and a 12 from September mean different things. If the scale must change, rescore everything open and annotate the change.

Summing expected cost and calling it contingency. Covered in §3.2, and worth stating twice because it is the most common quantitative error in the discipline. The sum of expected values is a mean. Contingency is set from a distribution at a confidence level the project has chosen and can defend.

Opportunities absent. A register with no opportunities in it is a statement about the culture, not about the project. The same fields work with the sign reversed, and the strategies are Exploit, Enhance, Share and Accept.

6. Adapting it

Safe to change. The category taxonomy, the band thresholds, the severity convention, the number of scoring levels, and the currency. Add columns for anything your governance requires — a portfolio reference, a contract package, an insurance position, a regulatory flag. Add a second schedule impact against a different milestone if the project has more than one date that carries a consequence.

Safe to simplify. On a small project, columns AL to AO can collapse into a single note field, and the post-response block can be omitted until responses are actually being planned. Nothing is lost as long as the three statement fields survive.

Do not change. The separation of cause, event and effect. The requirement that the scale is agreed, written down and quoted alongside every score. The rule that the risk owner is a named individual. And the distinction between the pre-response and post-response assessment — a register with one assessment cannot show what the response bought, which means it cannot justify the response.

6.1 Before the risk review meeting

- Every open row has all three statement fields populated, and none of them contains two of the others.
- Every open row has a named risk owner who is in the room or has sent a position.
- The scoring scale is printed on the agenda, in the version currently in force.
- Every risk with a proximity inside the next reporting period has been reviewed since the last meeting.
- Every action past its due date has either a completion date or an explanation, and no action has been silently re-dated.
- The top ten by pre-response severity and the top ten by post-response severity have both been produced — they are usually different lists, and the difference is the agenda.
- Contingency treatment is stated for every risk scoring 12 or above, and the amounts reconcile to the contingency actually held.
- Any risk realised since the last meeting has been closed with an outcome recorded in column AR and, where it changed the scope or the cost, a corresponding entry in the change order log.

Related

- `BPG-16 — Risk registers that work` — the method behind this instrument, including how to run an identification workshop that produces events rather than worries
- `BPG-10 — Contingency and management reserve` — how a register becomes a defensible contingency figure, and why the sum of expected values is not it
- `BPG-17 — Quantitative schedule risk analysis` — what happens to the schedule-impact columns once they reach a simulation
- `TPL-11 — Quantitative schedule risk analysis input sheet` — the sheet the discrete events in this register feed into, and the double-counting check between the two
- `TPL-15 — Project controls health check` — the risk dimension of the health check assesses whether this register is being used or merely maintained

Sources and standards

This is an original instrument developed by the Institute. It reproduces no third-party template, form, matrix or scoring scale. The three-part risk statement, the probability–impact convention and the four response strategies are general practice in the discipline, described here in the Institute's own words. The principle that risk is the effect of uncertainty on objectives is the organising idea of ISO 31000, which is named here as a framework and not quoted; no edition of it was consulted for this template and nothing in it is reproduced. The scales in §2.2 are conventions offered for adaptation, not standards, and are labelled as such wherever they appear.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-11 · FREE TEMPLATES

Quantitative schedule risk analysis input sheet

The inputs a simulation runs on, elicited so that the answer means something

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

The three sheets a quantitative schedule risk analysis actually needs: duration uncertainty by activity with three-point estimates and a stated distribution, discrete risk events mapped from the risk register, and a run configuration that makes the result reproducible. Every input records who gave it, when, by what method and what would have to be true for the extremes — because an unattributed number cannot be challenged, and an unchallenged number is where anchoring hides.

About this document

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Quantitative schedule risk analysis input sheet

In one paragraph. The three sheets a quantitative schedule risk analysis actually needs: duration uncertainty by activity with three-point estimates and a stated distribution, discrete risk events mapped from the risk register, and a run configuration that makes the result reproducible. Every input records who gave it, when, by what method and what would have to be true for the extremes — because an unattributed number cannot be challenged, and an unchallenged number is where anchoring hides.

Who this is for. Planners, schedulers, risk analysts and project controls managers who run or commission a quantitative schedule risk analysis; and the project managers who have to act on a P80 date they did not produce.

1. When to use this

A quantitative schedule risk analysis — from here, QSRA — samples uncertain activity durations and discrete risk events many times over the schedule network, and reports the distribution of completion dates that results. This sheet is where its inputs are captured, argued over and recorded.

Use it at four points. **At sanction**, to state the confidence attached to the date being committed to. **At baseline**, to size the schedule contingency and to identify which activities deserve the effort of tighter control. **At a major re-plan**, because a re-plan changes the driving path and therefore changes which uncertainties matter. **Before a completion date is contractually committed**, which is the one occasion where the difference between the deterministic date and the P80 date has a price attached.

Do not run one on a schedule that has not passed a quality review. A simulation propagates whatever logic it is given: with open ends, the uncertainty on an activity with no successor cannot reach the completion milestone, and the model will report a reassuringly tight distribution because most of the network is not connected to the answer. Run `TPL-14 — Schedule quality review checklist` first and record the outcome in the run configuration.

A QSRA is not a forecast of what will happen. It is a statement about the schedule as modelled, under the uncertainty as elicited. Both halves of that sentence belong in the report.

2. How to complete it

2.1 Decide what is in scope before you elicit anything

Model the driving paths and everything that could become one. A common and defensible approach is to model every activity with any float below an agreed threshold, plus every long-duration activity regardless of float, plus every activity a risk event attaches to. State the threshold and the reason for it in the run configuration. Modelling every activity in a ten-thousand-line schedule is usually a way of spending three weeks to produce a less trustworthy answer than modelling four hundred of them well.

For activities in progress at the data date, model the **remaining** duration, not the original. Column H does this automatically from the status in column E.

2.2 Elicit the three-point estimate in a way that does not manufacture the answer

Anchoring is the dominant failure mode of duration elicitation, and it is not a character flaw in the estimator. Show someone a number and their subsequent estimates cluster around it. The deterministic duration in the schedule is a number, and it is usually on the screen. Six practices counter it, and they cost nothing but discipline.

Hide the deterministic duration during the elicitation. If the estimator can see 40 days, the answers will orbit 40 days. Ask for the three-point estimate cold, and only then reveal the schedule duration and discuss the difference. Column X flags every row where the most likely equals the schedule duration exactly, which is the signature of an anchored input.

Ask for the pessimistic value first, then the optimistic, then the most likely. Asking for the most likely first plants the anchor yourself. Starting at the extremes produces wider and better-calibrated ranges.

Define the extremes as frequencies, not adjectives. "Worst case" has no upper bound, so people either give an unusable number or an embarrassed one. Ask instead: if this activity were run ten times under the conditions we expect, what duration would you beat in nine of them? That is a question an experienced person can answer. Do the same at the optimistic end.

Require a written answer to "what would have to be true". Columns M and N are mandatory. An extreme with no story behind it is a number, not an estimate, and it cannot be reviewed by anyone else. This single requirement removes more bad inputs than any amount of statistical checking.

Elicit individually before the group converges. Ask each estimator separately and record the spread of answers before any discussion. Where individual answers differ by a factor of two, that disagreement is the most useful information in the whole exercise, and a workshop will destroy it in four minutes.

Name the person, not "the team". Column O takes a name and a role. Attribution is not about blame; it is about being able to go back and ask a specific person a specific question when the model produces something surprising.

Two further traps. If a discrete risk from the register is being modelled as an event on sheet 2, it must not also be baked into the pessimistic duration on sheet 1 — column L on sheet 2 flags candidates for that double count. And if the estimator's most likely duration is materially longer than the schedule duration, column AA will show it: that is not a risk finding, it is a schedule finding, and it should be dealt with as one before the simulation runs.

2.3 Choose the distribution deliberately

Three shapes cover almost everything, and the choice affects the answer.

Triangular takes the three points literally and gives real weight to the extremes. It is the honest default when the estimator has genuine views about how bad things can get, and it produces a wider spread than the alternatives.

BetaPERT weights the most likely value four times as heavily as the extremes, producing a smoother, tighter distribution. It suits activities where the most likely value is well evidenced and the extremes are softer judgements.

Uniform says every duration between the optimistic and pessimistic values is equally likely. That is a strong statement of ignorance and is occasionally the correct one — a permit determination with a statutory window and no other information, for instance. It should be rare, and it should be justified in column AE.

Record the choice per activity in column L. A model in which every activity carries the same distribution because that was the software default is not a modelling decision; it is an absence of one.

2.4 Set correlation, and record why

Activities that share a crew, a supplier, a design package, a weather window or a permitting authority do not vary independently. If the crew is slow on one, it is slow on the others. Modelling them as independent lets the simulation cancel the variation out, which narrows the distribution and produces a P80 that sits implausibly close to the P50 — the classic symptom of an uncorrelated model.

Assign a correlation group in column AB and a coefficient in column AC. Assigning correlation is judgement, not measurement, and the honest treatment is to record the reason for each group in column AE, run the model at the assigned coefficients, then re-run at a materially different set and report how much the answer moved. If it moves a lot, correlation is the dominant assumption in your analysis and the report should say so.

3. The template

Three sheets. Headers in row 1, data from row 2, formulas written for row 2 and filled down.

3.1 Sheet 1 — `Durations`: input columns

Col	Field	What goes in it
A	Activity ID	Exactly as it appears in the schedule, so the mapping can be verified
B	Activity name	As in the schedule
C	WBS or discipline	For grouping the results
D	Calendar ID	The activity's calendar in the schedule. Durations are in that calendar's working days
E	Status	Not started · In progress · Complete
F	Deterministic duration	The original duration in the current schedule, in working days
G	Remaining duration at the data date	For in-progress activities only
I	Optimistic duration (O)	The duration that would be beaten only about one time in ten
J	Most likely duration (M)	The single most probable duration
K	Pessimistic duration (P)	The duration that would be beaten about nine times in ten
L	Distribution	Triangular · PERT · Uniform
M	Basis of the optimistic value	What would have to be true. Mandatory
N	Basis of the pessimistic value	What would have to be true. Mandatory
O	Elicitation source	Name and role of the person who gave the estimate
P	Elicitation date	
Q	Elicitation method	Individual interview · Facilitated workshop · Reference class · Analyst assumption
R	Input confidence	High · Medium · Low. Drives the review order, not the model
AB	Correlation group	A short group code, e.g. `CIVILS-CREW-A`
AC	Correlation coefficient	The coefficient assigned within the group, as a decimal
AD	Risk drivers mapped	Risk IDs from the register that drive this activity's uncertainty
AE	Notes and assumptions	Including the reason for the correlation group and any unusual distribution choice

3.2 Sheet 1 — `Durations`: calculated columns

Col	Field	Formula in words	Spreadsheet expression
H	Duration modelled	The remaining duration if the activity is in progress, otherwise the deterministic duration	`=IF($E2="In progress",$G2,$F2)`
S	Order check	Confirms the three points are in ascending order; blank until all three are entered	`=IF(COUNT($I2:$K2)<3,"",IF(AND($I2<=$J2,$J2<=$K2),"OK","Check order"))`
T	Spread ratio	Pessimistic divided by most likely; blank if the most likely is zero or empty	`=IF(N($J2)=0,"",$K2/$J2)`
U	Downside days	Pessimistic less most likely	`=IF(COUNT($J2:$K2)<2,"",$K2-$J2)`
V	Upside days	Most likely less optimistic	`=IF(COUNT($I2:$J2)<2,"",$J2-$I2)`
W	Skew ratio	Downside days divided by upside days; reported as text where no upside was modelled	`=IF(OR($U2="",$V2=""),"",IF($V2=0,"No upside modelled",$U2/$V2))`
X	Anchoring flag	Flags rows where the most likely duration equals the duration in the schedule exactly	`=IF(OR($H2="",$J2=""),"",IF($J2=$H2,"Most likely equals schedule – challenge",""))`
Y	Distribution mean	Triangular: the three points averaged. PERT: the three points averaged with the most likely weighted four times. Uniform: the midpoint of the two extremes	`=IF($L2="Triangular",($I2+$J2+$K2)/3,IF($L2="PERT",($I2+4*$J2+$K2)/6,IF($L2="Uniform",($I2+$K2)/2,"")))`
Z	Distribution standard deviation	Triangular: the closed-form standard deviation of a triangular distribution. PERT: the classical approximation, the range divided by six. Uniform: the range divided by the square root of twelve	`=IF($L2="Triangular",SQRT(($I2^2+$J2^2+$K2^2-$I2*$J2-$I2*$K2-$J2*$K2)/18),IF($L2="PERT",($K2-$I2)/6,IF($L2="Uniform",($K2-$I2)/SQRT(12),"")))`
AA	Mean less modelled duration	The distribution mean less the duration currently in the schedule	`=IF(OR($Y2="",$H2=""),"",$Y2-$H2)`

The mean and standard deviation columns are **sense checks on the inputs, not the model**. The simulation samples the distribution itself; it does not use these figures. The PERT standard deviation in column Z is the classical approximation and differs slightly from the exact beta-PERT value — it is here to let a reviewer spot an input whose spread is implausible for the work described, which it does perfectly well.

Column AA is the one to read first at a review. If the sum of column AA over the driving path is large and positive, the schedule's durations are systematically shorter than the people doing the work believe. That is a planning problem masquerading as a risk problem, and no amount of simulation will fix it.

3.3 Sheet 2 — `Risk events`

Discrete events from the risk register that are modelled as occurrences rather than as duration spread.

Col	Field	Input or calculated	What goes in it
A	Risk ID	Input	Matching the ID in `TPL-10 — Risk register`, so the two can be reconciled
B	Risk title	Input	The event, in the register's words
C	Modelled as	Input	Discrete event · New logic branch · Both
D	Probability of occurrence	Input	As a decimal. Where it differs from the register, state why in column N
E	Optimistic added duration	Input	Working days
F	Most likely added duration	Input	Working days
G	Pessimistic added duration	Input	Working days
H	Distribution	Input	Triangular · PERT · Uniform
I	Attaches to	Input	The activity ID or logic path the impact applies to
J	New logic created	Input	Describe any rework loop, re-tender or re-mobilisation the event introduces
K	State modelled	Input	Pre-response · Post-response. Which of the register's two assessments this row represents
L	Double-count check	Calculated	Flags where the same risk ID also appears against a duration input
M	Expected added duration	Calculated	Sense check only
N	Notes	Input	Including the reason for any divergence from the register

Col	Formula in words	Spreadsheet expression
L	Flags the row if this risk ID also appears in the risk-driver column of the durations sheet	`=IF(COUNTIF(Durations!$AD:$AD,"*"&$A2&"*")>0,"Also mapped to a duration input – check for double counting","")`
M	The probability multiplied by the mean of the impact distribution	`=IF(OR($D2="",$H2=""),"",$D2*IF($H2="Triangular",($E2+$F2+$G2)/3,IF($H2="PERT",($E2+4*$F2+$G2)/6,IF($H2="Uniform",($E2+$G2)/2,""))))`

Column L is a prompt, not a proof — it catches the case where the same identifier appears in both places, not the case where the same uncertainty has been described twice in different words. That one is found by reading, at the review.

Column M is a sense check with a caveat that should be printed next to it: a discrete event does not add its expected duration to anything. It either happens, in which case it adds its sampled impact, or it does not, in which case it adds nothing. The expected value is useful for ranking the events and for nothing else.

3.4 Sheet 3 — `Run configuration and outputs`

Reproducibility is the point of this sheet. A QSRA that cannot be re-run to the same numbers cannot be defended.

Field	What goes in it
Schedule file name and version	The exact file analysed
Data date	
Schedule quality review reference and date	The `TPL-14` review this schedule passed, and any check it failed
Activities in schedule / activities modelled	Both numbers, with the in-scope rule stated
Float threshold used for scope selection	And the reason for it
Sampling method	Latin hypercube · Simple random
Iterations	With the stability check below
Random seed	Recorded so the run can be repeated exactly
Correlation matrix reference	Where the coefficients and their reasons are held
Calendars used	And any calendar overridden for the analysis, with the reason
Exclusions	What was left out of the model and why. The most-read line on this sheet
Analyst	Name and date
Reviewer	Name and date

Outputs record, one row per milestone:

Field	What goes in it
Milestone ID and name	
Deterministic date	The date in the schedule as analysed
P50 date · P80 date · P90 date	
Working days from deterministic to P80	The number the project is actually being asked to fund or absorb
Confidence at the committed date	The percentile at which the currently committed date sits
Top five drivers	By sensitivity or correlation to the milestone, named
The sentence this supports	One sentence stating what the analysis does and does not show

Stability check. Run the model twice with different random seeds and record both P80 dates. The absolute difference between them, in working days, must be smaller than the precision you intend to report — if you are going to report a date to the day, a two-day swing between runs means the iteration count is too low. Increase iterations and re-run. State both runs and the difference on this sheet rather than reporting a single number and hoping.

3.5 Pasting it into a spreadsheet

Copy each header line into cell A1 of its own sheet and split on the pipe character.

```
Activity ID|Activity name|WBS or discipline|Calendar ID|Status|Deterministic duration|Remaining duration|Duration modelled|Optimistic|Most likely|Pessimistic|Distribution|Basis of optimistic|Basis of pessimistic|Elicitation source|Elicitation date|Elicitation method|Input confidence|Order check|Spread ratio|Downside days|Upside days|Skew ratio|Anchoring flag|Distribution mean|Distribution SD|Mean less modelled|Correlation group|Correlation coefficient|Risk drivers mapped|Notes and assumptions
```

```
Risk ID|Risk title|Modelled as|Probability|Optimistic added|Most likely added|Pessimistic added|Distribution|Attaches to|New logic created|State modelled|Double-count check|Expected added duration|Notes
```

4. Worked fragment

Illustrative figures. Durations in working days on a five-day calendar. Three activities from a `Durations` sheet, showing the checks doing their work.

Activity	Modelled	O	M	P	Distribution	Spread ratio	Down	Up	Skew	Mean	SD	Mean less modelled	Anc flag
A-1420 Install and terminate switchgear	20	18	22	34	PERT	1.55	12	4	3.00	23.33	2.67	+3.33	—
A-1510 Commission and energise	15	12	16	30	Triangular	1.88	14	4	3.50	19.33	3.86	+4.33	—
A-1180 Civil works to substation base	40	38	40	44	Triangular	1.10	4	2	2.00	40.67	1.25	+0.67	Mos equa sche — chal

The substitutions.

A-1420, PERT mean: `(18 + 4 × 22 + 34) ÷ 6 = (18 + 88 + 34) ÷ 6 = 140 ÷ 6 = 23.33` working days. Classical PERT standard deviation: `(34 − 18) ÷ 6 = 16 ÷ 6 = 2.67` working days. Spread ratio: `34 ÷ 22 = 1.55`. Mean less modelled duration: `23.33 − 20 = +3.33` working days.

A-1510, triangular mean: `(12 + 16 + 30) ÷ 3 = 58 ÷ 3 = 19.33` working days. Triangular standard deviation: `SQRT((12² + 16² + 30² − 12×16 − 12×30 − 16×30) ÷ 18) = SQRT((1,300 − 1,032) ÷ 18) = SQRT(14.889) = 3.86` working days.

A-1180, triangular mean: `(38 + 40 + 44) ÷ 3 = 122 ÷ 3 = 40.67` working days. Standard deviation: `SQRT((1,444 + 1,600 + 1,936 − 1,520 − 1,672 − 1,760) ÷ 18) = SQRT(28 ÷ 18) = SQRT(1.556) = 1.25` working days.

What the checks found. A-1180 is the row to challenge, and two columns say so independently. The most likely duration is exactly the schedule duration, so the anchoring flag fires. The spread ratio is 1.10, meaning the estimator believes a forty-day civils activity will finish within four days of plan nine times out of ten — a claim that would be remarkable on a site with no weather, no ground and no other trades. The standard deviation of 1.25 days on a forty-day activity says the same thing in a different unit. Almost certainly the estimator was shown the schedule and asked to put a range around it. That row should be re-elicited before the model is run, hiding the deterministic duration and asking for the pessimistic value first.

Compare A-1510, which looks like a real estimate: a skew ratio of 3.5 says the estimator sees far more ways for commissioning to run long than to run short, which is what commissioning is like. The basis fields on that row should say what those ways are.

What the fragment does not show. The three activities have a combined mean-less-modelled figure of `3.33 + 4.33 + 0.67 = 8.33` working days. That is **not** eight days of schedule slip. Uncertainty combines through the network rather than adding along a list: some of these activities may not be on the driving path, paths merge, and merge bias pushes the completion distribution later than any single path suggests. Only the simulation over the network can produce the completion distribution, which is exactly why this sheet is an input and not an answer.

5. Common mistakes

Running the model on an unreviewed schedule. Open ends, missing successors and constraints that override logic all suppress the propagation of uncertainty. The result is a tight, confident and meaningless distribution. This is the single most common way a QSRA misleads.

Anchored three-point estimates. Symptoms: most likely values that equal the schedule durations, spread ratios near 1, and empty basis fields. Columns M, N, T and X are all pointed at this failure because it is the one that quietly determines the answer.

Symmetric ranges. Duration risk is almost never symmetric — there are more ways for work to take longer than to finish early, and the ways it can finish early are bounded by physics and the supply chain while the ways it can run long are not. A register of symmetric triangles usually means the analyst applied a percentage rather than eliciting anything.

Percentage uncertainty applied globally. Applying a blanket "minus 10 %, plus 30 %" to every activity is fast, defensible-sounding and worthless: it says nothing the schedule did not already say, and it produces a ranking of drivers that is simply a ranking of durations.

Double counting. A risk modelled as a discrete event on sheet 2 and also written into the pessimistic duration on sheet 1 is counted twice. Column L catches the flagged case; the rest is found by reading the basis fields against the risk descriptions.

Ignoring correlation. Independent sampling of activities that share a crew, a supplier or a weather window produces a distribution that is too narrow. If your P80 is only a handful of days beyond your P50 on a multi-year project, correlation is the first place to look.

Reporting a P-value without the exclusions. "P80 completion is 14 March" is not a finding until it is followed by what was modelled, what was excluded, and which state — pre-response or post-response — the risk events were modelled in. A P80 that excludes the consent risk and a P80 that includes it are different numbers with the same name.

Treating the P80 as a target. It is a confidence statement about a model. Handing it to the delivery team as the new date converts a risk analysis into a schedule extension and guarantees that the next one will be gamed.

6. Adapting it

Safe to change. The distribution options, the in-scope selection rule, the correlation coding, the addition of columns for anything your tool imports. If your software takes distribution parameters rather than three points, add the parameter columns and keep the three points as the human-readable record of what was elicited.

Safe to add. A resource-uncertainty block where labour availability, not duration, is the driver. A weather block that models seasonal calendars as a distribution rather than an activity. A column recording the individual answers before group convergence, which is often the most informative data the exercise produces.

Do not change. The mandatory basis fields for both extremes; the named elicitation source and date; the recording of the seed and the iteration count; and the separation between duration uncertainty and discrete events. Each of those exists because a QSRA is an argument, and every one of them is a place where the argument is either supported or is not.

6.1 Before the model is run

- The schedule has passed a quality review, and any failed check is recorded with its effect on the analysis.
- The in-scope rule is written down, and the count of activities modelled against activities in the schedule is stated.
- Every modelled activity has all three points, a stated distribution, and both basis fields completed.
- The order check reports OK on every row.
- Every anchoring flag has been either cleared by re-elicitation or annotated with why the estimate stands.
- Every spread ratio below an agreed floor has been reviewed and either corrected or justified.
- In-progress activities are modelled on remaining duration, and the data date matches the schedule.
- Every discrete risk event names a register ID, states which response state it models, and has been checked for double counting against the duration inputs.
- Correlation groups have a written reason each, and a sensitivity run at different coefficients is either done or scheduled.
- The seed, the iteration count and the stability check between two seeds are recorded.
- The exclusions line is complete and has been read out loud to whoever will act on the result.

Related

- `BPG-17 — Quantitative schedule risk analysis` — the method: what the simulation does, how merge bias arises, and how to read a tornado chart without over-reading it
- `BPG-05 — Schedule quality — a practical review` — why an unreviewed schedule cannot carry a simulation
- `TPL-14 — Schedule quality review checklist` — the review that must precede the run, and the record the run configuration references
- `TPL-10 — Risk register` — the source of the discrete events on sheet 2, and the register that must reconcile to them
- `BPG-10 — Contingency and management reserve` — how the output distribution becomes a schedule contingency someone will actually approve

Sources and standards

This is an original instrument developed by the Institute. It reproduces no third-party template, form or worked example. The triangular, PERT and uniform distributions and their closed-form means and standard deviations are standard results in probability, computed here rather than quoted, and every figure in §4 was independently recomputed before publication. The classical PERT standard-deviation approximation is identified as an approximation wherever it is used. The elicitation guidance in §2.2 describes practice in the Institute's own words and does not reproduce any published protocol.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-12 · FREE TEMPLATES

Change order log

Every change, its status, its dates — and whether it is in the forecast

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A change order log that tracks each change from first identification to agreement, records the date of every status transition, keeps the submitted, assessed and best-estimate values apart, and carries the one column most logs omit: whether the change is in the cost forecast, and for how much. The difference between the best estimate and the amount in the forecast is unrecognised exposure, and this log adds it up.

About this document

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Change order log

In one paragraph. A change order log that tracks each change from first identification to agreement, records the date of every status transition, keeps the submitted, assessed and best-estimate values apart, and carries the one column most logs omit: whether the change is in the cost forecast, and for how much. The difference between the best estimate and the amount in the forecast is unrecognised exposure, and this log adds it up.

Who this is for. Quantity surveyors, commercial managers, cost engineers and project controls managers who run the change process; and the project managers and finance partners who rely on a forecast that only holds if this log is complete.

1. When to use this

Open it on day one, before the first change arrives, because the fields you cannot reconstruct are the early ones — the date something was first identified, and whether a notice was served in time.

It does three jobs that nothing else does.

It protects entitlement. Most contracts make some relief conditional on notice, and some make it a strict condition. The notice block in columns I to N is the part of this log that has to be right, and it is the part most often filled in retrospectively, which is to say wrongly.

It keeps the forecast honest. A change with a real cost that is not in the forecast is a loss that has already happened and has not yet been reported. Columns AE to AH exist to make that visible before the month closes rather than at the final account.

It is the audit trail. When a change is disputed a year later, the argument is almost always about sequence: what was known, when, and what was said about it. A log with a date against every status transition answers that in one row.

It is not a substitute for the baseline change control process. A change to scope may or may not change the baseline, and the two are separate decisions with separate approvals — `TPL-04 — Baseline change request` is where that lives. This log tracks the commercial event; the baseline change request tracks what happens to the measurement of performance.

2. How to complete it

2.1 Register it the day it appears, not the day it is agreed

The trigger for a row is the moment somebody becomes aware that the work may change — not the instruction, not the quotation, not the agreement. Registering late is how notice periods are missed and how the log ends up describing an outcome instead of a process. If it turns out not to be a change, close the row as Withdrawn with a reason; a log with withdrawn rows in it is a log somebody is actually using.

2.2 Get the notice block right first

Column I asks whether notice is required, column J records the date of the trigger, and column K records the notice period. Two points make the difference between a useful record and a misleading one.

Know what starts the clock in your contract. Some contracts run the period from the occurrence, some from the date the party became aware, and some from the date the party ought reasonably to have become aware. These are different dates and they lead to different answers. Record in column J the date that starts the clock under your contract, and note in column AS which test you applied.

Record the position honestly, including when it is bad. Column N calculates whether the notice was served inside the period. If it was not, the log should say so. A commercial team that discovers a late notice in month four has options; one that discovers it in the final account has an argument it will probably lose. Whether a late notice defeats entitlement, and what relief remains if it does, depends on the contract and the law governing it — this log records the fact, not the consequence.

2.3 Keep three cost figures apart

Column W is what was submitted. Column X is what the other party has assessed. Column Y is the current best estimate — the number the commercial team genuinely expects to settle at, which is usually neither of the other two.

Merging them destroys the log. If you record only the submitted value, the forecast is optimistic and the gap in column Z is invisible. If you record only the assessed value, you have adopted the other side's position in your own forecast. If you record only the best estimate, you cannot show anyone how far apart the parties are, which is the information a negotiation strategy is built from.

The same logic applies to time. Column AA is the days claimed, column AB the days agreed, and column AJ the days actually reflected in the current schedule — three different numbers that are routinely assumed to be one.

2.4 Answer the forecast question every month

Column AE takes Yes, No or Partial. Column AF takes the amount actually included. Column AG records the basis — agreed value, assessed value, best estimate, risk-adjusted, or nil.

This is the column that saves people, and it saves them in a specific way. The failure it prevents is not dishonesty; it is a handover gap. The commercial team knows a change is worth 240,000 and is confident of recovering it. The cost engineer does not include it in the forecast because it is not agreed. Neither is wrong, and both are behaving reasonably. The result is a forecast that is 240,000 light and a commercial position that nobody has priced. The gap lives in the space between two competent people, and a single column closes it.

State the convention the project uses for what goes into the forecast, and apply it to every row. Any convention is defensible — include the assessed value, include the best estimate, include a risk-adjusted value — provided it is written down, applied consistently, and the unrecognised exposure in column AH is reported alongside the forecast rather than instead of it.

2.5 Use a report date, not today

Every date-difference formula below uses a named cell called `Report_Date` rather than `TODAY()`. This matters more than it sounds: a log built on `TODAY()` produces different numbers every time it is opened, so the log attached to March's report no longer agrees with March's report by April. Set `Report_Date` to the reporting cut-off, and the log is reproducible.

3. The template

Header row in row 1, data from row 2. Formulas are written for row 2 and fill down. `Report_Date` and `Original_Contract_Value` are named cells on a `Parameters` sheet.

3.1 Input columns

Col	Field	What goes in it
A	Change ID	Permanent, never reused, e.g. `CHG-018`
B	Date first identified	When somebody first became aware the work might change
C	Origin	Client instruction · Design development · Site condition · Statutory or consent · Supply chain · Error or omission · Opportunity · Scope clarification
D	Originator	Named individual and organisation
E	Description	One sentence saying what changes. Not why, not who is to blame
F	Affected WBS or control account	Where the cost lands
G	Affected activity IDs	Where the time lands, in schedule identifiers
H	Contract mechanism relied on	The mechanism in general terms — variation, compensation event, change order, claim. Record the clause reference from your executed contract, not from memory of a standard form
I	Notice required?	Yes · No · Unknown
J	Notice trigger date	The date that starts the clock under your contract
K	Notice period (days)	From the contract. Note in column AS whether these are calendar or working days
M	Notice served date	
O	Status	Notified · Quoted · Instructed · Agreed · Disputed · Withdrawn · Rejected
P–T	Date notified · quoted · instructed · agreed · disputed	One date per transition, left blank until it happens. Never overwrite an earlier one
W	Cost — submitted	The value submitted
X	Cost — assessed	The value assessed by the other party
Y	Cost — current best estimate	What the commercial team expects to settle at
AA	Schedule — days claimed	
AB	Schedule — days agreed	
AD	Commercial position	Agreed · In negotiation · Gap noted · Reserved · Referred
AE	In the cost forecast?	Yes · No · Partial
AF	Amount included in the forecast	Currency
AG	Basis of inclusion	Agreed value · Assessed value · Best estimate · Risk-adjusted · Nil
AI	In the schedule?	Yes · No · Partial
AJ	Days reflected in the current schedule	
AL	Approval reference	Instruction number, minute reference, signed order reference
AM	Approved by	Named individual with the authority to approve
AN		

Col	Field	What goes in it
	Payment application first included in	
AO	Linked risk ID	Where the change is a realised risk, the ID from `TPL-10`
AP	Owner	The named person progressing this change
AQ	Next action	
AR	Next action date	
AS	Notes	Including which notice test applied, whether the period is calendar or working days, and any reservation

3.2 Calculated columns

Col	Field	Formula in words	Spreadsheet expression
L	Notice due date	The trigger date plus the notice period, where notice is required	`=IF(OR($I2<>"Yes",$J2="",$K2=""),"",$J2+$K2)`
N	Notice position	Not required where no notice applies; otherwise whether it was served, served in time, or the period has expired unserved	`=IF($I2<>"Yes","Not required",IF($M2="",IF(AND($L2<>"",Report_Date>$L2),"Not served – period expired","Not served"),IF($M2<=$L2,"Served within the period","Served late")))`
U	Date of latest status change	The most recent of the five transition dates; blank if none has occurred	`=IF(MAX($P2:$T2)=0,"",MAX($P2:$T2))`
V	Days in current status	Days from the latest transition to the report date	`=IF($U2="","",Report_Date-$U2)`
Z	Cost gap	Submitted less assessed; blank until both exist	`=IF(COUNT($W2:$X2)<2,"",$W2-$X2)`
AC	Schedule gap	Days claimed less days agreed; blank until both exist	`=IF(COUNT($AA2:$AB2)<2,"",$AA2-$AB2)`
AH	Unrecognised cost exposure	Best estimate less the amount included in the forecast, treating a blank inclusion as zero	`=IF($Y2="","",$Y2-N($AF2))`
AK	Unrecognised time exposure	Days claimed less days reflected in the current schedule, treating a blank as zero	`=IF($AA2="","",$AA2-N($AJ2))`

Summary cells, to be placed above the log and read at every monthly review:

Field	Formula in words	Spreadsheet expression
Total current best estimate	Sum of the best-estimate column	`=SUM($Y:$Y)`
Total in the forecast	Sum of the amounts included	`=SUM($AF:$AF)`
Total unrecognised cost exposure	Sum of the exposure column	`=SUM($AH:$AH)`
Share of change value not in the forecast	Unrecognised exposure divided by total best estimate; blank if there is no change value	`=IF(SUM($Y:$Y)=0,"",SUM($AH:$AH)/SUM($Y:$Y))`
Change value as a share of the original contract	Total best estimate divided by the original contract value; blank if that value is unset	`=IF(N(Original_Contract_Value)=0,"",SUM($Y:$Y)/Original_Contract_Value)`
Notices at risk	Count of rows where the notice period has expired without service	`=COUNTIF($N:$N,"Not served – period expired")`
Total unrecognised time exposure	Sum of the time-exposure column	`=SUM($AK:$AK)`

3.3 Pasting it into a spreadsheet

Copy the header line into cell A1 and split on the pipe character. Format the date columns, set `Report_Date` on the `Parameters` sheet, then apply the §3.2 formulas to row 2 and fill down.

```
Change ID|Date first identified|Origin|Originator|Description|Affected WBS|Affected activity IDs|Contract mechanism|Notice required?|Notice trigger date|Notice period (days)|Notice due date|Notice served date|Notice position|Status|Date notified|Date quoted|Date instructed|Date agreed|Date disputed|Latest status change|Days in current status|Cost submitted|Cost assessed|Cost best estimate|Cost gap|Days claimed|Days agreed|Schedule gap|Commercial position|In the cost forecast?|Amount in forecast|Basis of inclusion|Unrecognised cost exposure|In the schedule?|Days reflected in schedule|Unrecognised time exposure|Approval reference|Approved by|Application first included|Linked risk ID|Owner|Next action|Next action date|Notes
```

In Markdown, split it into three blocks sharing the change ID: identification and notice (A to N), status and value (O to AD), and forecast and approval (AE to AS).

4. Worked fragment

Illustrative figures. Currency-neutral units. Report date 30 April 2026. Notice periods in calendar days. Original contract value 24,000,000.

Identification and notice

Change ID	Origin	Description	Notice required?	Trigger date	Period	Due	Served	Notice position
CHG-011	Design development	Revised louvre specification to the plant room façade	No	—	—	—	—	Not required
CHG-018	Client instruction	Additional fire dampers to the level 3 riser	No	—	—	—	—	Not required
CHG-023	Site condition	Buried services encountered on the southern access route	Yes	4 Mar 26	14	18 Mar 26	16 Mar 26	Served within the period

Status and value

Change ID	Status	Latest transition	Days in status	Submitted	Assessed	Best estimate	Cost gap	Days claimed	Days agreed	Schedu gap
CHG-011	Agreed	13 Feb 26	76	62,000	62,000	62,000	0	0	0	0
CHG-018	Instructed	6 Mar 26	55	148,000	96,000	120,000	52,000	5	0	5
CHG-023	Notified	16 Mar 26	45	—	—	240,000	—	12	—	—

Forecast position

Change ID	In the forecast?	Amount in forecast	Basis	Unrecognised cost exposure	In the schedule?	Days in schedule	Unrecognised time exposure
CHG-011	Yes	62,000	Agreed value	0	Yes	0	0
CHG-018	Partial	96,000	Assessed value	24,000	No	0	5
CHG-023	No	0	Nil	240,000	No	0	12

The substitutions.

CHG-023 notice due date: `4 March 2026 + 14 days = 18 March 2026`. Served 16 March, which is on or before 18 March, so the position is "Served within the period".

CHG-018 days in current status: latest transition 6 March 2026, report date 30 April 2026, so `30 April − 6 March = 55 days`. CHG-011: `30 April − 13 February = 76 days`. CHG-023: `30 April − 16 March = 45 days`.

CHG-018 unrecognised cost exposure: `120,000 − 96,000 = 24,000`. CHG-023: `240,000 − 0 = 240,000`. CHG-011: `62,000 − 62,000 = 0`.

Totals: current best estimate `120,000 + 240,000 + 62,000 = 422,000`. Amount in the forecast `96,000 + 0 + 62,000 = 158,000`. Unrecognised cost exposure `24,000 + 240,000 + 0 = 264,000`, and as a check, `158,000 + 264,000 = 422,000`. Share of change value not in the forecast: `264,000 ÷ 422,000 = 0.626`, or 62.6 %. Total unrecognised time exposure: `0 + 5 + 12 = 17` working days.

What the fragment is telling you. Every one of these three changes is being competently managed. Each has an owner, a status and a next action, and anyone reviewing the log would say the change process is working. And the cost forecast is understating this project by 264,000 — nearly two-thirds of the change value on the log — while the schedule carries none of the 17 days claimed. Neither of those two facts appears anywhere in a change log without columns AE to AK. That is the entire argument for the template.

The second reading is about CHG-018: it has been Instructed for 55 days with a 52,000 gap between what was submitted and what was assessed, and no date in the Agreed column. A change that has been instructed but not agreed is work being executed at a price nobody has settled, and 55 days is long enough for the record of what was actually done to start degrading. That row is the one for the commercial meeting.

5. Common mistakes

Registering changes when they are instructed. By then the notice period may have run and the early factual record — what was known and when — is already being reconstructed from memory.

One cost column. The submitted, assessed and best-estimate figures are three different pieces of information and they are needed for three different purposes. A log with one column has thrown two of them away.

Overwriting status dates. Columns P to T are a history, not a status field. Overwriting the notified date with the instructed date destroys the only record of how long each stage took, which is exactly what is argued about later.

Blank forecast columns. The most dangerous state of column AE is empty, because empty reads as "not yet considered" and behaves as "no". Every row should carry Yes, No or Partial by the monthly cut-off, even if the answer is No with a reason.

Adopting the other side's assessment as your forecast. Putting the assessed value into the forecast and leaving column Y unpopulated makes the gap disappear from your own reporting. Keep the best estimate, report the exposure, and let the negotiation be about a number that is visible.

A log that never closes rows. Agreed, Withdrawn and Rejected are terminal statuses with dates. A log where nothing ever leaves is one nobody trusts, and the summary totals become meaningless because settled changes are still counted as exposure.

Time impacts tracked only in days claimed. Column AJ — days actually reflected in the schedule — is the time equivalent of the forecast column, and it is omitted even more often. A schedule that does not carry the agreed extensions will show a completion date the project cannot achieve and cannot explain.

Using `TODAY()`. Covered in §2.5. A saved report should say the same thing next month as it said this month.

6. Adapting it

Safe to change. The origin taxonomy, the status list, the currency, and the addition of columns your contract or governance requires — a package reference, a client reference number, a sub-contract flow-down identifier, a retention or bond consequence. If your contract has more status stages than the five here, add a date column for each rather than reusing one.

Safe to add. A margin block, showing the value, the cost and the margin on each change, which is how a contractor finds out that the changes it fought hardest for were the least profitable. A cumulative chart of best estimate against amount in the forecast over time, which shows whether the gap is being closed or grown. A column for the party bearing the risk, where a change flows down to a subcontractor.

Do not change. The separation of submitted, assessed and best estimate. The dated status transitions. The notice block. And columns AE to AH — if you take one thing from this template into a log you already have, take those.

6.1 Before the monthly cut-off

- Every change identified this period has a row, including the ones that may turn out not to be changes.
- Every row where notice is required has a trigger date, a period, and a served date or an explanation.
- No row shows "Not served — period expired" without an entry in the notes and an escalation.
- Every row has a current best estimate, including rows that have not been quoted.
- Every row carries Yes, No or Partial in the forecast column, with a basis.
- The unrecognised cost exposure total has been given to whoever owns the cost forecast, in writing.
- The unrecognised time exposure total has been given to whoever owns the schedule, in writing.
- Any row instructed more than one reporting period ago and still not agreed is on the commercial agenda.
- Terminal rows have been closed with a date, and the summary totals exclude them or are reported both ways.
- `Report_Date` is set to this period's cut-off before the log is issued.

Related

- `BPG-11 — Change orders and variations` — the method: identification, valuation, negotiation and the behaviours that make a change process work
- `BPG-04 — Baselining and baseline change control` — why a commercial change and a baseline change are separate decisions with separate approvals
- `TPL-04 — Baseline change request` — the instrument that moves the baseline, once a change is agreed
- `TPL-13 — Claim and extension-of-time narrative structure` — where a disputed change goes when the commercial position does not resolve

- `BPG-09 — Estimate at completion — choosing and defending a method` — how the unrecognised exposure in column AH reaches the forecast

Sources and standards

This is an original instrument developed by the Institute. It reproduces no third-party template, form or contract wording. The status sequence, the notice mechanics and the valuation stages described here are general commercial practice expressed in the Institute's own words. No standard form of contract is quoted and no clause number is cited: the mechanism, the notice test and the consequence of late notice are determined by the contract in front of you and by the law governing it, and nothing in this template should be read as advice on either.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-13 · FREE TEMPLATES

Claim and extension-of-time narrative structure

The skeleton of a defensible submission — contractual basis, notice, chronology, causation, method, effect, quantum, records

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

The structure of a claim or extension-of-time submission that can survive being read by someone looking for a reason to reject it: the contractual basis element by element, the notice record including where it is weak, a dated and sourced chronology, the causal chain event by event, the delay analysis method stated and justified, the effect on each contractual date, the quantum with its arithmetic, the records relied on, and the assumptions. This is a structure, not legal advice.

About this document

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Claim and extension-of-time narrative structure

In one paragraph. The structure of a claim or extension-of-time submission that can survive being read by someone looking for a reason to reject it: the contractual basis element by element, the notice record including where it is weak, a dated and sourced chronology, the causal chain event by event, the delay analysis method stated and justified, the effect on each contractual date, the quantum with its arithmetic, the records relied on, and the assumptions. This is a structure, not legal advice.

Who this is for. Commercial managers, quantity surveyors, planners and project controls managers who prepare or assess submissions; and the project directors who have to decide whether one is ready to go.

This is a structure, not legal advice. Whether an event gives rise to relief, what relief it gives, what notice is required and what happens if it is late, whether concurrent delay reduces or extinguishes entitlement, and which heads of loss are recoverable are all determined by the contract in front of you and by the law governing it. Those things differ between contracts and between jurisdictions, and nothing here is a substitute for advice from a qualified adviser in the relevant jurisdiction. Extension of time is abbreviated to EOT throughout.

1. When to use this

Use it in three situations, and the earlier of them is the one that pays.

While the events are happening, as the live skeleton the project fills in as it goes. A submission assembled from a structure that has been populated contemporaneously is a different document from one reconstructed eight months later, and the difference is visible on the first page. If you take one thing from this template, take the habit of opening Part D — the chronology — on the day the first event occurs.

When preparing a submission, as the table of contents and the completeness test. Every part below has a purpose; a part left empty is a question the assessor will ask.

When assessing somebody else's submission. The structure works in reverse. Read it looking for the parts that are missing, and you will find the weak point in about ten minutes: an unstated contractual basis, a chronology braided together with argument, a delay analysis whose method is never named, or a quantum with no path back to a record.

It is not a form to be filled in and sent. A claim is an argument, and the structure exists so the argument is visible, not so it can be replaced by headings.

2. How to complete it

2.1 Establish what the contract requires before writing anything else

Start at Part B, not Part D. The most common structural failure in submissions is a long, well-evidenced factual account that never connects to a mechanism in the contract. Identify the mechanism you rely on, read it in your executed contract — not in your memory of a standard form — and list the elements it requires to be established. That list becomes the specification for the rest of the document.

Do not cite clause numbers from a standard form from memory. Amendments are the norm, and a submission that relies on unamended wording that your contract does not contain damages every other part of it.

2.2 Separate fact from argument, absolutely

Part D is fact: dated, sourced, neutral, and written so that the other party could not reasonably object to any single line of it. Part E is argument: what those facts caused. Keeping them apart is not a stylistic preference. A chronology laced with characterisation invites the assessor to dispute the facts, which is a much better fight for them than disputing the causation. A neutral chronology forces the argument to happen where you want it.

The practical test: if a sentence in Part D contains an adjective about the other party's conduct, it belongs in Part E or nowhere.

2.3 Prove causation event by event, not in aggregate

The weakest common structure is a list of events followed by a global statement of delay. Part E requires, for each event: the activity or work affected, the physical mechanism by which it was affected, the effect on that activity, whether the affected path was already critical or became critical, and the effect on the completion date. That is more work and it is the difference between a submission that is assessed and one that is dismissed.

2.4 State the delay analysis method, and say what it is weak at

Part F names the method, justifies it against the records available and the timing of the analysis, and says plainly what the method is known to be poor at. Volunteering the limitation is not weakness; it is the single most credible thing a submission can do, and it removes the assessor's best line of attack.

The main families, described here in general terms:

As-planned versus as-built. Compare the planned programme with what actually happened and identify the differences. Simple, transparent and readable by a non-specialist. Weak at attributing cause: it shows that things happened differently, not why.

Impacted as-planned. Insert the delay events into the baseline logic and observe the movement of the completion date. Requires only a baseline and the events. Weak because it ignores everything else that happened — actual progress, other delays, changes in sequence — and so tends to overstate.

Time impact analysis. Insert the event into the schedule updated to just before it occurred, and measure the resulting movement. Prospective in character, closely matched to how a contract administrator would have assessed at the time. Requires good contemporaneous schedule updates, and is sensitive to their quality.

Collapsed as-built. Remove the events from the as-built programme and observe what the completion date would have been without them. Requires a reliable as-built with defensible logic, which is often the hardest input to produce. Sensitive to the logic the analyst inserts.

Windows or time-slice analysis. Divide the period into windows and determine what actually drove completion in each. Generally the most robust where good updates exist, because it follows the critical path as it moved. The most expensive, and dependent on the quality of the updates.

As-built critical path or longest-path analysis. Identify the path that actually determined completion and analyse the delays on it. Intuitive, and contentious where the as-built logic must be reconstructed.

The choice is a matter of professional judgement, and competent analysts differ. The grounds for choosing are: what records exist, the quality of the contemporaneous schedule updates, whether the analysis is being done prospectively or after the event, what the contract says about how EOT is to be assessed, and proportionality. Record all five in Part F. An analysis whose method was chosen because it produced the largest number, and whose justification was written afterwards, is usually detectable — the justification will not mention the records.

2.5 Validate the schedule you are relying on, and disclose every adjustment

A delay analysis is only as good as the schedule underneath it. Run a quality review over every programme relied on — baseline, updates, as-built — and record the outcome, including the failures. Use `TPL-14 — Schedule quality review checklist` and reference it in Part F.

Then disclose every adjustment made to the schedule for the purposes of the analysis: logic added, logic removed, constraints released, calendars amended, activities split. Each with a reason. An undisclosed adjustment discovered by the other side does more damage than the adjustment itself ever saved.

2.6 Build the quantum from records, and price the right period

Part H requires, for each head of claim: the principle, the period claimed, the basis of the rate, the records the figure comes from, and the arithmetic. Two disciplines matter most.

Price the period in which the delay occurred, not the period at the end of the job. Time-related costs differ by phase — the resources standing during a foundations delay are not the resources standing during commissioning — and pricing the wrong window produces a number that is wrong in a direction the assessor will find.

Derive rates from actual cost records where you have them. The tender allowance is what you hoped to spend. Where actual records are genuinely unavailable, say so, state what you used instead, and explain why it is a reasonable proxy.

Which heads of loss are recoverable — prolongation, disruption, acceleration, finance charges, head-office overhead contribution, loss of profit — is a matter of the contract and the governing law, and differs between jurisdictions. Set out the principle and the arithmetic; leave the question of recoverability to the parties and their advisers, and say in the submission that you have done so.

3. The template

Eleven parts. Each is a section of the submission, with the control table that goes in it.

Part A — Submission control

Field	What goes in it
Submission reference and revision	
Date of submission	
Contract	The parties, the contract title and the date of the executed contract
Relief sought	Time · Money · Both. State the number for each
Events relied on	A numbered list, each with a short label used consistently throughout
Prepared by	Names and roles of everyone who contributed, and what each contributed
Basis of preparation	Contemporaneous · Retrospective · Mixed, with dates
Statement of limitations	What this submission does not address, and why

Part B — Contractual basis

One row per element the mechanism requires.

Field	What goes in it
Mechanism relied on	Identified from the executed contract, with its reference as it appears there
Element the contract requires	One row per element, in the contract's own terms
Where it is established	The part and paragraph of this submission
Evidence reference	The document reference in Part I
Position	Established · Partly established · Not established, with a note where it is not

Part C — The notice record

One row per event.

Field	What goes in it
Event reference	Matching Part A
What the contract requires as a trigger	Occurrence · Awareness · Constructive awareness, as your contract provides
Date the trigger arose	And how that date is evidenced
Notice period	And whether calendar or working days
Notice due	
Notice served	Date, method of service, addressee, and the reference
Recipient's response	And its date
Notice position	Compliant · Arguably compliant · Late · Not served
Where the position is weak, what is said about it	Stated here, not omitted

Part D — Factual chronology

Facts only. One row per entry, ascending by date.

Field	What goes in it
Date	
Fact	What happened, stated so the other party could not reasonably object to the sentence
Source	Document reference, with its date and author
Event reference	Which of the events in Part A this entry relates to, if any

Part E — The causal chain

One block per event. Repeat the whole block for each.

Field	What goes in it
Event reference and description	
Work or activity affected	Named activities, with schedule identifiers
Mechanism of effect	Access denied · Information late · Instructed change · Resequencing · Resource diverted · Standing time · Rework, and how it operated in fact
Effect on the activity	Start delayed by, duration extended by, or both, in working days
Path status before the event	Critical · Near critical with stated float · Not critical
Path status after the event	And whether the path became critical because of the event
Effect on the completion date	In working days, with the analysis reference in Part F
Concurrency	Whether any other delay ran concurrently, whose it was, and the position taken — noting that the treatment of concurrent delay depends on the contract and the governing law
Mitigation	What was done to mitigate, what it cost, and whether it was instructed

Part F — Delay analysis

Field	What goes in it
Method used	Named, from §2.4 or otherwise
Why this method	Against the records available, the quality of the updates, the timing of the analysis, what the contract requires, and proportionality
What this method is weak at	Stated plainly
Methods considered and not used	And why
Schedules relied on	File names, revisions, dates, data dates
Schedule validation	The quality review reference, its date, and every check it failed
Adjustments made	Every logic change, constraint release, calendar amendment or activity split, each with a reason
Analysis windows	Where windows are used, their boundaries and why they were drawn there
Results	Per window or per event: delay identified, to which path, and to which milestone
Reviewer	Independent of the analyst, named, with date

Part G — The effect

Field	What goes in it
Milestone or sectional date	
Contractual date before this submission	
Extension claimed	In working days, and the calendar used
Revised date sought	
Basis	Reference to the Part F result that produces it
Monetary consequence attached to the date	Whether the date carries damages, a bonus, or a payment trigger, in general terms

Part H — Quantum

Field	What goes in it
Head of claim	Prolongation · Disruption · Acceleration · Additional resource · Finance · Other, named
Principle	What this head compensates, in one sentence
Period claimed	The specific window, with dates
Basis of the rate	Actual cost records · Rate in the contract · Reasoned proxy, with the reason
Source records	Reference into Part I
Arithmetic	The substitution shown, not just the result
Amount	
Recoverability	Noted as a matter for the contract and the governing law, not asserted here

Part I — Records relied on

Field	What goes in it
Reference	The citation used throughout the submission
Document	Title, date, author, recipient
Custodian and location	Where the original is held
What it establishes	The specific proposition, not "background"
Corroborated by	Another reference, or "single source"

The corroboration column is worth the effort. A submission that says plainly which propositions rest on a single document, and which are supported by two or more independent records, is far harder to attack than one that presents everything at the same weight. It also tells your own team where to go looking before the submission is issued.

Part J — Assumptions and reservations

Field	What goes in it
Assumption	Stated as an assumption
Why it is necessary	What record is missing or what question is open
Effect if it is wrong	Quantified where possible
How it could be resolved	What document or confirmation would settle it

Any reservation of rights should be in wording the party's own advisers have approved. Do not draft it from a template.

Part K — What is not claimed

A short section stating what has been excluded and why: events assessed and found not to give rise to relief, periods excluded as the claiming party's own delay, heads of loss not pursued, and any part of the period where the records are insufficient to support a claim. It costs a paragraph and it changes how the whole document is read.

4. Worked fragment

Illustrative figures. Currency-neutral units. Working days on a five-day Monday-to-Friday calendar, no public holidays assumed — a real analysis uses the contract calendar. The example assumes the contract provides for both an extension of time and the recovery of time-related costs for this category of event; whether any particular contract does so, and under what conditions, is a matter for that contract and its governing law.

Part G extract

Milestone	Contractual date	Extension claimed	Revised date sought	Basis
Sectional completion — substation energisation	Wednesday 30 September 2026	22 working days	Friday 30 October 2026	Part F, window 3

The revised date is the contractual date advanced by 22 working days on the stated calendar: `WORKDAY(30 September 2026, 22)` gives Friday 30 October 2026. Counting confirms it — 1 and 2 October are the first two working days, and the twenty-second falls on Friday 30 October. State the calendar every time you state a working-day figure; the same 22 days on a six-day calendar lands on a different date and the difference will be found.

Part H extract — prolongation

Head	Period claimed	Basis of rate	Rate	Weeks	Amount
Prolongation — time-related site costs	1 October to 30 October 2026	Actual cost records for the period in which the delay ran	47,500 per week	4.4	209,000

The substitutions: the extension is 22 working days, and `22 ÷ 5 = 4.4` weeks on a five-day calendar. The weekly rate of 47,500 is the sum of the time-related costs actually incurred in that window — site establishment, supervision, temporary services, site accommodation and standing small plant

— taken from the cost ledger for the period and reconciled to it. Prolongation is therefore `47,500 × 4.4 = 209,000`.

Three things this small table is doing deliberately. The rate comes from the actual records for **that window**, not from the tender allowance and not from an average across the job. The period claimed is stated as dates, so an assessor can go to the ledger and check it. And the conversion from working days to weeks names the calendar, because 22 working days is 4.4 weeks or 3.67 weeks depending on a fact the reader otherwise has to guess.

What the table does not do is assert that this amount is recoverable. That depends on the contract and the governing law, and the submission should say so in the recoverability column rather than leaving the reader to infer an assertion the preparer is not qualified to make.

5. Common mistakes

No stated contractual basis. A submission that describes events and asks for relief without identifying the mechanism and its elements is asking the assessor to build the claim for them. They will not.

Clause numbers quoted from memory of a standard form. Amendments are normal. Cite the executed contract.

Chronology and argument braided together. It converts a dispute about causation into a dispute about facts, which is a fight you did not need to have.

Global causation. "The above events caused a delay of 22 weeks" without an event-by-event chain is the single most common reason a well-founded claim is rejected.

The method not named. An analysis presented as self-evident is an analysis whose weaknesses have not been thought about. Name it, justify it, and say what it is poor at.

The schedule not validated. Relying on a programme with open ends, hard constraints and out-of-sequence progress, without disclosing them, means the whole analysis fails when the other side runs the checks — and they will.

Undisclosed adjustments. Every change you make to a programme for the purposes of analysis must be disclosed with a reason. This is the fastest way to lose credibility across an entire submission.

Quantum from the tender. Rates taken from the tender allowance rather than the cost records invite the response that the claim is a recovery of a pricing error, and that response is often right.

The wrong window priced. Time-related costs from the wrong phase produce a number that does not reconcile to the ledger for the period claimed.

Concurrency ignored. A submission silent on concurrent delay, where concurrency plainly existed, reads as either careless or evasive. State the position and the grounds, and note that the treatment depends on the contract and the governing law.

Notice weaknesses hidden. A late notice found by the other party is worse than a late notice disclosed and addressed. Part C exists to make that decision consciously rather than by omission.

No statement of what is not claimed. Everything in the submission is read as maximal. A short exclusions section makes the rest credible.

6. Adapting it

Safe to change. The order of Parts G and H, the granularity of the chronology, the number of analysis windows, and the addition of parts your contract's procedure requires. Add an executive summary at the front if the submission is long — but write it last, and make sure every sentence in it is supported by a part below.

Safe to add. A disruption analysis as a separate part where the claim includes loss of productivity, as it needs a different structure from a delay analysis. An interface part where several contracts are involved. A without-prejudice commercial position, kept clearly separate from the submission itself and handled on your advisers' instructions.

Do not change. The separation of Part D from Part E. The requirement to name and justify the method. The disclosure of schedule adjustments. The records-to-proposition mapping in Part I. And the position that recoverability and entitlement are matters for the contract and the governing law, not assertions made by the preparer.

6.1 Before the submission is issued

- The mechanism is identified from the executed contract, and every element it requires has a row in Part B with a stated position.
- Every event in Part A has a notice row in Part C, including the events where the notice position is weak.
- Part D contains no adjectives about the other party's conduct and every entry has a source.
- Every event has its own causal block in Part E, with the path status before and after.
- The delay analysis method is named, justified on all five grounds, and its weakness stated.
- Every programme relied on has a quality review reference, and every failed check is disclosed.
- Every adjustment to a programme is listed with a reason.
- Every quantum figure has its arithmetic shown, its period stated as dates, and a record reference.
- Every working-day figure names the calendar it is counted on.
- Every proposition in Part I is marked corroborated or single source.
- The assumptions in Part J each state what would change if they were wrong.
- Part K states what is not claimed.
- Someone who did not prepare the submission has read it cold and been asked where they would attack it.

Related

- `BPG-12 — Claims and extension of time` — the method behind this structure, including how entitlement, causation and quantum interact and where each is typically lost
- `BPG-11 — Change orders and variations` — the process a change goes through before it becomes a claim
- `TPL-12 — Change order log` — the contemporaneous record that supplies Part C and much of Part D

- `TPL-14 — Schedule quality review checklist` — the validation Part F must reference for every programme relied on
- `BPG-05 — Schedule quality — a practical review` — why an unvalidated programme cannot carry a delay analysis

Sources and standards

This is an original instrument developed by the Institute. It reproduces no third-party template, protocol, form or contract wording, and cites no clause number, case or published guidance. The delay analysis method families in §2.4 are described in the Institute's own words as the general shapes in professional use; the descriptions are not attributed to, and should not be read as summaries of, any particular published protocol. Nothing in this template is legal advice, and no jurisdiction's law, procedure or accounting treatment is presented as universal.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-14 · FREE TEMPLATES

Schedule quality review checklist

Fourteen checks, each with what is measured, how to measure it, and a place for the evidence

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A schedule quality review in fourteen checks covering logic, open ends, constraints, negative float, lags and leads, out-of-sequence progress, duration distribution, critical path credibility, resourcing, calendars, float integrity, baseline and progress integrity, milestones and interfaces, and coding. The Institute supplies the measurement; the project supplies the threshold, because thresholds are conventions to be agreed and not universal truths. Every check carries a result, an evidence field and an owner.

About this document

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Schedule quality review checklist

In one paragraph. A schedule quality review in fourteen checks covering logic, open ends, constraints, negative float, lags and leads, out-of-sequence progress, duration distribution, critical path credibility, resourcing, calendars, float integrity, baseline and progress integrity, milestones and interfaces, and coding. The Institute supplies the measurement; the project supplies the threshold, because thresholds are conventions to be agreed and not universal truths. Every check carries a result, an evidence field and an owner.

Who this is for. Planners and schedulers reviewing their own work before issue; project controls managers and assurance reviewers reviewing someone else's; and the risk analysts who must not run a simulation on a schedule that has not passed.

1. When to use this

Before a baseline is accepted. A baseline is a commitment, and accepting one that cannot pass these checks means committing to a network that will not behave predictably when it is updated.

Every reporting period, on the update. Most of these checks take minutes once the reports are built, and the failures that matter — out-of-sequence progress, actual dates beyond the data date, constraints added quietly — arise in updates rather than in baselines.

Before a quantitative schedule risk analysis. A simulation propagates whatever logic it is given. On a network with open ends the uncertainty never reaches the completion milestone and the model reports a confident, narrow and worthless distribution. `TPL-11 — Quantitative schedule risk analysis input sheet` requires a reference to this review in its run configuration for exactly that reason.

Before a schedule is relied on in a submission. A delay analysis built on an unvalidated programme fails the moment the other party runs these checks themselves.

The review answers one question: can this schedule be relied on for the purpose it is about to be used for? That purpose belongs in the header, because the answer changes with it. A schedule good enough to manage a fortnight's work is not necessarily good enough to support a claim.

2. How to complete it

2.1 Separate the measurement from the threshold

This is the design principle of the whole instrument, and it is why the checks below give you formulas and not pass marks.

The measurement is arithmetic. How many activities have no successor. What proportion of activities carry a date constraint. What the minimum total float is. These are facts about a file and two competent reviewers will get the same answer.

The threshold is a convention. Whether three per cent of activities without successors is acceptable is a judgement about this project, its size, its stage, its contract and what the schedule is being used for. There is no universal figure, and any number presented as one should be treated with suspicion until its provenance is produced.

So: agree the thresholds at baseline, with the planner in the room, and record them in the parameter block. Write them down before the first review, not after the first result. A threshold set after the measurement is not a threshold; it is a negotiation.

Where this template offers a number, it is labelled **convention to be agreed** and it is offered as a starting point for that conversation. No threshold here is attributed to any published standard, because none of them comes from one.

2.2 Define the population before you count anything

Every proportion in this checklist has a denominator, and an undefined denominator makes the whole review unreproducible. Agree and record:

- Whether milestones are included in the activity count. Usually they are not, because a milestone has zero duration and no predecessor requirement of the same kind.
- Whether level-of-effort, hammock or summary activities are included. Usually they are not, and they are usually the reason a first review reports alarming open-end figures.
- Whether completed activities are included. For an update review, completed activities should generally be excluded from the forward-looking checks and included in the progress-integrity checks.
- Whether subcontractor or supplier fragnets integrated into the schedule are in scope.

Record the population as a number in the parameter block, and record the filter used to produce it. Someone must be able to reproduce your denominator.

2.3 Grade honestly, in five states

Result	What it means
Pass	The check is met, and the evidence field names what was inspected
Observation	The check is not met, but the effect is understood and contained. Recorded with an owner and a date
Fail	The check is not met and the schedule cannot be relied on for the stated purpose until it is corrected
Not applicable	The check does not apply, with the reason stated
Not tested	The reviewer could not test it — no access, no report, no time. Stated, not omitted

Not tested is the state that makes a checklist honest. The usual failure of schedule reviews is silent omission: a check nobody ran appears indistinguishable from a check that passed. If you could not test it, say so, and the reader can decide what to do about it.

2.4 Make the evidence field reproducible

The evidence field should let another reviewer get the same answer without asking you anything. That means: the file name and revision, the data date, the report or filter used, the count returned, and where the output is stored. "Reviewed logic" is not evidence. "Filter TF < 0 on rev 12, data date 31 July 2026, returned 87 activities, output at `\reviews\2026-07\SQ-04.csv`" is.

3. The template

3.1 Parameter block

Held above the checklist or on a `Parameters` sheet, completed before the first review.

Parameter	Value	Note
Schedule file and revision		The exact file reviewed
Data date		
Purpose of the review		Baseline acceptance · Monthly update · Pre-QSRA · Pre-submission · Assurance
`Total_Activities`		The agreed population per §2.2
Population filter used		So the denominator is reproducible
Reporting period length, in working days		Drives the duration check
Agreed thresholds		One per check, agreed at baseline
Reviewer and date		
Planner and date received		

3.2 The checks

Each row of the instrument carries: check ID, check, what is measured, how to measure it, the agreed threshold, the result, the evidence, the owner and a due date. The measurements are set out below; the thresholds are yours.

SQ-01 — Logic completeness. What is measured: the number of activities with no predecessor and the number with no successor, excluding the single project start and the single project finish. Why it matters: an activity with no predecessor can start at any time the software likes; an activity with no successor cannot pass delay to anything, which means it can slip indefinitely without the completion date moving. Measure it: count from the network report and express each as a proportion of the population.

Formula in words: activities missing a predecessor, divided by the agreed population. Spreadsheet: `=IF(N(Total_Activities)=0,"",Missing_Predecessors/Total_Activities)` and `=IF(N(Total_Activities)=0,"",Missing_Successors/Total_Activities)`. Threshold: convention to be agreed. A common starting position is zero, on the grounds that every open end should have a written reason.

SQ-02 — Undriven starts and finishes. What is measured: activities whose only successor relationship is start-to-start, and activities whose only predecessor relationship is finish-to-finish. Why it matters: these pass SQ-01 and are still open ends. An activity whose only successor is start-to-start has an undriven finish — it can run as long as it likes without consequence. This is the open end that survives the first review. Measure it: count from the relationship report, filtered by relationship type per activity. Threshold: convention to be agreed.

SQ-03 — Constraint use. What is measured: the number and type of date constraints, the proportion of the population carrying one, and how many are of a type that can override logic. Why it matters: a constraint that overrides logic stops the network telling you the truth. Delay stops propagating, float becomes fictional, and the schedule reports a date it is holding rather than a date it is calculating. Also record: whether each constraint has a written reason, a named owner and a review date, and whether it comes from the contract or from convenience.

Formula in words: constrained activities divided by the agreed population. Spreadsheet: `=IF(N(Total_Activities)=0,"",Constrained_Activities/Total_Activities)`. Threshold: convention to be agreed. Contract-imposed dates are usually accepted; convenience constraints usually are not, and the count of logic-overriding constraints is often agreed at zero.

SQ-04 — Negative float. What is measured: the minimum total float in the schedule, the number of activities carrying negative float, and the paths they sit on. Why it matters: negative float is not a status, it is a message — the plan as drawn does not work. It usually means a constraint is fighting the logic, and it must be either resolved by re-planning or accepted explicitly by someone with the authority to accept it. Measure it: filter on total float below zero; report the count, the minimum value and the milestone each path drives. Threshold: convention to be agreed. Zero is the usual starting position for an accepted baseline; an update carrying negative float needs a recovery position, not a threshold.

SQ-05 — Lags and leads. What is measured: the number of relationships carrying a lag, the number carrying a negative lag, the longest lag, and whether each has a written reason. Why it matters: a lag is a promise that nothing happens for a period, and nothing can be progressed, resourced or reported during it. A lag longer than a reporting period is usually work — curing, drying, approval, delivery — and should be an activity so it can be tracked. A negative lag almost always means either the logic is wrong or the activity is too coarse to describe what actually happens. Measure it: count from the relationship report. Threshold: convention to be agreed. Negative lags are often set at zero; a maximum lag duration is worth agreeing at the same time.

SQ-06 — Out-of-sequence progress. What is measured: the number of activities progressed ahead of their predecessors, and the effect on the completion date under retained logic versus progress override. Why it matters: out-of-sequence progress means the network no longer describes how the work is being done, and the two calculation settings can give materially different completion dates from the same file. Measure it: the software's out-of-sequence report, plus a calculation under each setting with both dates recorded. Threshold: convention to be agreed. The count matters less than whether the logic has been corrected to reflect what is actually happening.

SQ-07 — Duration distribution. What is measured: the number of activities with an original duration longer than the reporting period, the number below a very short floor, and the shape of the distribution. Why it matters: an activity longer than a reporting period cannot be progressed meaningfully — its percent complete will be an opinion — and a schedule full of one-day activities generates noise without adding control. Measure it: count from the duration report against the agreed bounds.

Formula in words: activities longer than the reporting period, divided by the agreed population. Spreadsheet: `=IF(N(Total_Activities)=0,"",Long_Duration_Activities/Total_Activities)`. Threshold: convention to be agreed. The principle worth agreeing is that an activity that cannot be measured within one reporting cycle is a candidate for breakdown; the tolerated proportion is a project decision.

SQ-08 — Critical path credibility. What is measured: whether the longest path, traced end to end, describes how the job will actually be built. Why it matters: this is the only check on the list that cannot be automated, and it is the one that finds the serious problems. Measure it: trace the longest path from completion back to the data date and read it aloud to someone who knows the work. Then ask three questions. Does it pass through procurement, design approvals, commissioning and handover, or does it run only through construction? Does it change wildly between updates, which usually means near-critical paths are separated by almost nothing? How many near-critical paths are there, and what is the float separation between them? Record: the path itself, the answer to each question, and the number of paths within the agreed near-critical band. Threshold: not a numeric check. The result is a reviewer's judgement, and the evidence is the traced path.

SQ-09 — Resourcing. What is measured: whether activities carry resources, whether resource assignments reconcile to the cost baseline, and whether the resulting histogram is achievable. Why it matters: an unresourced schedule cannot tell you whether the plan is possible, only whether it is arithmetically consistent. Measure it: count unresourced activities in scope, reconcile total resourced value to the cost baseline, and inspect the histogram for peaks nobody has agreed to staff. Threshold: convention to be agreed, and often "not applicable" where the project does not resource-load — in which case say so rather than passing the check by default.

SQ-10 — Calendar sanity. What is measured: the number of distinct calendars, which activities sit on the default calendar, whether non-working periods are reflected, and whether milestone calendars match the activities that drive them. Why it matters: a seven-day calendar on work that happens five days a week is a silent accelerator that shortens every affected path. A milestone on a different calendar from its driving activity produces dates that do not reconcile. Measure it: activity count by calendar; inspect the non-working periods in each; compare each contract milestone's calendar with its driver's. Threshold: convention to be agreed. Every calendar in use should have a stated purpose.

SQ-11 — Float integrity. What is measured: the distribution of total float, the count of activities with float above an agreed high bound, and whether free float and total float are being used correctly in reporting. Why it matters: a large block of activities carrying very high float usually means missing successors or a constraint holding the end of the network, not genuine slack. Measure it: a histogram of total float across the population, with the high-float tail investigated rather than reported. Threshold: convention to be agreed, and the high bound needs setting alongside it.

SQ-12 — Baseline and progress integrity. What is measured: whether the current schedule reconciles to the approved baseline; whether every difference traces to an approved change; whether the data date is correct; whether any activity is progressed beyond the data date; whether any actual date is in the future; and whether any completed activity carries remaining duration. Why it matters: these are the errors that make a schedule report a position that cannot be true, and they are all detectable in one pass. Measure it: activity-count and date reconciliation to the baseline; filters for actual dates after the data date and for remaining duration on completed activities. Threshold: the mechanical checks are usually agreed at zero. The baseline reconciliation is a count of unexplained differences.

SQ-13 — Milestones and interfaces. What is measured: whether every contract milestone is present and correctly dated, whether interface milestones with other parties exist and are agreed, and whether each has a named owner. Why it matters: interfaces are where projects fail and they are the part of the schedule nobody owns by default. Measure it: reconcile the milestone list against the contract and against each interface agreement. Threshold: usually complete presence, agreed at baseline.

SQ-14 — Coding and structure. What is measured: whether the WBS coding is complete, whether activity codes are populated for every report the project is contractually or internally required to produce, and whether the schedule maps to the cost breakdown structure. Why it matters: incomplete coding is discovered at the worst possible moment, which is when a report is due. Measure it: count of activities with a blank value in each required code field. Threshold: usually zero blanks in the required fields.

3.3 Summary calculations

Field	Formula in words	Spreadsheet expression
Checks tested	Count of results that are Pass, Observation or Fail	`=COUNTIF($F:$F,"Pass")+COUNTIF($F:$F,"Observation")+COUNTIF($F:$F,"Fail")`
Fails	Count of Fail results	`=COUNTIF($F:$F,"Fail")`
Pass rate of checks tested	Passes divided by checks tested; blank if nothing was tested	`=IF(COUNTIF($F:$F,"Pass")+COUNTIF($F:$F,"Observation")+COUNTIF($F:$F,"Fail")=0,"",COUNTIF($F:$F,"Pass")/(COUNTIF($F:$F,"Pass")+COUNTIF($F:$F,"Observation")+COUNTIF($F:$F,"Fail")))`
Not tested	Count of checks the reviewer could not test	`=COUNTIF($F:$F,"Not tested")`
Open actions past due	Actions with a due date before the review date and no closure	`=COUNTIFS($I:$I,"<"&Review_Date,$J:$J,"<>Closed")`

The pass rate deliberately excludes Not applicable and Not tested from its denominator, and the count of Not tested is reported beside it. A single figure that quietly counts an untested check as a pass is how a review becomes reassurance.

Read the fails, not the percentage. The pass rate is a communication device for a report page. The verdict is the list of Fails and what they prevent the schedule being used for, and that verdict belongs in a sentence, not a number.

3.4 Pasting it into a spreadsheet

Copy the header line into cell A1 and split on the pipe character. Column F is the result column referenced by the summary formulas.

```
Check ID|Check|What is measured|Measured value|Agreed threshold|Result|Evidence|Owner|Action due|Action status|Notes
```

4. Worked fragment

Illustrative figures. Three checks from a monthly update review. Schedule revision 12, data date 31 July 2026. Agreed population: 1,240 activities, excluding 46 milestones and 8 level-of-effort activities, filter recorded in the parameter block. Reporting period: one calendar month, taken as 21 working days for the duration check.

Check	What is measured	Measured value	Agreed threshold	Result	Evidence
SQ-01 Logic completeness	Activities with no predecessor / no successor, excluding project start and finish	30 (2.4 %) / 43 (3.5 %)	Zero, other than project start and finish	Fail	Network report, rev 12, DD 31 Jul 26; filters "no predecessor" and "no successor"; counts 31 and 44 before excluding the project start and finish; output at `\reviews\2026-07\SQ-01.csv`
SQ-04 Negative float	Minimum total float; count of activities with total float below zero	−18 days; 87 activities	Zero	Fail	Total float report, rev 12, DD 31 Jul 26, filter TF < 0; 87 rows; all on the path to the substation energisation milestone; output at `\reviews\2026-07\SQ-04.csv`
SQ-07 Duration distribution	Activities with original duration greater than the reporting period	96 (7.7 %)	5 %, agreed at baseline as a project convention	Observation	Duration report, rev 12, filter OD > 20 working days; 96 rows; 61 of them in the commissioning phase; output at `\reviews\2026-07\SQ-07.csv`

The substitutions. SQ-01: missing predecessors `31 − 1 = 30`, and `30 ÷ 1,240 = 0.0242`, reported as 2.4 %. Missing successors `44 − 1 = 43`, and `43 ÷ 1,240 = 0.0347`, reported as 3.5 %. SQ-07: `96 ÷ 1,240 = 0.0774`, or 7.7 %, against a threshold of 5 % agreed at baseline.

How to read this fragment. Two Fails and one Observation, and they are not equal. SQ-04 is the one that stops work: 87 activities at up to eighteen days of negative float on a single milestone path means the schedule is asserting that a contractual date cannot be met, and either the plan changes or somebody accepts that in writing. SQ-01 is the one that invalidates other uses: with 43 activities unable to pass delay to anything, this file cannot support a risk simulation and cannot support a delay analysis, whatever it says about the completion date. SQ-07 is a genuine observation — 7.7 % against an agreed 5 %, concentrated in commissioning, where long activities are common and often defensible. It gets an owner and a date, not an escalation.

Note what the threshold column is doing. The 5 % in SQ-07 is not a standard and is not presented as one; it is what this project agreed at baseline, recorded before the first review, and it appears on the face of the result so that anyone reading it knows what the measured 7.7 % is being judged against.

5. Common mistakes

A pass mark with no evidence. A review reporting 86 % with an empty evidence column cannot be repeated, challenged or trusted. The evidence field is the check on the checker.

Thresholds invented after the measurement. Setting the bar once you know the score is the most common way a review becomes decorative. Agree thresholds at baseline and record the date they were agreed.

Borrowed thresholds presented as standards. A number picked up from another project, or half-remembered from a course, and then reported as though it came from a published standard, is a claim that will not survive being asked for a source. If it is a convention, label it a convention.

An undefined denominator. A proportion computed on "all activities" one month and "incomplete activities excluding milestones" the next produces a trend that is an artefact of the filter.

Silent omission. The check nobody had time to run, reported as nothing at all, reads as a pass. Use Not tested.

Running only the automated checks. SQ-08 cannot be automated and is where the serious findings are. A review that reports twelve clean metrics and never traced the longest path has not looked at the schedule.

Reviewing the baseline once and never the updates. Most of these defects arrive in updates: constraints added under pressure, out-of-sequence progress, actual dates typed into the wrong column. A baseline-only review checks the schedule at the one moment it was most likely to be clean.

Treating Observations as a filing system. An Observation without an owner and a date is a Fail that has been made comfortable.

Confusing schedule quality with schedule realism. Every check here can pass on a schedule whose durations are fantasy. Quality is about whether the network behaves correctly; realism is a separate question, and it is answered by the duration elicitation in `TPL-11` and by asking the people doing the work.

6. Adapting it

Safe to change. Every threshold — that is the point. The population definition, the check IDs, the addition of checks your organisation or contract requires: earned value fields populated, a required update narrative, a specific coding scheme, a client's own reporting requirement.

Safe to add. A trend block carrying each measured value across the last six periods, which turns the review from a snapshot into a control and shows whether the schedule is getting better or worse. A severity weighting where some checks matter more for the stated purpose. A second result column recording the position at the previous review, so closure is visible.

Do not change. The separation of measurement from threshold. The evidence field. The Not tested state. The requirement to record the purpose of the review in the header, because the verdict depends on it. And SQ-08 — the moment the critical path trace comes off the list, the review stops looking at the schedule and starts looking at the file.

6.1 Issuing the review

- The parameter block is complete, including the population, the filter and the purpose of the review.
- Every threshold in use was agreed before this review, and the agreement date is recorded.
- Every check has one of the five results — none is blank.
- Every Fail and Observation has a named owner and a due date.
- Every evidence field names the file, the revision, the data date, the filter and the count.
- The longest path has been traced and read to someone who knows the work, and the trace is attached.
- The Not tested count is reported next to the pass rate, not buried.

- The verdict is stated as a sentence answering the header's question: whether this schedule can be relied on for the purpose it is about to be used for.
- The planner has seen the findings before anyone else did.

Related

- `BPG-05 — Schedule quality — a practical review` — the reasoning behind each check and how to fix what the review finds
- `BPG-17 — Quantitative schedule risk analysis` — why a simulation on an unreviewed network produces a confident wrong answer
- `TPL-11 — Quantitative schedule risk analysis input sheet` — the run configuration that must reference this review
- `TPL-15 — Project controls health check` — where schedule quality sits within a whole-function assessment
- `BPG-04 — Baselining and baseline change control` — what a baseline commits to, and why SQ-12 reconciles to it

Sources and standards

This is an original instrument developed by the Institute. It reproduces no third-party checklist, template, metric set or scoring scheme, and no threshold in it is attributed to any published standard — because none of them comes from one. Every check is described in the Institute's own words and every numeric value that appears is labelled as a convention to be agreed by the project. Where a project is contractually required to meet a specific published schedule-quality specification, that specification governs and this checklist is a supplement to it, not a substitute.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-15 · FREE TEMPLATES

Project controls health check

Eleven dimensions, evidence-based maturity levels, and an action plan with names against it

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A health check across eleven dimensions of a project controls function — structures and coding, baseline integrity, progress measurement, cost capture and cut-off, forecasting, change control, risk, reporting, systems and data, AI governance, and capability. Levels are awarded against defined descriptors and only where evidence was seen: absent evidence caps the level, whatever the interview said. The output is not a score but a ranked action plan with named owners and target levels.

About this document

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Project controls health check

In one paragraph. A health check across eleven dimensions of a project controls function — structures and coding, baseline integrity, progress measurement, cost capture and cut-off, forecasting, change control, risk, reporting, systems and data, AI governance, and capability. Levels are awarded against defined descriptors and only where evidence was seen: absent evidence caps the level, whatever the interview said. The output is not a score but a ranked action plan with named owners and target levels.

Who this is for. Project controls managers and heads of function assessing their own operation; PMO leads and assurance reviewers assessing someone else's; and the project directors and sponsors who commission the review and have to fund the actions.

1. When to use this

At mobilisation, as a design specification rather than an assessment. Run it on the function you intend to build, use the level descriptors to define what "done" means, and you have a set-up plan instead of a wish list.

At a stage gate or before a major commitment. A sanction decision rests on numbers the controls function produced. If the function cannot demonstrate how it produced them, the decision is resting on less than it appears to be.

When the numbers have started arguing with each other. A forecast that moves without explanation, a cost report that will not reconcile to the ledger, a schedule that nobody quotes — these are symptoms, and a structured review across all eleven dimensions is a faster route to the cause than chasing each symptom.

Annually on a long programme, so that improvement is visible and reversal is caught.

Do not run it as an audit unless you are prepared to say so. A health check is a diagnostic done with the function; an audit is done to it. Both are legitimate and they need different framing, different access and different reporting. Say which one this is in the first line of the report.

2. How to complete it

2.1 Who runs it

Not the person who owns the control being assessed. Self-assessment has a place — it is fast, it builds ownership, and it is a reasonable first pass — but its output is a view, not a finding, and the report must say which it is. Where possible use a reviewer from outside the project who understands the discipline: an insider who does not know controls will accept plausible answers, and an outsider who does not know the project will misread deliberate choices as failures.

2.2 Ask for the evidence pack before you arrive

Request it in advance, by artefact and not by topic. The request itself is diagnostic — a function that can produce all of it in two days is already telling you something, and so is one that cannot.

The pack: the work breakdown structure and cost breakdown structure with the mapping between them; the approved baseline and the log of baseline changes; the rules of credit and the last three progress assessments; the last three cost reports with their reconciliation to the ledger; the current forecast with its basis of estimate; the change log; the risk register and the last risk review minutes; the report distribution list and the last three reports issued; a list of systems with their owners and interfaces; any record of where AI is used in the controls process; and the team structure with roles, names and the competence held.

2.3 Award levels against the descriptors, and only on evidence

Five levels. They are deliberately coarse, and there are no half levels — a half level is how a review avoids a conversation.

Level	Name	Descriptor
1	Absent	The control does not exist, or exists only as one individual's habit
2	Ad hoc	It is done, inconsistently. It depends on who is present. It is not defined in writing
3	Defined	It is documented, owned and followed for the main scope. The gaps are known and named
4	Consistent	It is followed across the whole scope, evidenced, with checks that catch failures, and it survives the absence of any one person
5	Improving	As level 4, and the output is measured, challenged and used to change how the project is run. Deficiencies are found by the process itself, not by a review

The evidence rule: no evidence caps the level at 2. If the reviewer has not seen the artefact named in the evidence column, the dimension cannot be scored above 2, regardless of how convincingly it was described. This single rule is what separates a health check from a conversation about intent, and it should be stated to the function before the review starts so that nobody is ambushed by it.

Two consequences worth being ready for. Functions that do good work but document none of it will score lower than their delivery deserves — and that is the correct result, because a control that lives in one person's head fails the moment that person is on leave. And a reviewer will occasionally be shown an artefact that exists but is not used; level 3 requires that it is followed, not that it exists, so ask for the last three instances of it being applied.

2.4 Weight the dimensions, and set targets to risk rather than ambition

Equal weights are a defensible default. Where the project's risk profile is lopsided — a heavily subcontracted job, a reimbursable contract, a first-of-a-kind design — weight accordingly and record the reason. Set the weights **before** the assessment, so they cannot be tuned to the answer.

Target levels are a project decision, not an aspiration. Not every dimension needs to reach level 4 on every project, and a target of 5 everywhere is a statement that nobody has thought about cost. Set the target where the consequence of the control failing justifies the effort, and write the justification down.

If the review finds something that changes the weighting argument — an AI tool generating forecast commentary that reaches the board unvalidated, on a dimension weighted at 0.5 — the reviewer should recommend a change to the weight and say so explicitly in the report. What they must not do is quietly re-weight to make the answer come out differently.

2.5 Write findings that name the effect

A finding with no effect is an opinion. Every one should state what is true, what it costs or risks, and what the evidence for it was. "Progress is self-assessed by the delivery team without independent check" is a fact. "Progress is self-assessed by the delivery team without independent check, so the earned value underpinning the forecast is unverified, and the last three periods show progress claimed at over 98 % of plan while cost ran at 106 % of plan" is a finding.

Show every finding to the person who owns the control before it is reported. Not for approval — for correction. A review that is factually wrong on one point loses the argument on all of them.

3. The template

Two sheets: the assessment and the action plan.

3.1 Sheet 1 — `Assessment`

Col	Field	Input or calculated	What goes in it
A	Dimension ID	Input	`HC-01` to `HC-11`
B	Dimension	Input	From §3.2
C	What good looks like here	Input	The level-4 statement for this dimension, agreed before the review
D	Weight	Input	Set before the assessment, with the reason in column K
E	Assessed level	Input	1 to 5, whole numbers only
F	Evidence seen	Input	The specific artefacts inspected, with dates and versions. Blank caps column E at 2
G	Evidence not available	Input	What was asked for and not produced
H	Target level	Input	Set to the project's risk, with the justification in column K
I	Gap to target	Calculated	
J	Priority	Calculated	
K	Basis and notes	Input	The reason for the weight, the target and the level awarded

Col	Formula in words	Spreadsheet expression
I	Target level less assessed level; blank until both are entered	`=IF(OR($E2="",$H2=""),"",$H2-$E2)`
J	Gap multiplied by weight	`=IF(OR($I2="",$D2=""),"",$I2*$D2)`

Summary cells:

Field	Formula in words	Spreadsheet expression
Weighted score	The sum of level multiplied by weight, divided by the sum of the weights; blank if no weights are set	`=IF(SUM($D$2:$D$12)=0,"",SUMPRODUCT($E$2:$E$12,$D$2:$D$12)/SUM($D$2:$D$12))`
Unweighted mean	The average of the assessed levels; blank if none is set	`=IF(COUNT($E$2:$E$12)=0,"",AVERAGE($E$2:$E$12))`
Dimensions at level 2 or below	The count of dimensions assessed at 1 or 2	`=COUNTIF($E$2:$E$12,"<=2")`
Dimensions capped by absent evidence	The count where a level above 2 was claimed but no evidence was recorded	`=COUNTIFS($F$2:$F$12,"",$E$2:$E$12,">2")`

The last of those should always read zero in a completed review. If it does not, the evidence rule has been broken and the report is not ready.

3.2 The eleven dimensions

HC-01 Structures and coding. Whether a work breakdown structure and a cost breakdown structure exist, whether they map to each other and to the schedule and the ledger, whether the code of accounts is complete, and whether new scope can be coded without a debate. Evidence: the structures themselves, the mapping, and a sample transaction traced from the ledger to the control account to the activity.

HC-02 Baseline integrity. What is baselined — scope, cost, schedule — who approved it, whether every change since is traceable to an approval, and whether the current baseline is the one being reported against. Evidence: the approved baseline, the change log, and a reconciliation from the original to the current.

HC-03 Progress measurement. Whether rules of credit exist, whether they are objective, who assesses progress, whether that person is independent of the person whose progress it is, and whether the method suits the work. Evidence: the rules of credit, the last three assessments, and a sample walked back to the physical work.

HC-04 Cost capture and cut-off. Whether commitments are captured at the point of commitment, whether accruals are made on a defined basis, whether the cut-off date is enforced, and whether the cost report reconciles to the ledger without manual adjustment. Evidence: the cut-off calendar, the accrual basis, and the last three reconciliations including the unreconciled differences.

HC-05 Forecasting. Whether the forecast has a stated method, whether the method suits the cause of the variance, who owns the forecast, whether it is challenged by anyone, and whether previous forecasts are compared against outturn. Evidence: the current forecast with its basis of estimate, the challenge record, and the last three forecasts against what actually happened.

HC-06 Change control. Whether changes are identified early, whether notice obligations are tracked, whether changes reach the cost forecast and the schedule, and whether anyone reconciles the change log to the forecast. Evidence: the change log, and the reconciliation between unrecognised exposure and the forecast — see `TPL-12 — Change order log`.

HC-07 Risk. Whether the register is used or merely maintained, whether risks have named owners who attend, whether responses are completed, whether the register feeds quantification, and whether contingency is linked to it. Evidence: the register, the last three review minutes, the contingency basis, and the drawdown record.

HC-08 Reporting. Who the report is for, what decisions it has actually driven, how long it takes from cut-off to issue, whether it says anything a reader could disagree with, and whether the numbers in it reconcile to the systems they came from. Evidence: the last three reports, the distribution list, the issue dates against the cut-off dates, and a decision traced back to the report that prompted it.

HC-09 Systems and data. What tools are in use, who owns each, how they interface, how much manual re-keying happens between them, who has access to change what, and whether there is an audit trail. Evidence: the system list with owners, the interface map, and a sample of a figure traced across systems.

HC-10 AI governance. Where AI is used in the controls process — drafting, classification, forecasting, document review, anomaly detection — what it is permitted to decide and what it is not, how outputs are validated before use, whether the human decision is recorded, whether the data going into it is permitted to be there, and whether anyone could explain a given output to a reviewer. The Institute's position is that AI proposes and the professional disposes: an output that reaches a decision must be explainable, validated and owned by a competent human. Evidence: the record of where AI is in use, the validation records, and a sample output traced to the human who accepted it.

HC-11 Capability. Whether the roles required by the other ten dimensions exist and are filled, whether the people in them hold the competence the role needs, whether there is cover for absence, and whether development is happening. Assess against a defined competency set rather than an impression — `CMP-03 — PCL-AI: the fourteen competencies` is one such set. Evidence: the team structure with named roles, the competence held against each, and the cover arrangements.

3.3 Sheet 2 — `Findings and actions`

Col	Field	What goes in it
A	Finding ID	`F-01` onward
B	Dimension ID	Links to sheet 1
C	Finding	What is true, stated as a fact
D	Evidence	What was inspected that establishes it
E	Effect	What it costs or risks, quantified where possible
F	Severity	High · Medium · Low, defined by effect and not by tidiness
G	Recommendation	What to do, specifically
H	Action owner	A named individual, not a function
I	Target level for the dimension	
J	Due date	
K	Status	Open · In progress · Complete · Accepted risk
L	Verification evidence	What was seen at re-verification
M	Date verified	
N	Owner acceptance	The control owner's response, including where they disagree

Column N matters. A finding the owner disputes, recorded with their reason, is more useful than a finding they were talked out of disputing — and if they turn out to be right, it is on the record that they said so.

3.4 Pasting it into a spreadsheet

Copy each header line into cell A1 of its own sheet and split on the pipe character.

```
Dimension ID|Dimension|What good looks like here|Weight|Assessed level|Evidence seen|Evidence not available|Target level|Gap to target|Priority|Basis and notes
```

```
Finding ID|Dimension ID|Finding|Evidence|Effect|Severity|Recommendation|Action owner|Target level|Due date|Status|Verification evidence|Date verified|Owner acceptance
```

4. Worked fragment

Illustrative figures. A complete assessment sheet, weights set before the review, target level 4 across all eleven dimensions on this project's agreed basis.

Dimension	Weight	Assessed level	Target	Gap	Priority
HC-01 Structures and coding	1.0	4	4	0	0.0
HC-02 Baseline integrity	1.5	3	4	1	1.5
HC-03 Progress measurement	1.5	2	4	2	3.0
HC-04 Cost capture and cut-off	1.5	3	4	1	1.5
HC-05 Forecasting	1.5	2	4	2	3.0
HC-06 Change control	1.5	2	4	2	3.0
HC-07 Risk	1.0	3	4	1	1.0
HC-08 Reporting	1.0	3	4	1	1.0
HC-09 Systems and data	1.0	2	4	2	2.0
HC-10 AI governance	0.5	1	4	3	1.5
HC-11 Capability	1.0	3	4	1	1.0

The substitutions. Sum of weights: `1.0 + 1.5 + 1.5 + 1.5 + 1.5 + 1.5 + 1.0 + 1.0 + 1.0 + 0.5 + 1.0 = 13.0`.

Sum of level multiplied by weight: `(1.0×4) + (1.5×3) + (1.5×2) + (1.5×3) + (1.5×2) + (1.5×2) + (1.0×3) + (1.0×3) + (1.0×2) + (0.5×1) + (1.0×3) = 4 + 4.5 + 3 + 4.5 + 3 + 3 + 3 + 3 + 2 + 0.5 + 3 = 33.5`.

Weighted score: `33.5 ÷ 13.0 = 2.58` to two decimal places. Unweighted mean: `28 ÷ 11 = 2.55`. Dimensions at level 2 or below: HC-03, HC-05, HC-06, HC-09 and HC-10, so `5`.

How to read this. The weighted score of 2.58 is the least useful number on the page, and it is the one that will be quoted. The report is the priority column: HC-03 progress measurement, HC-05 forecasting and HC-06 change control all sit at 3.0, and they are not three separate problems. You cannot forecast what you cannot measure, and you cannot forecast what change control has not told you about. Progress measurement is the one to fix first, because the other two consume its output. That sentence — not the average — is the finding.

HC-10 deserves a second look for the opposite reason. It has the largest raw gap on the sheet, three levels, and the lowest priority score, because the weight of 0.5 was set before the review. If the review found AI being used to draft forecast commentary that reaches the board without a recorded human validation, then the weight is wrong, and the honest response is to say so in the report and recommend re-weighting for the next cycle — not to quietly change the number now and present a different score.

The two averages — 2.58 weighted and 2.55 unweighted — being almost identical is itself informative: the weighting is doing very little work on this profile, which means the argument about weights can be short and the argument about progress measurement can be long.

5. Common mistakes

Scoring the intent rather than the artefact. The most common failure, and the reason the evidence rule exists. A well-run interview will produce a description of a level-4 process that is not happening.

Half levels. A 3.5 is a way of not deciding. It also destroys the arithmetic's meaning, because the descriptors do not have midpoints.

Reporting the average. A function scoring 2.58 with one dimension at 1 has a specific problem, and the average is designed not to show it. Report the count at level 2 or below and the priority ranking; put the average in an appendix if it must appear at all.

Weights set after the assessment. Once the levels are known, weighting becomes score management. Set them first and record the date.

Targets of 5 everywhere. A target is a commitment to spend effort. Setting every target at the top says either that the reviewer has not costed the recommendations or that nobody intends to act on them.

No named owner. "The PMO" cannot be chased. Column H takes a person.

Findings without effects. "The rules of credit are not documented" invites a shrug. "The rules of credit are not documented, three assessors are applying different bases to the same work type, and the resulting earned value feeds a forecast that has moved twice this quarter without explanation" invites a decision.

No re-verification. A health check with no verification column becomes an annual event that produces the same findings each year with different dates. Columns L and M are what make it a control rather than a report.

AI governance assessed as a technology question. HC-10 is not about which tools are licensed. It is about what a machine is permitted to decide, who validated the output that reached a decision, and whether that validation is recorded. A function with no AI in use scores this dimension as such and explains how it knows — which requires having asked, because the answer is frequently that AI is in use and nobody had been asked.

6. Adapting it

Safe to change. The weights, the targets, the dimension names, and the addition of dimensions your organisation needs — earned value specifically, interface management, subcontractor controls, benefits tracking, health and safety reporting. If a dimension does not apply, mark it not applicable with a reason rather than scoring it low.

Safe to add. A trend column carrying the previous cycle's level, which turns the instrument into a control. A cost-to-remedy estimate against each recommendation, which makes the action plan fundable. A split of the assessment by project within a portfolio, which usually reveals that the function is not the problem and one project is.

Do not change. The evidence rule. The whole-number levels. The requirement that the action owner is a named individual. And the practice of showing every finding to the control owner before reporting it — that one costs a day and buys the credibility of the whole exercise.

6.1 Before the report is issued

- The report states whether this was a health check or an audit, and who ran it.
- Weights and targets were set before the assessment, with dates recorded.
- Every dimension has a level, and every level above 2 has an evidence entry — the capped-by-absent-evidence count reads zero.

- Everything requested and not produced is recorded in column G, not omitted.
- Every finding states an effect, quantified where the data allowed.
- Every finding has been shown to the control owner, and their response is recorded, including disagreement.
- Every action has a named individual, a target level and a date.
- The priority ranking, not the average, leads the report.
- Any recommendation to change a weight for the next cycle is stated openly rather than applied silently.
- A re-verification date is in the diary before the report is circulated.

Related

- `BPG-19 — Project controls assurance and health checks` — the method behind this instrument, including how to run the review and how to report it without losing the room
- `BPG-01 — Building a project controls function from zero` — the level-4 descriptions used as the specification when this template is run at mobilisation
- `AIG-08 — Governing AI on a project — the control framework` — what HC-10 is assessing against
- `TPL-16 — Lessons learned and closeout register` — where the findings go when the project ends, and the route by which they change the organisation rather than the project
- `BPG-14 — Monthly reporting that gets read` — the standard HC-08 assesses reporting against

Sources and standards

This is an original instrument developed by the Institute. It reproduces no third-party maturity model, assessment framework, questionnaire or scoring scheme. The five level descriptors, the eleven dimensions and the evidence rule are the Institute's own formulation, offered for adaptation rather than as a standard. The competency set referenced in HC-11 is the Institute's own, published in `CMP-03`. Where an organisation is required to assess against a specific published maturity model, that model governs and this template is a supplement to it, not a substitute.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

TPL-16 · FREE TEMPLATES

Lessons learned and closeout register

The fields that make a lesson actionable, and the closeout checklist that captures what the next project needs

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A lessons register built around the two fields most registers omit — the artefact that must change, and the process owner outside the project who owns that artefact — plus the implementation status that makes the whole thing measurable. It comes with a four-part closeout checklist covering commercial, financial and records closeout, and the benchmarking data capture that has to happen before the team disperses, because after that it is unrecoverable.

About this document

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Lessons learned and closeout register

In one paragraph. A lessons register built around the two fields most registers omit — the artefact that must change, and the process owner outside the project who owns that artefact — plus the implementation status that makes the whole thing measurable. It comes with a four-part closeout checklist covering commercial, financial and records closeout, and the benchmarking data capture that has to happen before the team disperses, because after that it is unrecoverable.

Who this is for. Project controls managers, project managers and commercial managers closing a project or a phase; the process owners in the wider organisation who receive the lessons; and the estimators and planners who will need the benchmarking data on the next bid.

1. When to use this

From the start of the project, not at the end of it. A lesson written eleven months after the event has lost the two things that made it useful: the detail of what actually happened, and the person who saw it. Open the register at mobilisation and add to it as things happen. The end-of-project workshop then becomes a review of a register rather than a memory test conducted under time pressure by people who have already been assigned elsewhere.

At every phase transition, because a lesson from the design phase is worth most to the design phase of the next project, and it will be forgotten by the time this one finishes.

At closeout, for the checklist in §3.3 and — most urgently — the benchmarking capture in §3.4. There is a window between the end of the work and the dispersal of the team in which the data that makes the next estimate better can still be collected. After that window closes, it can be reconstructed only expensively and badly.

The instrument answers one question: what will be different next time, and who is making it different? A register that cannot answer the second half is a record of experience, not a mechanism for improvement.

2. How to complete it

2.1 Separate context, event, cause and instruction

Four fields do the work, and merging them is why most registers are unreadable.

Context (column G) is what a stranger on a different project needs in order to decide whether this applies to them. Contract type, phase, scale, the arrangement that mattered. Without it every lesson reads as either universal or irrelevant, and both readings mean it is ignored.

What happened (column H) is the observable fact, dated. No characterisation.

Why (column I) separates the immediate cause from the condition that allowed it. This is the most valuable field in the register and the one most often filled with a restatement of the event. A divergence between two contract documents is the immediate cause; the absence of any step in the pre-contract review that reconciles them is the condition, and only the second one can be fixed. Most lessons that never get implemented failed here: they identified something that cannot be changed rather than something that can.

What to do differently (column L) is an instruction, testable by someone who was not there. "Improve communication with the design team" is not one. "Add a step to the pre-contract review requiring a documented reconciliation of the specification hierarchy, with divergences either priced or listed as exclusions" is.

2.2 Route it to an artefact and an owner

Columns M and N are the mechanism, and their absence is why lessons registers fail.

Column M — where it must land. Name the artefact that has to change: a procedure, a checklist line, an estimating norm, a standard scope of work, a tender question, an induction, a contract term, a schedule fragnet, a template. A lesson that does not change an artefact changes nothing, because the next project will be run by different people reading the same documents.

Column N — the process owner. The named role **outside the project** that owns that artefact. This is the field that makes a lesson real. A lesson routed to "the project team" dies with the project. A lesson routed to the person who owns the pre-contract review checklist arrives somewhere that outlives the job.

If no artefact can be named, the honest options are to record it as an observation rather than a lesson, or to work harder at column I until a changeable condition emerges. Both are better than a register entry that will never be actioned and will be counted as though it might be.

2.3 Record the positives, and mean it

Column K distinguishes what to repeat from what to avoid. A register containing only failures teaches the organisation that submitting to it is dangerous, and submissions stop. It also loses genuinely valuable information: the sequencing decision that saved six weeks is as transferable as the one that cost six, and considerably easier to get adopted.

2.4 Measure implementation, and publish the rate

The implementation rate in §3.2 is the only number that tells you whether the lessons process is real. A register with 140 entries and an implementation rate of four per cent is a filing cabinet with a search function. Publish the rate to the process owners quarterly. It is uncomfortable, and it is the only thing that reliably changes behaviour, because it moves the accountability from the project that raised the lesson to the function that received it.

2.5 Capture benchmarking data with its definitions attached

The benchmarking block in §3.4 is the part of closeout that pays for itself and the part most often skipped. The trap is that raw figures without their definitional metadata are worse than nothing: they look comparable, they are used as though they are, and they produce estimates that are confidently wrong.

Every captured figure needs its scope boundary, its currency and base date, its escalation treatment and the definition of its denominator recorded alongside it. A productivity figure without a definition of what was included in the hours is not data.

3. The template

Four parts: the lessons register, its summary measures, the closeout checklist, and the benchmarking capture.

3.1 Part 1 — the lessons register

Col	Field	What goes in it
A	Lesson ID	`LL-031` and so on. Permanent
B	Date raised	The date the entry was created, not the date of the event
C	Raised by	Named individual
D	Phase	When it occurred: business case · design · procurement · construction · commissioning · handover · closeout
E	Function	Cost · Schedule · Risk · Commercial · Procurement · Engineering · Construction · Commissioning · Digital and data · Health and safety
F	Category	The project's own taxonomy, agreed at mobilisation
G	Context	What a reader on a different project needs to judge whether it applies to them
H	What happened	The observable facts, dated, without characterisation
I	Why	The immediate cause, and separately the condition that allowed it
J	Effect	Quantified where possible: cost, working days, rework hours, incidents
K	Repeat or avoid	Repeat · Avoid. Both are lessons
L	What to do differently	An instruction, testable by someone who was not there
M	Where it must land	The named artefact that has to change
N	Process owner	The named role outside the project that owns that artefact
O	Change required	The specific edit to the named artefact
P	Owner response	Accepted · Rejected · Deferred, with the date and the reason
Q	Status	Proposed · Accepted · In progress · Implemented · Rejected · Superseded
R	Implementation evidence	The version of the artefact that now carries the change
S	Date verified	The date the change was confirmed in the artefact
T	Verified by	Named individual, not the person who raised it
U	Applicable to	Project types, contract types, sectors — the transferability statement
V	Confidence	Single observation · Repeated on this project · Seen across projects
W	Benchmarking record	Link to the Part 4 capture where the lesson has data behind it
X	Notes	

3.2 Part 2 — summary measures

`Report_Date` is a named cell holding the reporting cut-off, used instead of `TODAY()` so that a saved report still reads the same next quarter.

Elapsed days is a helper column — put it in column Y, immediately after the register.

Field	Formula in words	Spreadsheet expression
Elapsed days (column Y)	Days from raising to verification, or to the report date if not yet verified	`=IF($B2="","",IF($S2="",Report_Date-$B2,$S2-$B2))`
Lessons raised	Count of lesson IDs	`=COUNTA($A$2:$A$500)`
Implementation rate	Implemented lessons divided by the lessons that have a status recorded; blank if none has one	`=IF(COUNTIF($Q$2:$Q$500,"<>")=0,"",COUNTIF($Q$2:$Q$500,"Implemented")/COUNTIF($Q$2:$Q$500,"<>"))`
Rejection rate	Rejected lessons divided by the lessons that have a status recorded; blank if none has one	`=IF(COUNTIF($Q$2:$Q$500,"<>")=0,"",COUNTIF($Q$2:$Q$500,"Rejected")/COUNTIF($Q$2:$Q$500,"<>"))`
Median days to implementation	The median of the elapsed days across implemented lessons; blank if there are none	`=IF(COUNTIF($Q$2:$Q$500,"Implemented")=0,"",MEDIAN(IF($Q$2:$Q$500="Implemented",$Y$2:$Y$500)))`, entered as an array formula
Lessons with no named artefact	Count of rows that have a lesson ID but no entry in the artefact column	`=COUNTIFS($A$2:$A$500,"<>",$M$2:$M$500,"")`

The last measure is the health check on the register itself. A rising count of lessons with no named artefact means the register is filling with observations, and the implementation rate will follow it down.

A rejection rate of zero is not good news. It usually means process owners are accepting everything and implementing selectively, which produces an accepted-but-not-implemented backlog that nobody is counting. A rejection with a recorded reason is a decision; silent non-implementation is a leak.

3.3 Part 3 — the closeout checklist

Each item takes a status, an owner, a date and an evidence reference. The four groups are separate because they close at different times and involve different people.

Commercial closeout

- Final account position agreed, or the outstanding items listed with a route and a date for each.
- Every change on the change order log closed: agreed, withdrawn, rejected or formally in dispute.
- Claims resolved, withdrawn, or transferred to a named owner with the record set preserved.
- Retention release dates diarised, with the amounts and the trigger events recorded.
- Warranties, guarantees and bonds either released or diarised for release, with expiry dates held by a named owner.
- Subcontract and supply final accounts settled, and any contra-charges or back-charges concluded.
- Defects liability or rectification period start and end dates recorded, with the responsible owner named.
- Insurance position closed or converted to run-off cover as the contract requires.
- Outstanding notices either withdrawn in writing or deliberately preserved, with the decision recorded.

Financial closeout

- Final cost report reconciled to the general ledger, with any unreconciled difference explained in writing.
- Accruals reversed or converted, with none left standing against a closed cost code.
- Open commitments and purchase orders closed or transferred.
- Cost codes closed to further posting, with the date and the authoriser recorded.
- The accounting treatment of the completed asset or the completed contract concluded with the finance function — noting that recognition, capitalisation and tax treatment vary by jurisdiction and by accounting framework, and none of them should be assumed from another project.
- Final cash position stated, with outstanding receivables listed and owned.
- Currency and hedging positions closed with the finance function where any were held.
- Project bank account or ledger closed, or the closure date diarised.

Records closeout

- The record set indexed against a stated structure, with a named custodian.
- Retention period identified from the contract and the applicable law, which differ — record the period applied and its source, and do not carry over a period from another project.
- Access controls set, so that the records can be found by someone who was not on the project.
- The as-built programme and every schedule update preserved in native format, not only as reports.

- The correspondence and notice register preserved with its index.
- Where any dispute is live or reasonably anticipated, the relevant records preserved under a documented hold, and the hold's owner named.
- Personal data handled according to the applicable regime, which varies by jurisdiction — record what was retained, what was deleted, and on what basis.
- AI-generated artefacts identified as such, with their validation records preserved alongside them, so that a later reader can tell what was machine-produced and who accepted it.

Benchmarking capture — see Part 4. Its place in this checklist is a single item with a hard rule attached: it is complete before the team disperses, or it is not complete.

3.4 Part 4 — benchmarking data capture

Two blocks. The measures, and the metadata without which the measures are unusable.

Measure	Formula in words	Spreadsheet expression
Cost outturn ratio	Final outturn cost divided by the sanction budget; blank if the budget is unset	`=IF(N(Sanction_Budget)=0,"",Final_Outturn_Cost/Sanction_Budget)`
Schedule outturn ratio	Actual duration divided by baseline duration, in working days; blank if the baseline is unset	`=IF(N(Baseline_Duration_Days)=0,"",Actual_Duration_Days/Baseline_Duration_Days)`
Change value as a share of the original contract	Total agreed change value divided by the original contract value; blank if that value is unset	`=IF(N(Original_Contract_Value)=0,"",Total_Agreed_Change_Value/Original_Contract_Value)`
Contingency drawdown ratio	Contingency drawn divided by contingency held at sanction; blank if none was held	`=IF(N(Contingency_Held_At_Sanction)=0,"",Contingency_Drawn/Contingency_Held_At_Sanction)`
Indirect cost share	Indirect cost divided by total cost; blank if total cost is unset	`=IF(N(Total_Cost)=0,"",Indirect_Cost/Total_Cost)`
Design growth	Final quantity divided by tender quantity, per measured item; blank if the tender quantity is unset	`=IF(N($B2)=0,"",$C2/$B2)`
Productivity achieved	Output quantity divided by hours expended, per discipline; blank if hours are unset	`=IF(N($E2)=0,"",$D2/$E2)`
Confidence at outturn	The percentile at which the actual outturn sat in the sanction-stage risk analysis	Read from the analysis; record the analysis reference alongside it

Also capture, as raw figures rather than ratios: final quantities with their units; unit rates achieved; actual durations of standard activity types against the estimated durations; and the actual elapsed time between key milestones.

The metadata, recorded against every figure above:

Field	Why it is needed
Scope boundary	What is inside the number and what is outside — client costs, land, enabling works, escalation, contingency
Currency and base date	A rate with no base date cannot be escalated to today
Escalation treatment	Whether the figure is as-spent or rebased, and to when
Denominator definition	What "hours" includes: supervision, travel, rework, standing time
Measurement standard	The rule under which quantities were measured
Contract type	Lump sum, remeasurable, reimbursable, target cost — the ratios mean different things under each
Sector and geography	For like-for-like comparison
Data quality	Whether the figure is from the ledger, from a report, or estimated at closeout

The last row is the one that saves a future estimator. A figure reconstructed at closeout from memory and a figure taken from the ledger should not sit in a benchmarking database looking identical.

3.5 Pasting it into a spreadsheet

Copy the header line into cell A1 and split on the pipe character.

```
Lesson ID|Date raised|Raised by|Phase|Function|Category|Context|What happened|Why|Effect|Repeat or avoid|What to do differently|Where it must land|Process owner|Change required|Owner response|Status|Implementation evidence|Date verified|Verified by|Applicable to|Confidence|Benchmarking record|Notes
```

4. Worked fragment

Illustrative figures. Currency-neutral units. Working days on a five-day calendar. Report date 30 June 2026.

One lesson. Fields shown in the order of §3.1; `Raised by`, `Category`, `Verified by` and `Notes` are omitted here for space.

Field	Entry
Lesson ID	LL-031
Date raised	12 March 2026
Phase	Procurement
Function	Commercial
Context	A design-and-build package on a lump-sum contract where the employer's requirements referenced a performance specification for the fire alarm system and the contractor's proposals priced a prescriptive scope
What happened	The divergence was first identified at the design review eleven weeks after contract award. Resolution took a further nine weeks and required re-tendering of two related subcontract packages
Why	Immediate cause: a documentary divergence between two documents both forming part of the contract. Condition that allowed it: the pre-contract review had no step requiring the specification hierarchy and any divergence to be reconciled before execution
Effect	148,000 in additional design and abortive procurement cost; 15 working days of delay to package handover
Repeat or avoid	Avoid
What to do differently	Add a mandatory step to the pre-contract review requiring a documented reconciliation of the specification hierarchy, with every divergence either priced or listed as an exclusion, signed by a named reviewer
Where it must land	The pre-contract review checklist, section on documents forming the contract
Process owner	Head of Commercial, as owner of the pre-contract review checklist
Change required	New checklist item with a named reviewer and a mandatory output
Owner response	Accepted, 2 April 2026
Status	Implemented
Implementation evidence	Pre-contract review checklist version 4.2
Date verified	4 June 2026
Applicable to	Design-and-build and engineer-procure-construct packages where employer's requirements and contractor's proposals both form part of the contract
Confidence	Repeated on this project — a comparable divergence arose on the mechanical package

Elapsed days from raising to verification: 12 March 2026 to 4 June 2026 is **84 days**.

Note what makes this entry work. The condition in the "why" field is changeable, which is why there is a named artefact in the next field. The instruction is testable by someone who was not there. The process owner sits outside the project, so the change survives demobilisation. And the entry ends with an artefact version and a verification date, which is what allows it to be counted as implemented rather than accepted.

Benchmarking capture extract

Measure	Inputs	Result
Change value as a share of the original contract	2,640,000 ÷ 24,000,000	0.110, or 11.0 %
Contingency drawdown ratio	2,150,000 ÷ 1,800,000	1.194, or 119.4 %
Schedule outturn ratio	712 ÷ 640 working days	1.113, or 111.3 %
Cost outturn ratio	28,900,000 ÷ 26,500,000	1.091, or 109.1 %

Metadata for all four: figures at the sanction base date; no escalation adjustment applied; client-side costs and land excluded; single contract currency; lump-sum contract; figures taken from the ledger except the schedule outturn, which is taken from the as-built programme.

The substitutions. Change share: `2,640,000 ÷ 24,000,000 = 0.110`. Contingency drawdown: `2,150,000 ÷ 1,800,000 = 1.194` — the project drew about nineteen per cent more contingency than it held, which is a single sentence a benchmarking database should be able to produce and almost none can. Schedule outturn: `712 ÷ 640 = 1.1125`, reported as 1.113. Cost outturn: `28,900,000 ÷ 26,500,000 = 1.0906`, reported as 1.091.

Read the last two together. The project overran its sanction budget by 9.1 % and its baseline duration by 11.3 %, and it drew 19.4 % more contingency than it held. Those three figures are more use to the next estimate than any narrative, and all three become uncollectable within about six weeks of handover.

5. Common mistakes

Opening the register at the end. The lessons that survive to a closeout workshop are the ones that were still annoying somebody in the final month. Everything from the first year is gone.

Lessons that name no artefact. "Engage stakeholders earlier" cannot be implemented because there is nothing to change. Column M is the test, and the count of blanks in it is the register's health.

Routing to the project. A lesson owned by the project team is closed by demobilisation. The process owner must be outside the project.

A register of failures only. It teaches people that contributing is risky, and submissions dry up within two projects.

Restating the event in the "why" field. "The divergence was not identified" is not a cause; it is the event again. Keep asking until you reach a condition somebody can change.

Counting raised, not implemented. "We captured 140 lessons" is an input measure and it is the one that gets reported. The output measure is the implementation rate, and it is usually an order of magnitude less impressive.

Accepting everything. A rejection rate of zero with a low implementation rate means process owners are being polite. A recorded rejection with a reason is a better outcome than a silent acceptance.

Skipping the benchmarking capture. It is last on the list, it happens when everyone has been reassigned, and it is the item with the shortest window. Diarise it before the completion date, not after.

Benchmarking figures with no definitions. Numbers that look comparable and are not do more damage than no numbers, because they will be used with confidence.

Records closed on last project's rules. Retention periods, data protection obligations and accounting treatment differ by contract and by jurisdiction. Record the source of the rule you applied.

6. Adapting it

Safe to change. The phase and function taxonomies, the status list, and the closeout groups — add a technical or operational closeout group if the project hands over an asset that must be operated, and a stakeholder or consents group where permissions must be discharged or transferred.

Safe to add. A cross-project view that groups lessons by artefact rather than by project, which is how a process owner actually wants to receive them — several projects pointing at one checklist is a much stronger case than one project pointing at it. A quarterly implementation report to the process owners. A field linking a lesson to the risk that was realised, and another linking it to the change that resulted, so the three registers reconcile.

Do not change. The separation of the immediate cause from the condition. The named artefact and the external process owner. The implementation status and the verification evidence. And the rule that the benchmarking capture happens before the team disperses — every organisation that has skipped it has paid for the same data twice, once by not collecting it and again by estimating without it.

6.1 Closing the project

- The lessons register was open from mobilisation, and entries are dated across the whole project, not clustered in the last two months.
- Every lesson has a context a stranger could use, a changeable condition in the "why" field, and a testable instruction.
- Every lesson names an artefact and a process owner outside the project — and the count of blanks in the artefact column has been reported.
- Every process owner has responded, with rejections recorded and reasoned rather than left silent.
- The implementation rate has been calculated and sent to the process owners, not only to the project.
- All four closeout groups are complete, with an owner, a date and an evidence reference against each item.
- Retention periods, data protection treatment and accounting treatment each record the source of the rule applied, rather than carrying one over from another project.
- The benchmarking capture is complete, with metadata against every figure and a data-quality marker distinguishing ledger figures from reconstructions.
- The as-built programme, the schedule updates and the correspondence register are preserved in native format with a named custodian.
- Somebody who will run the next project has read the register and said which entries they will act on.

Related

- BPG-20 – Closeout, lessons learned and benchmarking — the method behind this instrument, including how to run a lessons workshop that produces conditions rather than complaints
- BPG-19 – Project controls assurance and health checks — the assurance view of whether a lessons process is real
- TPL-15 – Project controls health check — the review whose findings route into this register at project end
- BPG-10 – Contingency and management reserve — where the contingency drawdown ratio captured here goes to work on the next project
- BPG-09 – Estimate at completion – choosing and defending a method — the forecasting practice the outturn ratios are captured to improve

Sources and standards

This is an original instrument developed by the Institute. It reproduces no third-party template, form or checklist. The closeout groups, the register fields and the benchmarking measures are the Institute's own formulation, offered for adaptation. Retention periods, data protection obligations, revenue recognition, capitalisation and tax treatment all vary by contract, by jurisdiction and by accounting framework; this template requires that the source of the rule applied is recorded and presents no jurisdiction's treatment as universal.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.